

NON-WSQ COURSE

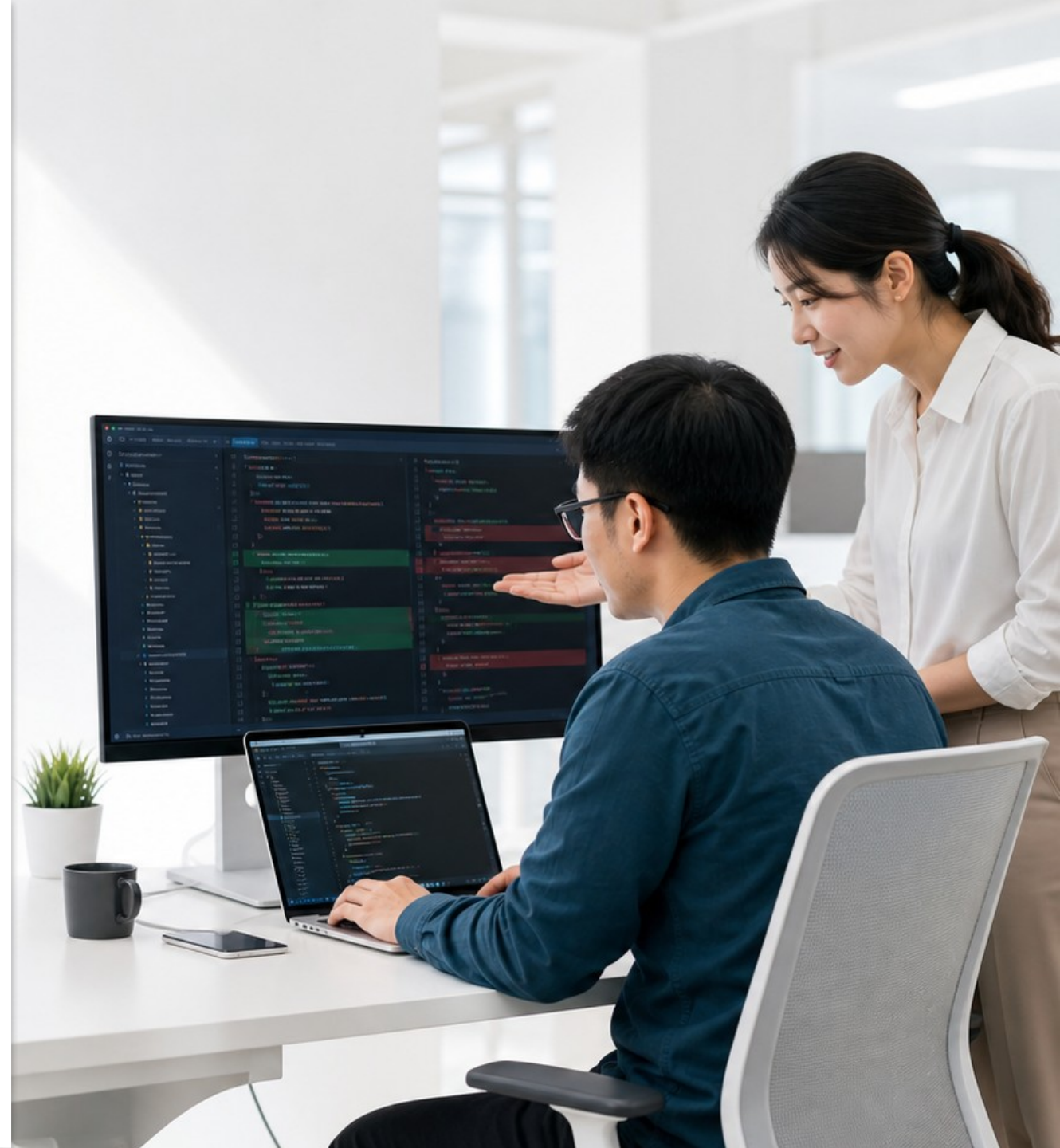
# Agentic AI Applications with Codex

Inspect → plan → act → verify

C695 | v1.0 | 1 day / 7.5 hours

Tertiary Infotech Academy Pte. Ltd.  
UEN 201200696W

Mechanism-led courseware • Ten self-contained Codex labs



# Trainer profile

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**Your trainer facilitates evidence-led AI application design and governed automation.**

1

**Agentic workflow architecture**

2

Codex harness and context engineering

3

Git, testing and browser verification

4

Technical evaluation and remediation

# Dr Alfred Ang

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AA

## Principal Trainer, Tertiary Infotech Academy

1

Doctorate-level applied technology educator

2

AI, data, cloud and automation curriculum leadership

3

Industry-oriented evaluation and implementation practice

4

Focus: explainable, governed technical outcomes

# Learner introduction

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**Connect your existing role to one software or knowledge-work automation problem.**

**1** Role and organisation context

**2** Current development or operations workflow

**3** One repeated process bottleneck

**4** One risk you want the agent to control

# Working agreements

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**RULE**

**Use governed experimentation throughout the course.**

**1**

**Use synthetic or approved data only**

**2**

**Never paste live secrets into prompts or files**

**3**

**Keep consequential actions behind approval**

**4**

**Challenge completion claims with observable evidence**

# Course facts and delivery context

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**C695**

## Commercial short course

1

Course code: C695

2

Duration: 1 day / 7.5 instructional hours

3

Delivery: facilitated workshop

4

Version: v1.0 / 20 September 2026

# Morning learning plan

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AM

**Applicability, agentic loops and a bounded Codex control harness.**

1

**Codex surfaces, task lifecycle and prompt contracts**

2

Strengths, limitations, accountability and fit

3

AGENTS.md, context, sandbox and approvals

4

Labs 01–05: baseline through worktree isolation

# Afternoon learning plan

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PM

**Extension surfaces, evaluation and governed adoption.**

1

Tools, browser QA, MCP, skills and subagents

2

Headless and cloud automation

3

Golden tasks and comparative effectiveness

4

Labs 06–10 and governed capstone

# Access the labs

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**Ten self-contained lab folders carry the complete end-to-end workflow.**

1

Open the individual lab README/PDF

2

Use only its supplied starter repository

3

Follow detailed steps in the Learner Guide

4

Record evidence in the folder checklist

<https://github.com/tertiarycourses/C695-Agentic-AI-Applications-with-Codex>

# 1

## TOPIC 1

# Codex Surfaces, Agentic Loops and Applicability

Analyse where Codex fits, how a task travels through the agentic loop, and where human judgment remains essential.

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1

CONTROL

2

EVIDENCE

3

AUTOMATION

4

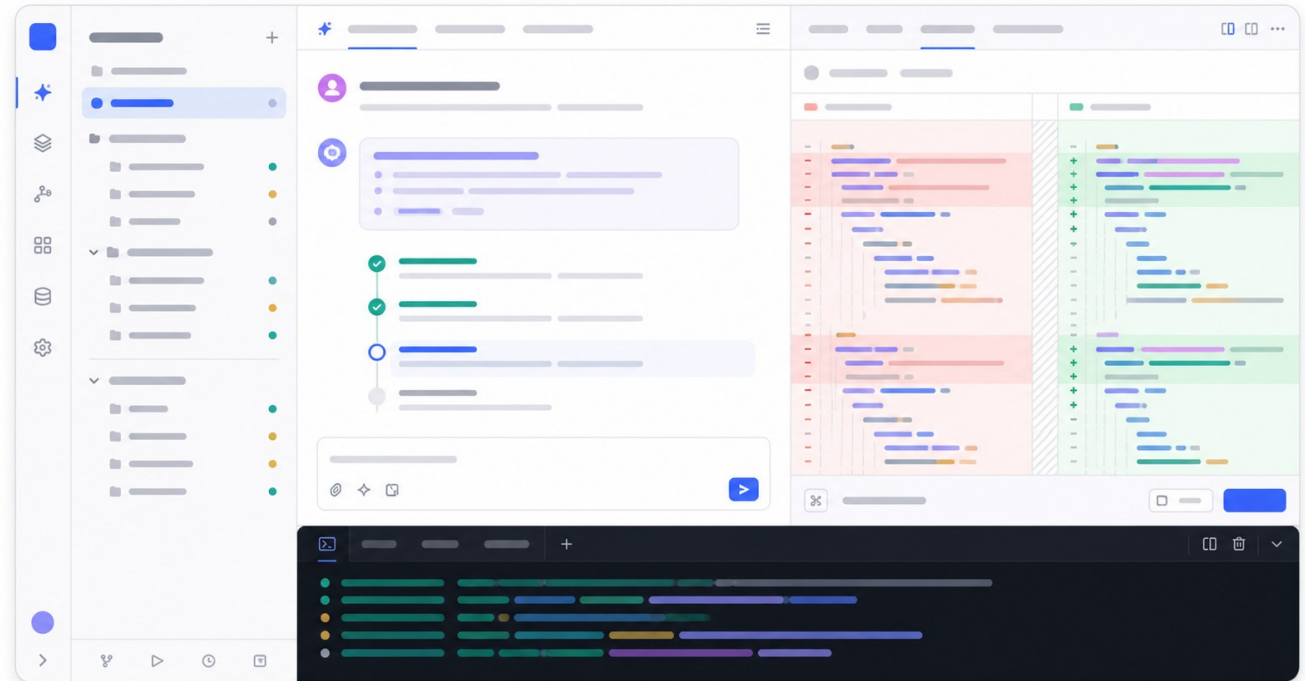
DECISION

Every mechanism ends in observable evidence—not an agent's assertion.

# 01 · The Codex product surface: execution path

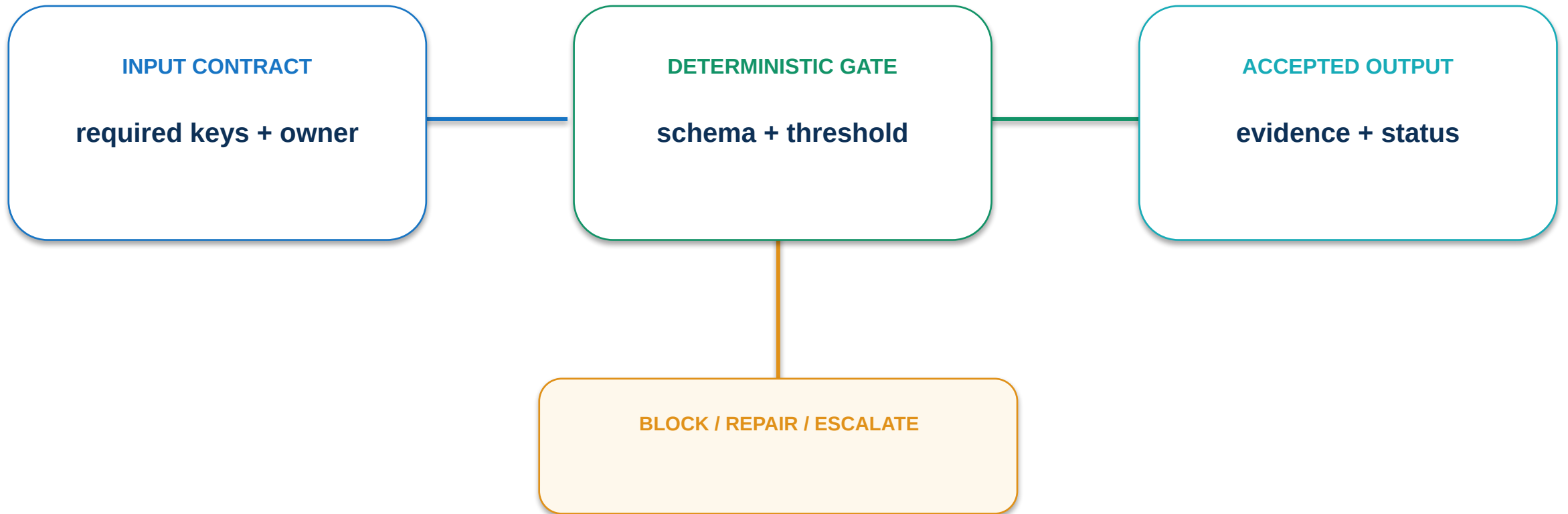
## CODEX COMMAND CENTRE

- 1 PROJECT** Repository and instruction scope
- 2 TASK** Conversation, plan and progress
- 3 CHANGE** Reviewable file diff
- 4 VERIFY** Terminal and test evidence



CONCEPT VISUAL • LABELLED

# 01 · The Codex product surface: contract and control



Choose by data location, duration, isolation and collaboration need.

## 01 · The Codex product surface: evidence and decision

### MEASURE

**Task-to-surface fit score across local, IDE, cloud and app**

### ACCEPTANCE EVIDENCE

Decision matrix plus selected surface rationale

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

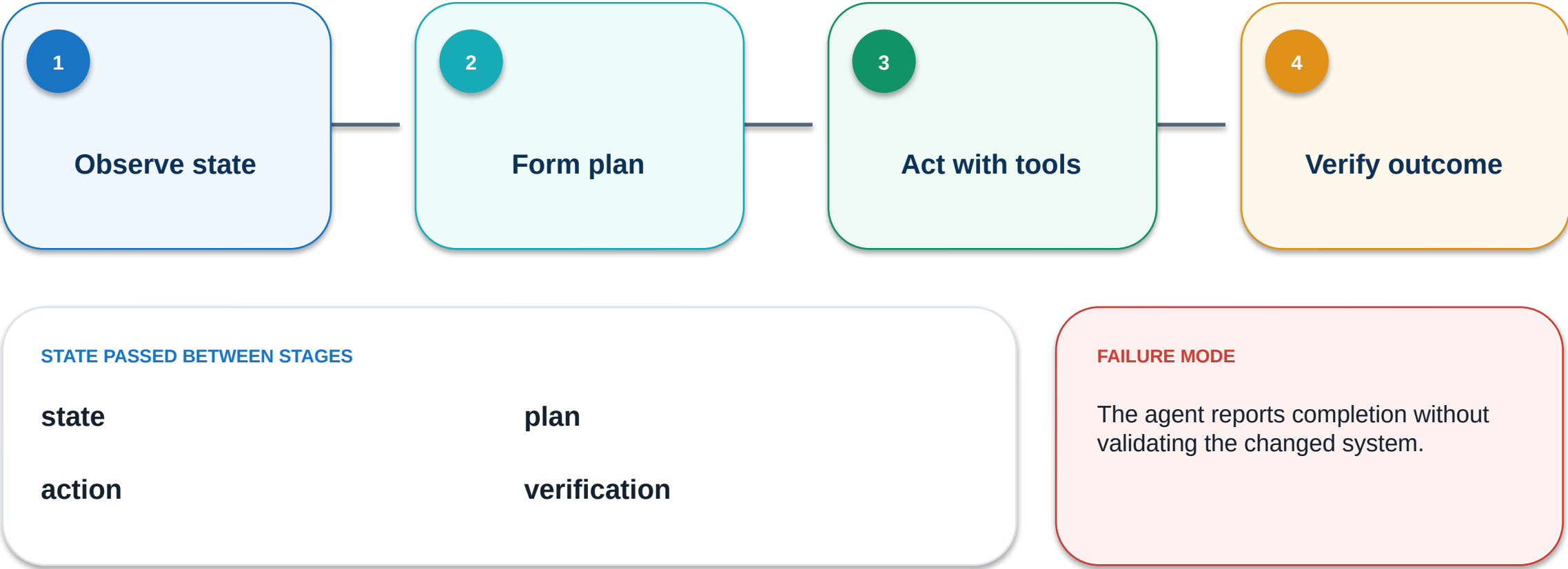
PASS / REPAIR

### OWNER

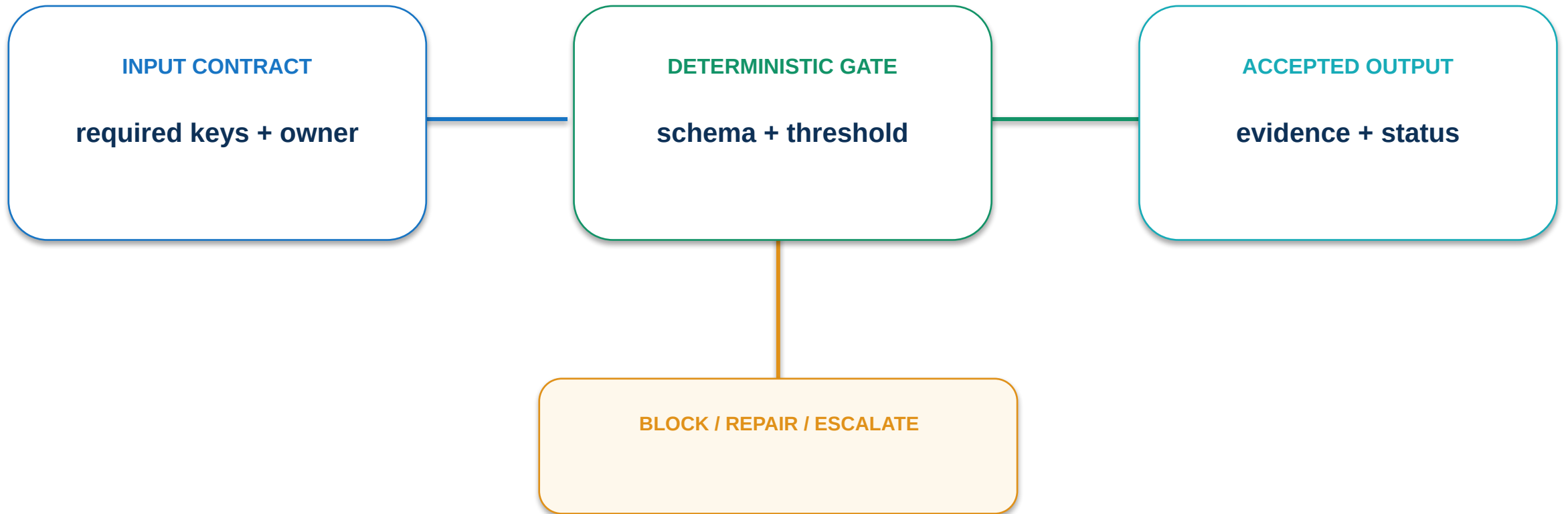
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 02 · Agentic task lifecycle: execution path



## 02 · Agentic task lifecycle: contract and control



**Make verification an explicit finish condition.**

## 02 · Agentic task lifecycle: evidence and decision

MEASURE

**Verified task transitions / total task transitions**

ACCEPTANCE EVIDENCE

Plan, command trace, test result and final diff

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

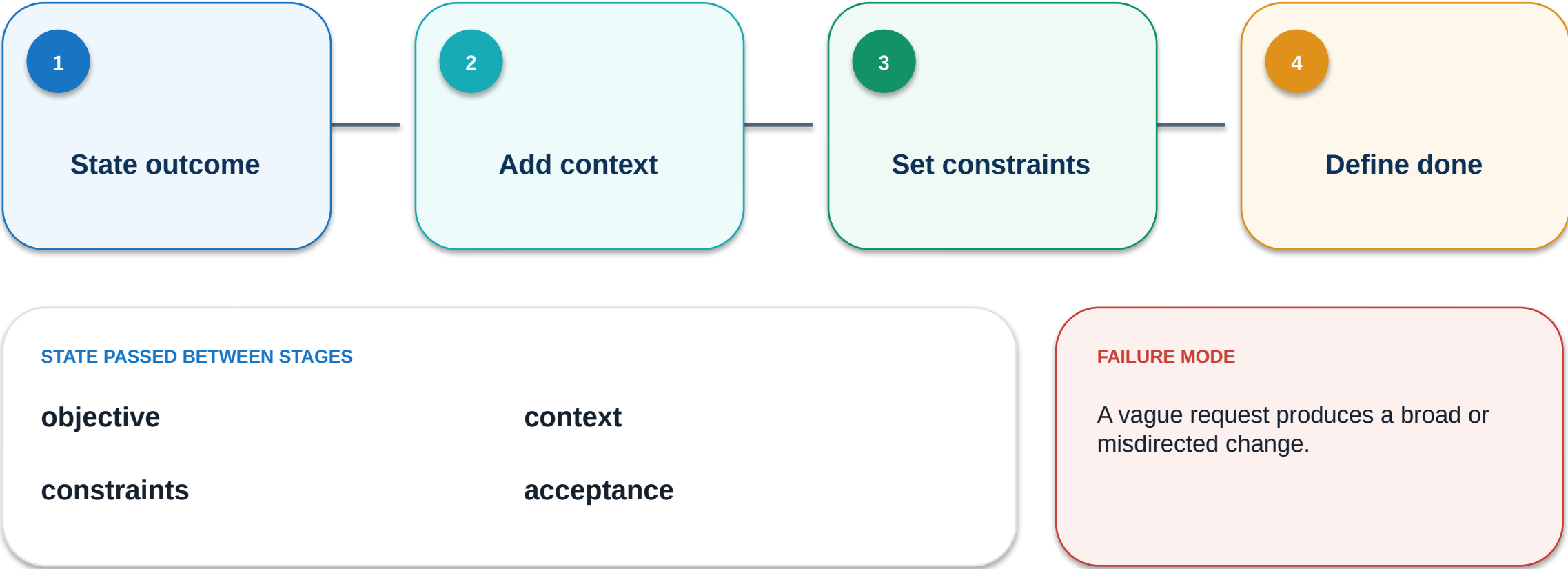
PASS / REPAIR

OWNER

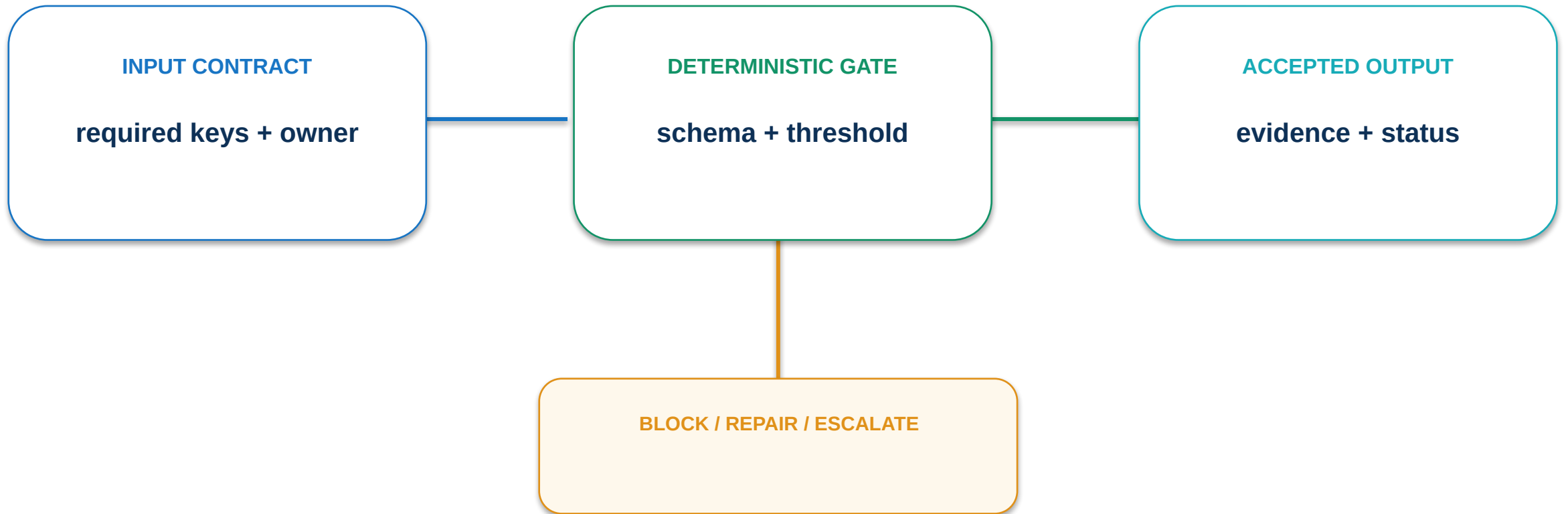
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 03 · Prompt as an execution contract: execution path



## 03 · Prompt as an execution contract: contract and control



Require observable acceptance checks and protected boundaries.

## 03 · Prompt as an execution contract: evidence and decision

### MEASURE

**Acceptance criteria satisfied on first review**

### ACCEPTANCE EVIDENCE

Prompt contract linked to passing checks

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

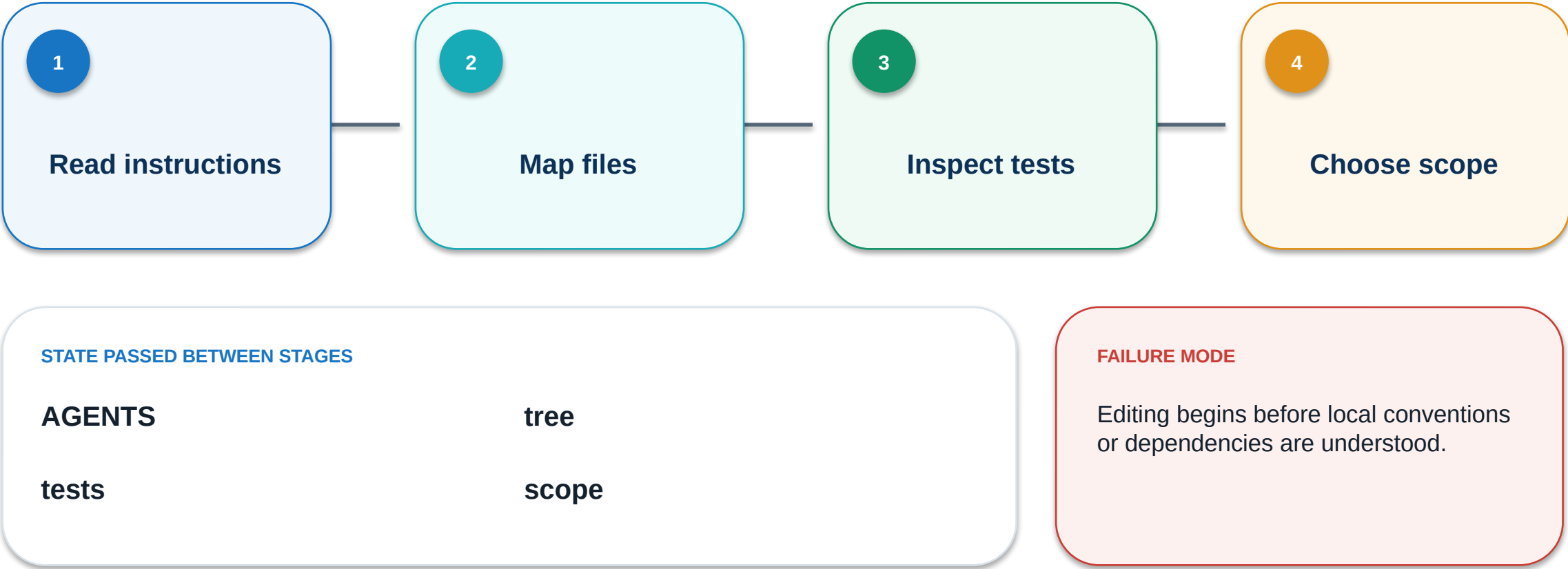
PASS / REPAIR

### OWNER

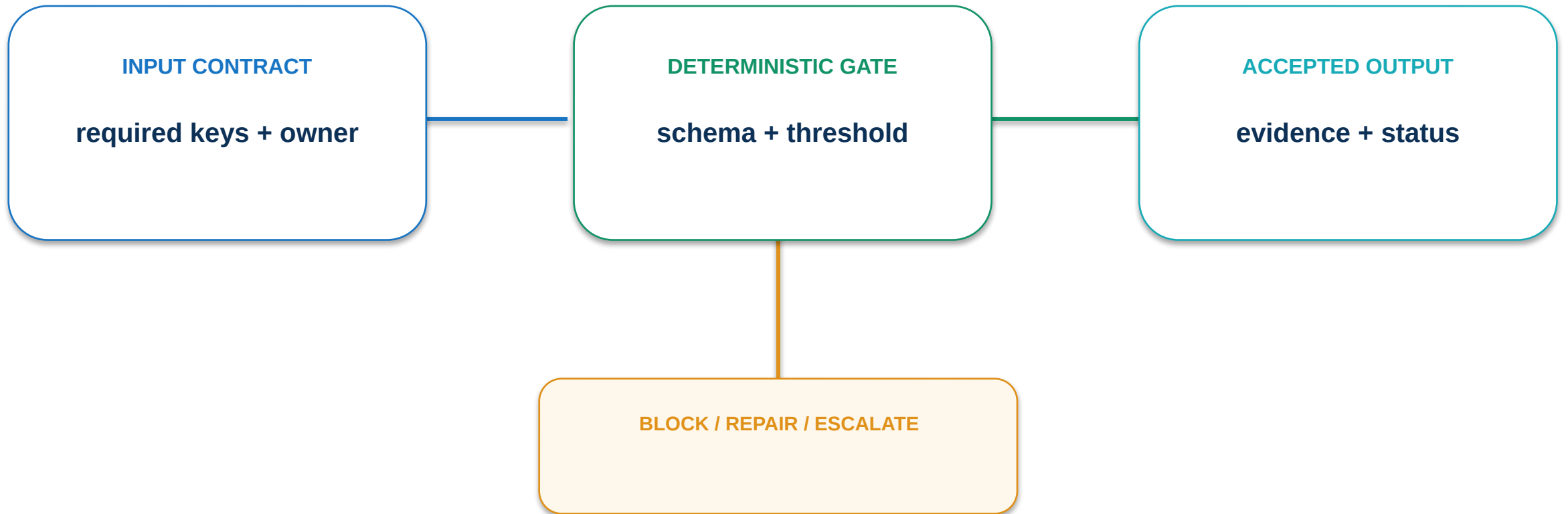
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 04 · Repository reconnaissance: execution path



## 04 · Repository reconnaissance: contract and control



Front-load read-only inspection and instruction discovery.

## 04 · Repository reconnaissance: evidence and decision

MEASURE

**Relevant files inspected before first edit**

ACCEPTANCE EVIDENCE

Inspection log and scoped file list

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

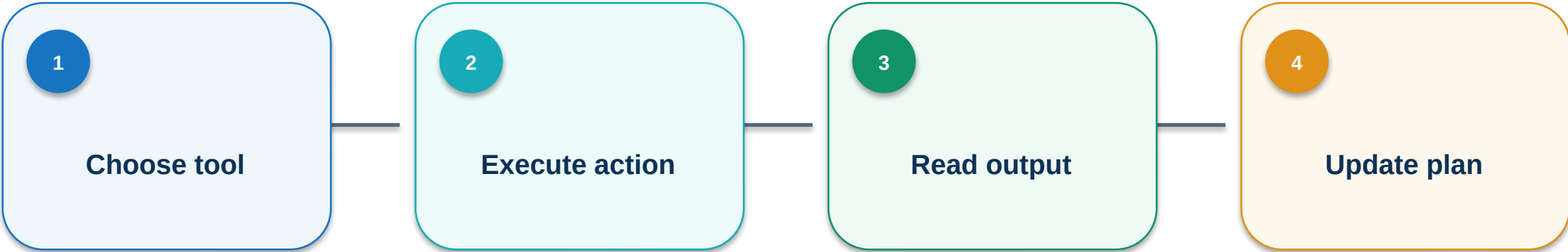
PASS / REPAIR

OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 05 · Tool-mediated reasoning: execution path



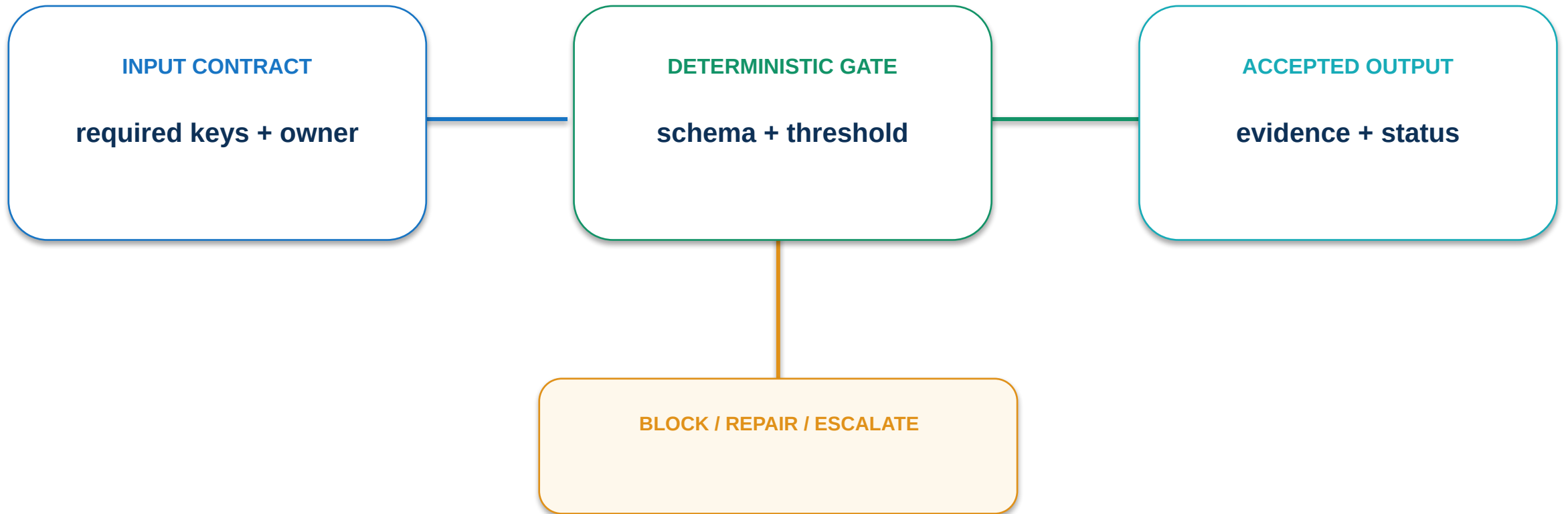
STATE PASSED BETWEEN STAGES

|               |                  |
|---------------|------------------|
| <b>tool</b>   | <b>command</b>   |
| <b>result</b> | <b>next_step</b> |

FAILURE MODE

Repeated tool calls add noise without reducing uncertainty.

## 05 · Tool-mediated reasoning: contract and control



Tie each call to one decision and stop when evidence is sufficient.

## 05 · Tool-mediated reasoning: evidence and decision

MEASURE

**Useful tool results / total tool calls**

ACCEPTANCE EVIDENCE

Tool result cited in the next decision

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

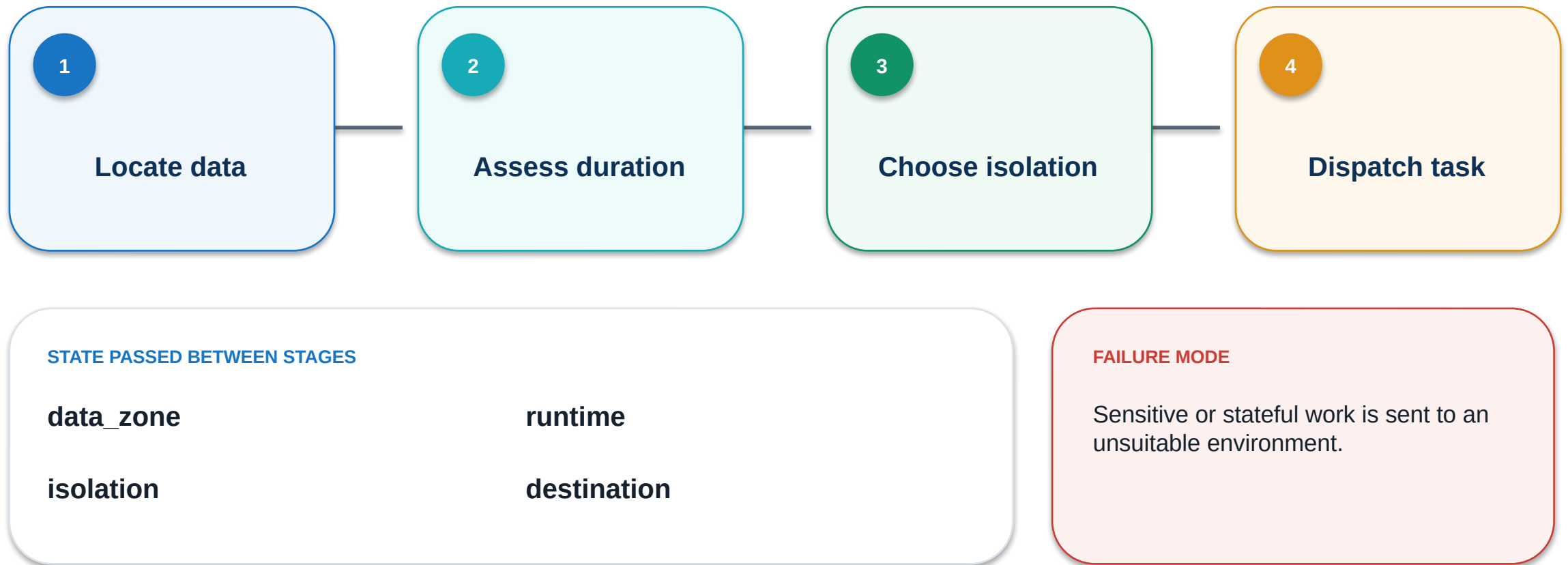
PASS / REPAIR

OWNER

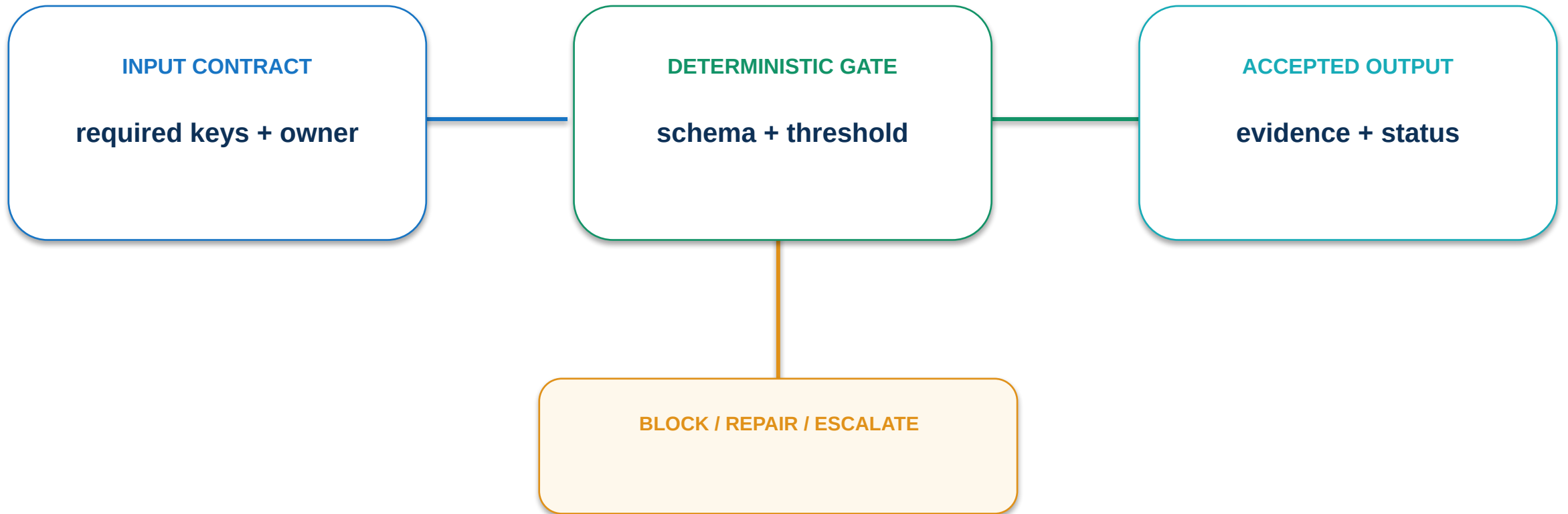
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 06 · Local versus cloud execution: execution path



## 06 · Local versus cloud execution: contract and control



Select the execution boundary before attaching the repository.

## 06 · Local versus cloud execution: evidence and decision

MEASURE

**Policy-compliant runs completed without relocation**

ACCEPTANCE EVIDENCE

Environment choice and policy check

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

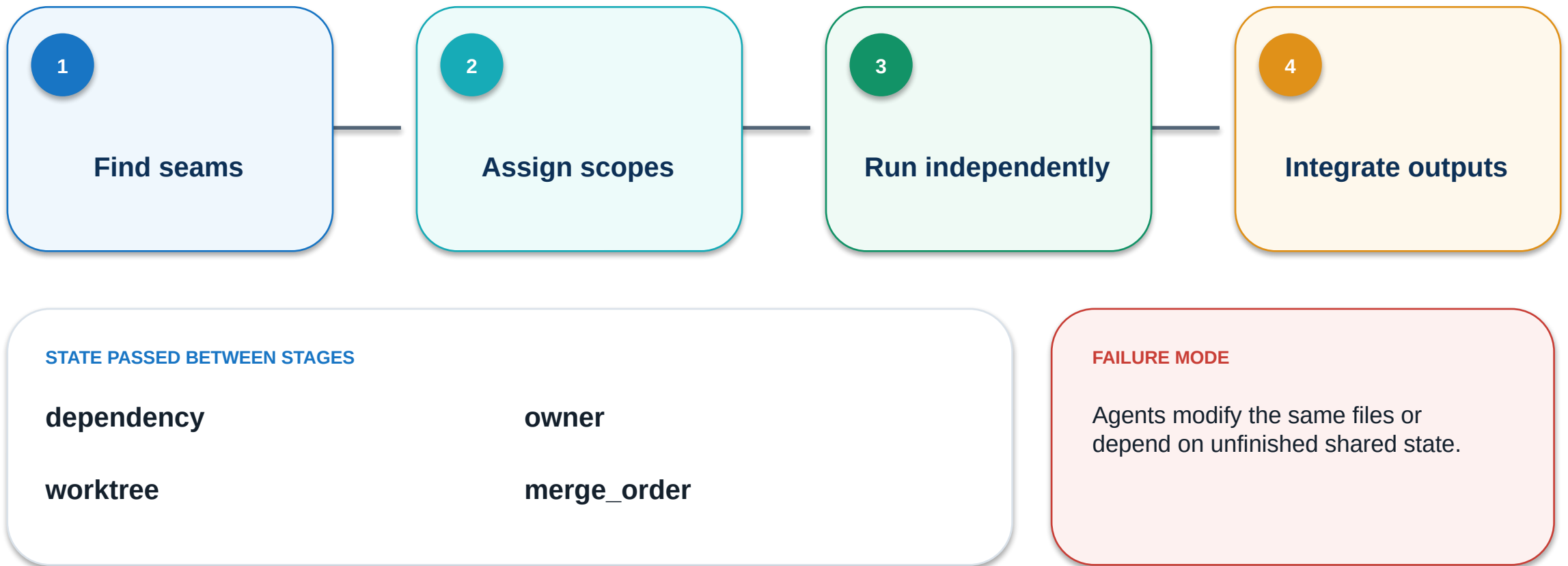
PASS / REPAIR

OWNER

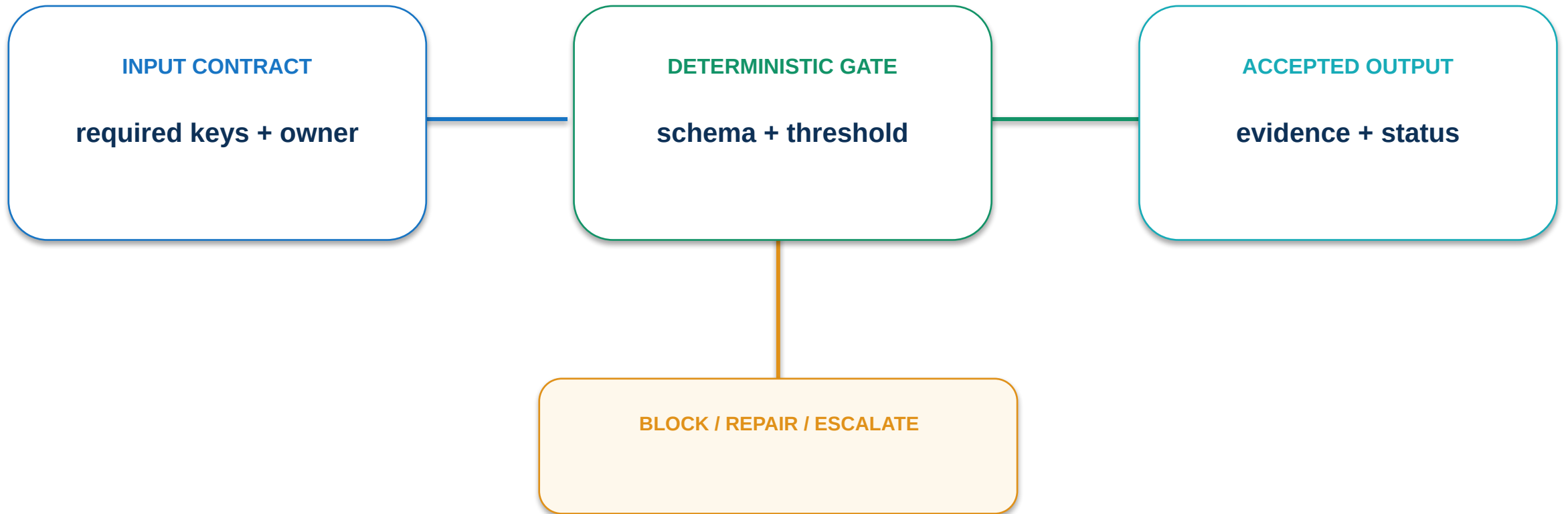
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 07 · Parallel task decomposition: execution path



## 07 · Parallel task decomposition: contract and control



**Decompose only at dependency-light boundaries.**

## 07 · Parallel task decomposition: evidence and decision

### MEASURE

**Parallel speed-up with zero overlapping edits**

### ACCEPTANCE EVIDENCE

Scope map, separate worktrees and integration checks

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

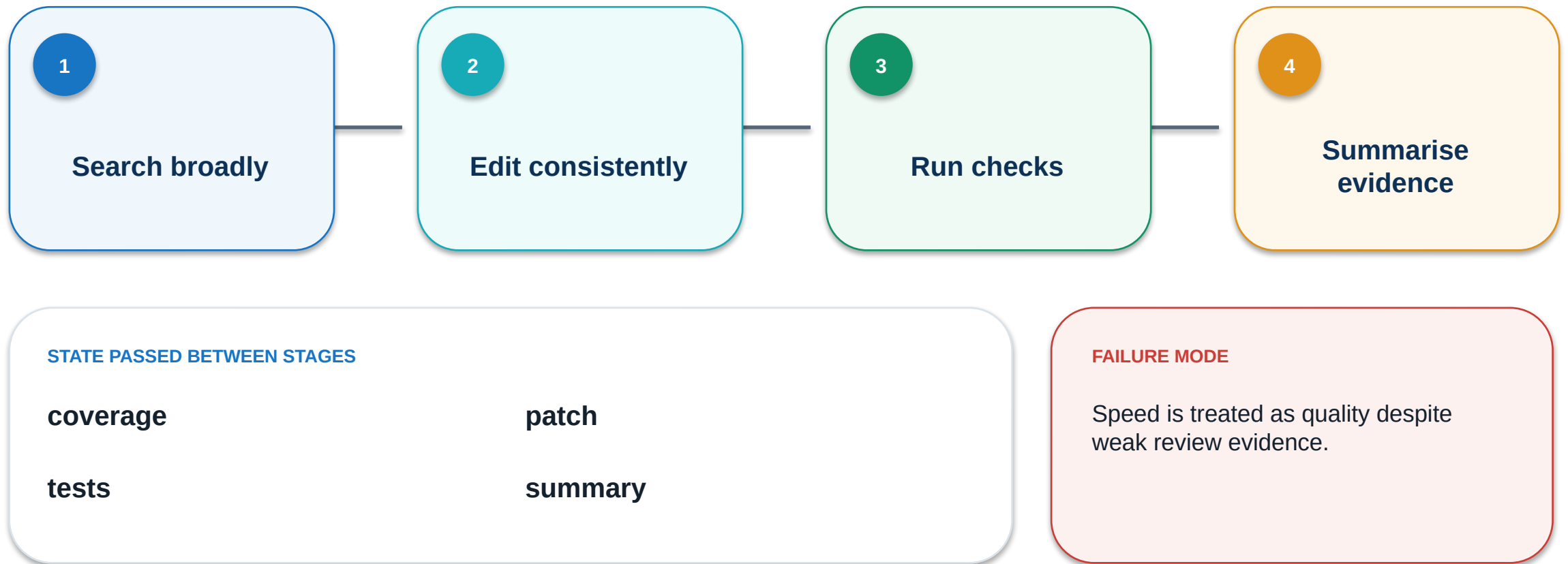
PASS / REPAIR

### OWNER

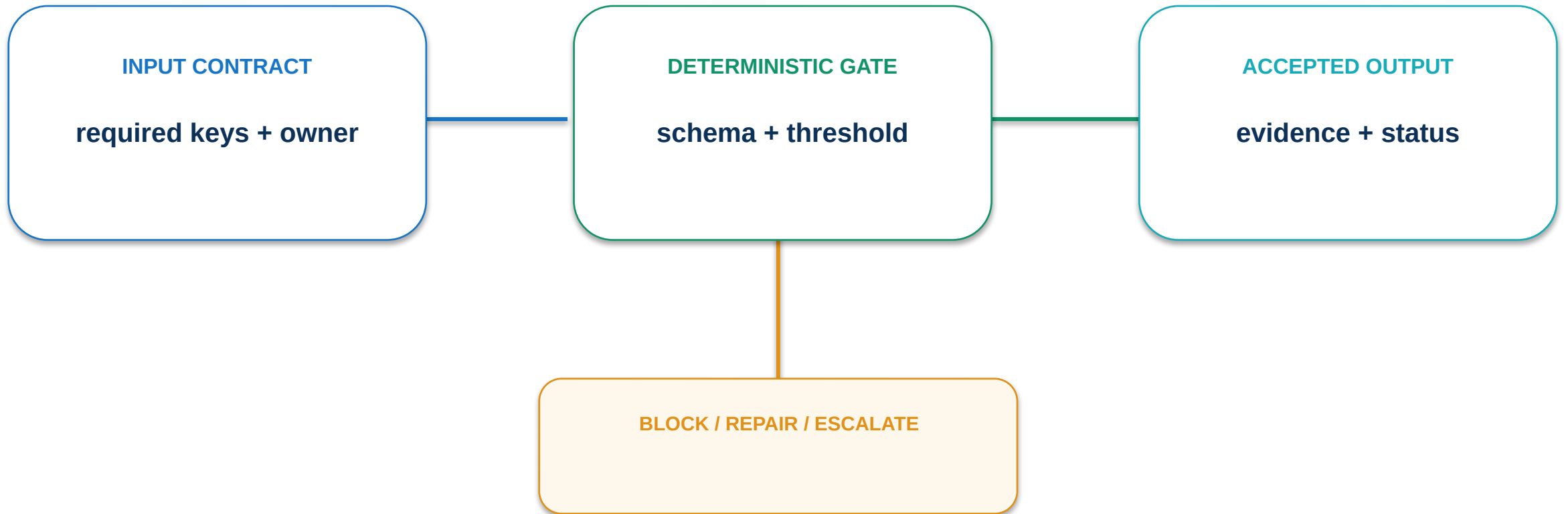
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 08 · Strengths of coding agents: execution path



## 08 · Strengths of coding agents: contract and control



Measure throughput and quality together.

## 08 · Strengths of coding agents: evidence and decision

### MEASURE

**Cycle time reduction at unchanged defect escape rate**

### ACCEPTANCE EVIDENCE

Baseline versus assisted task metrics

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

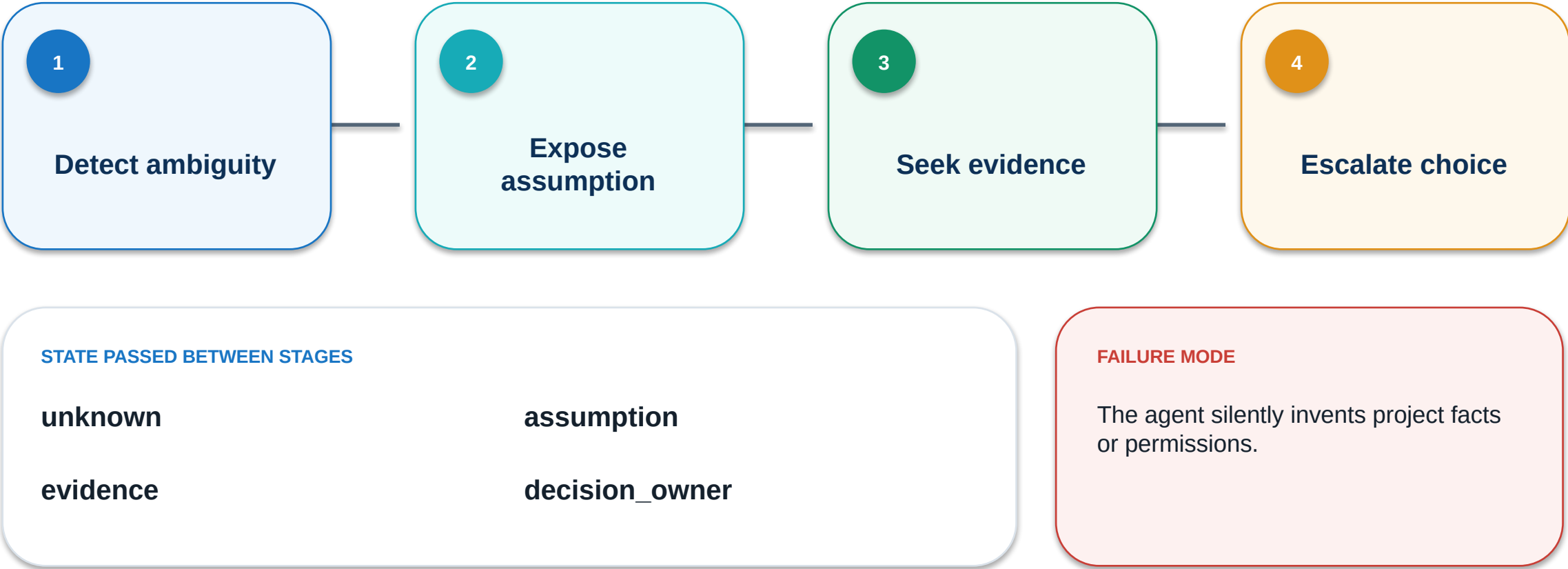
PASS / REPAIR

### OWNER

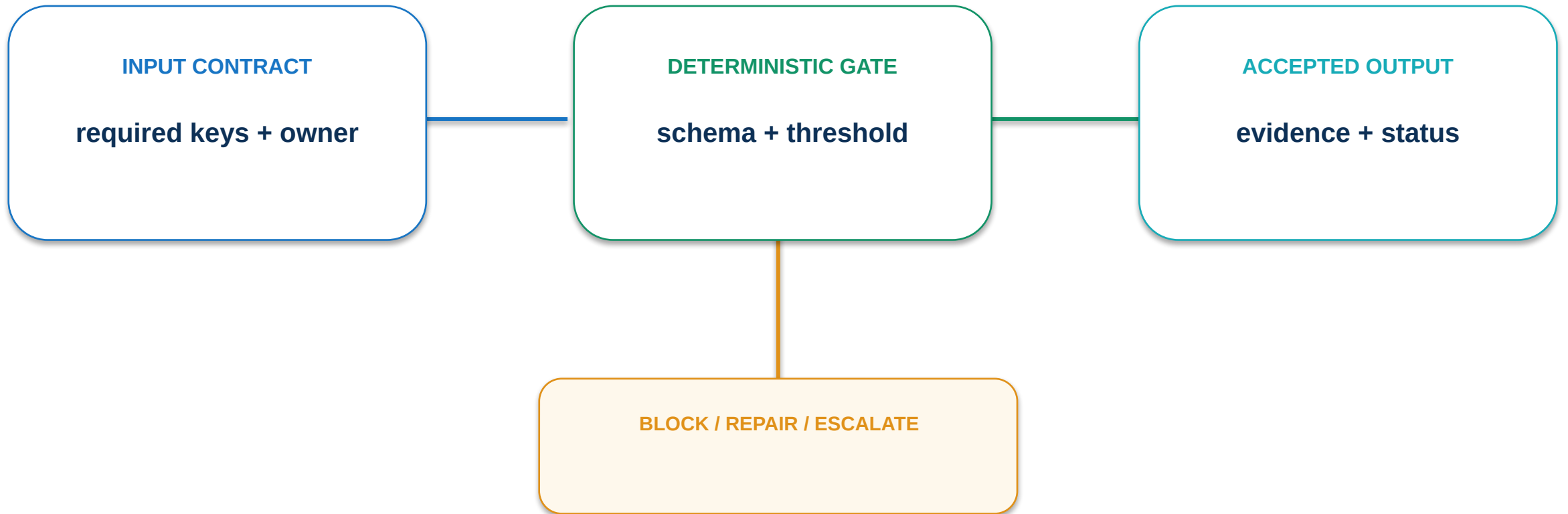
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 09 · Limitations and uncertainty: execution path



## 09 · Limitations and uncertainty: contract and control



Use uncertainty thresholds and explicit escalation points.

## 09 · Limitations and uncertainty: evidence and decision

### MEASURE

**Material assumptions surfaced before mutation**

### ACCEPTANCE EVIDENCE

Assumption register with resolved evidence

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

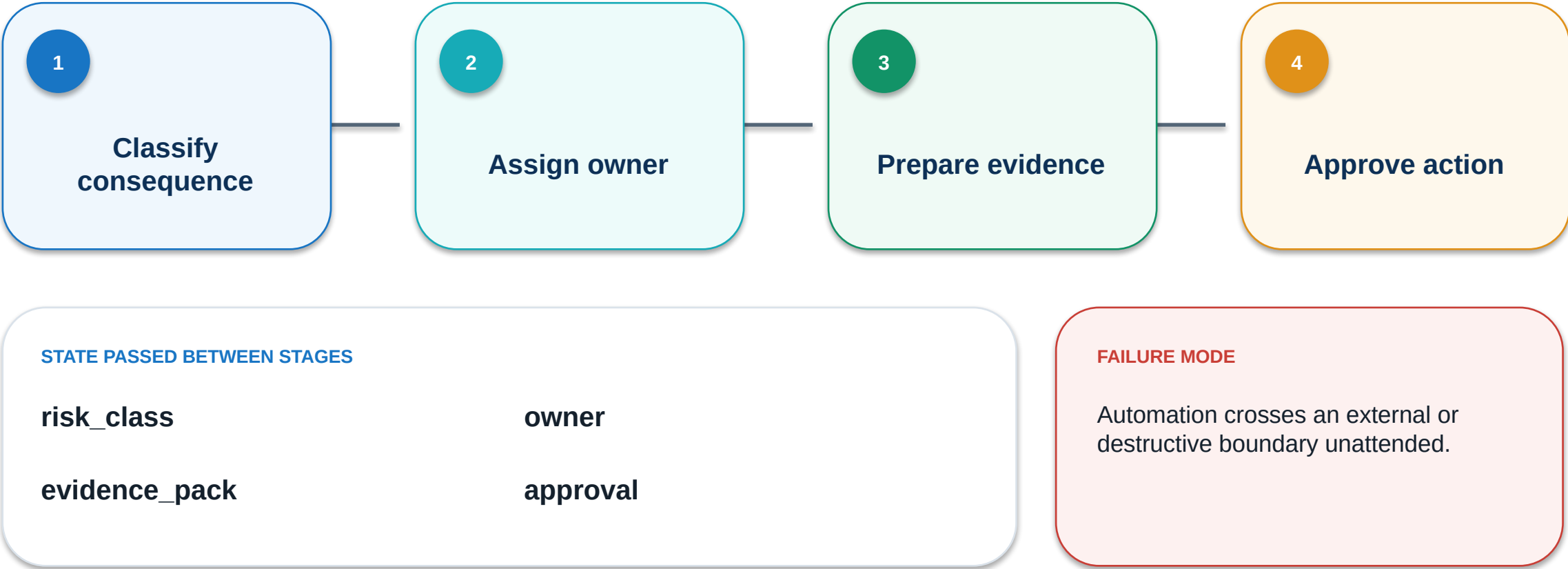
PASS / REPAIR

### OWNER

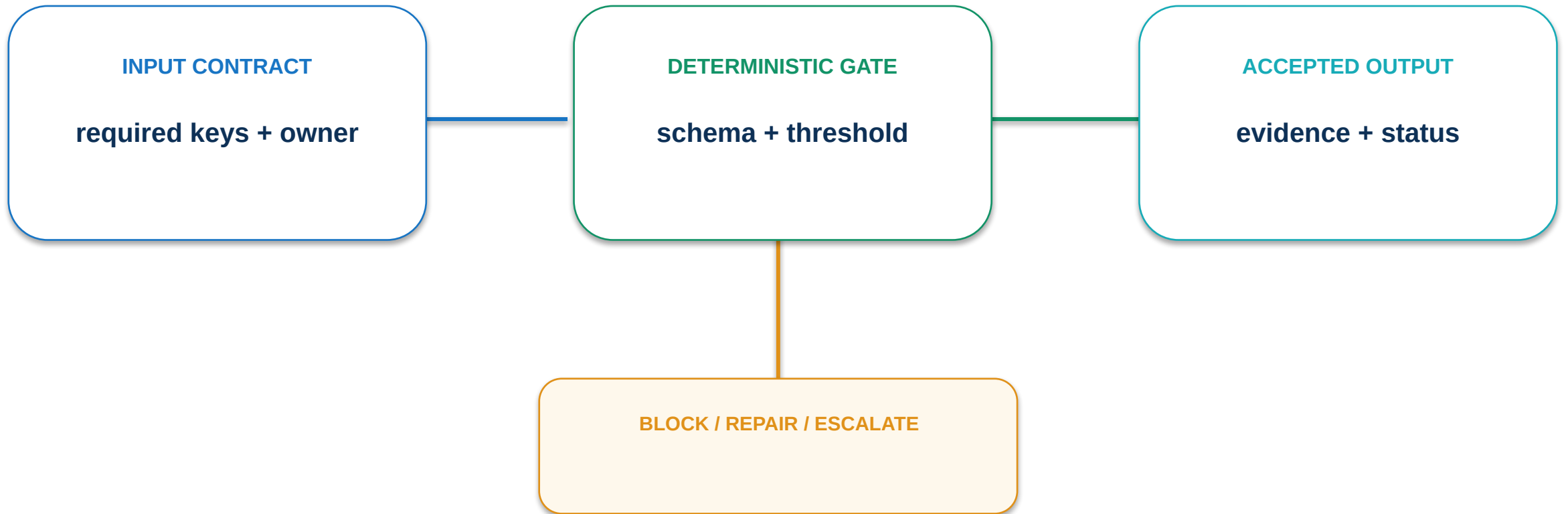
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 10 · Human accountability boundary: execution path



## 10 · Human accountability boundary: contract and control



Require current evidence and a named human before consequence.

## 10 · Human accountability boundary: evidence and decision

MEASURE

**Consequential actions with named approval**

ACCEPTANCE EVIDENCE

Approval record bound to the reviewed change

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

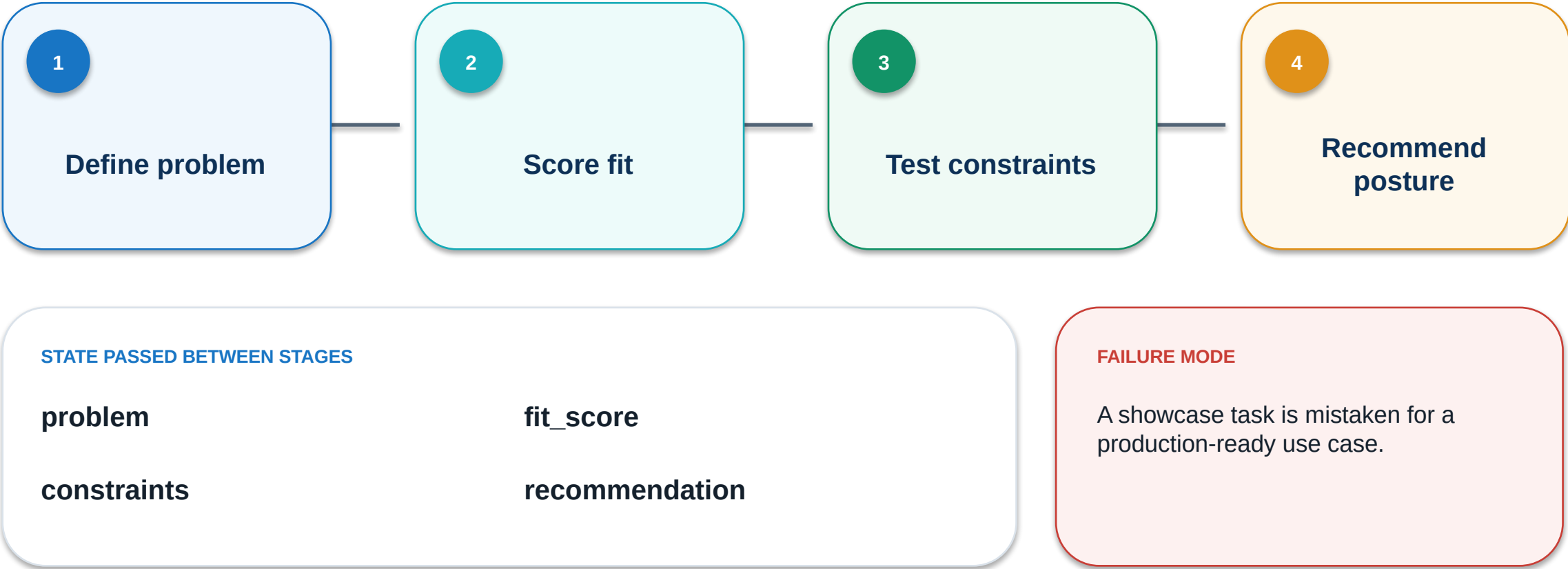
PASS / REPAIR

OWNER

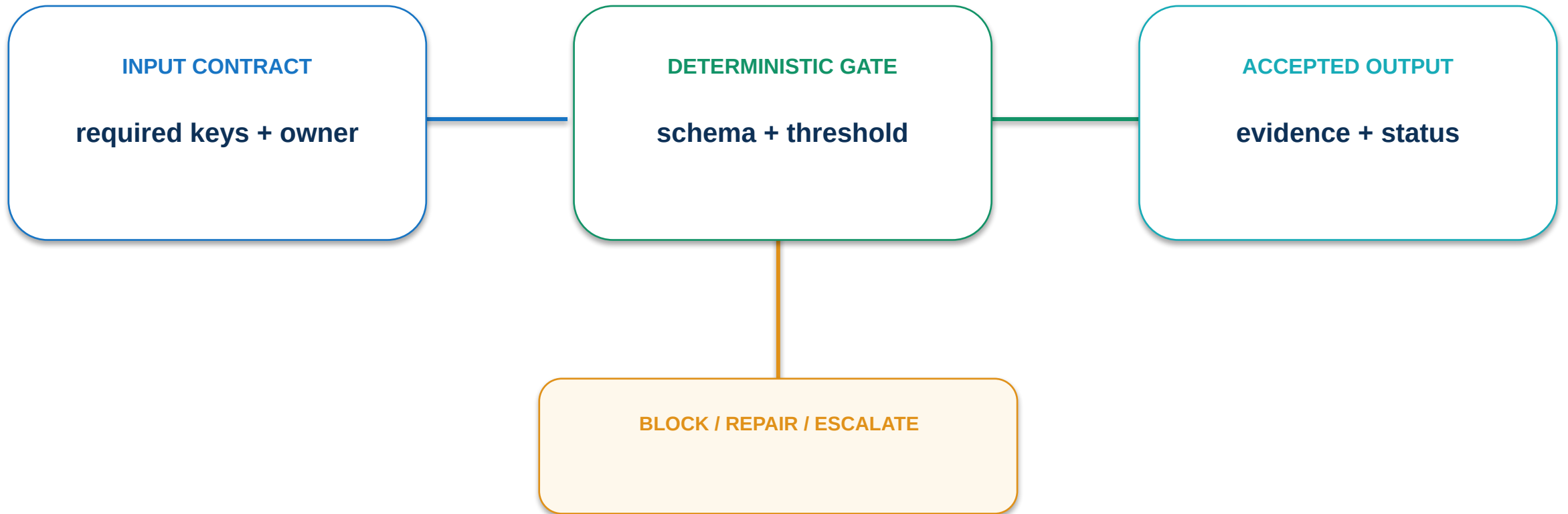
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 11 · Applicability decision matrix: execution path

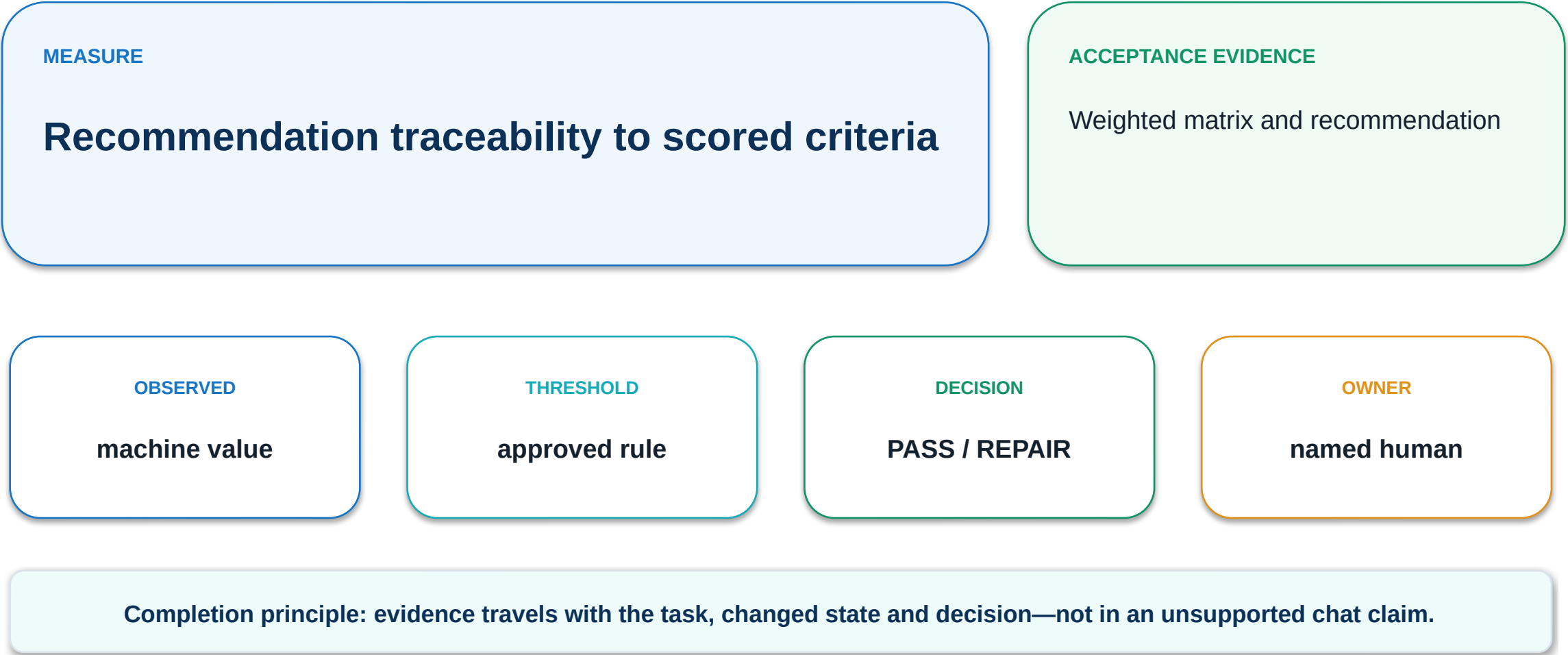


## 11 · Applicability decision matrix: contract and control



Score repeatability, verifiability, risk and integration cost.

# 11 · Applicability decision matrix: evidence and decision

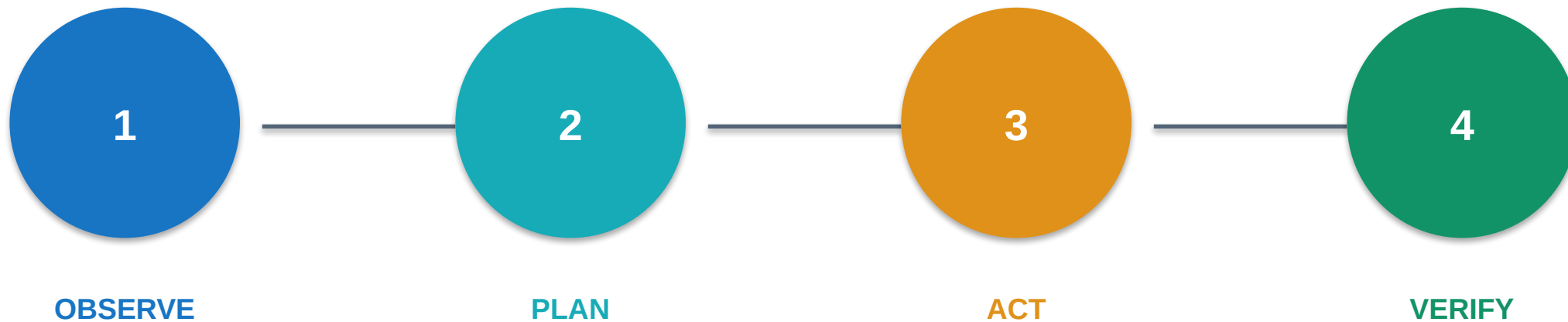


# Applicability boundary: Codex, validator and accountable human



# Task lifecycle: OBSERVE, PLAN, ACT and VERIFY

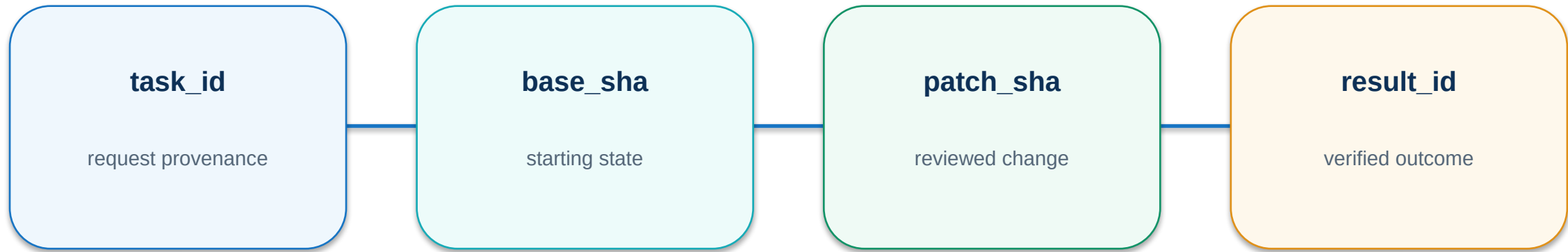
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**VERIFY** closes the loop only when the requested state and checks agree.

## Task evidence chain: request → base → diff → result

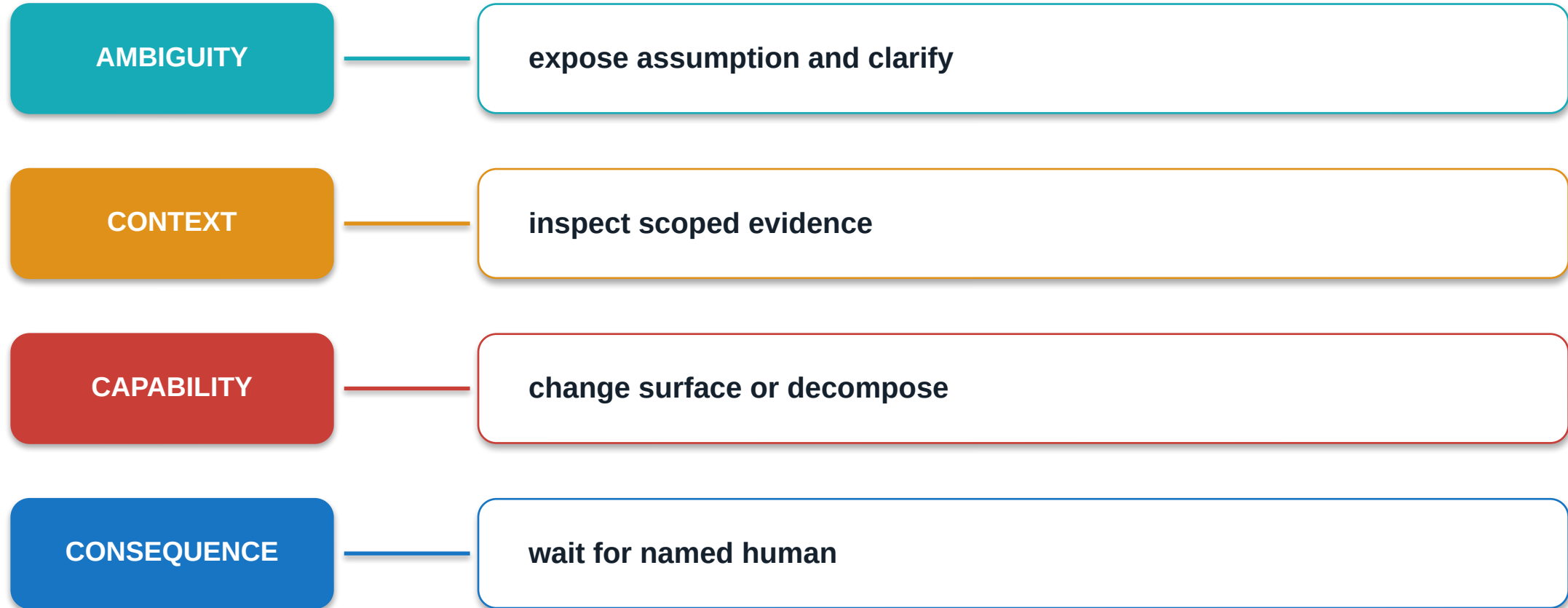
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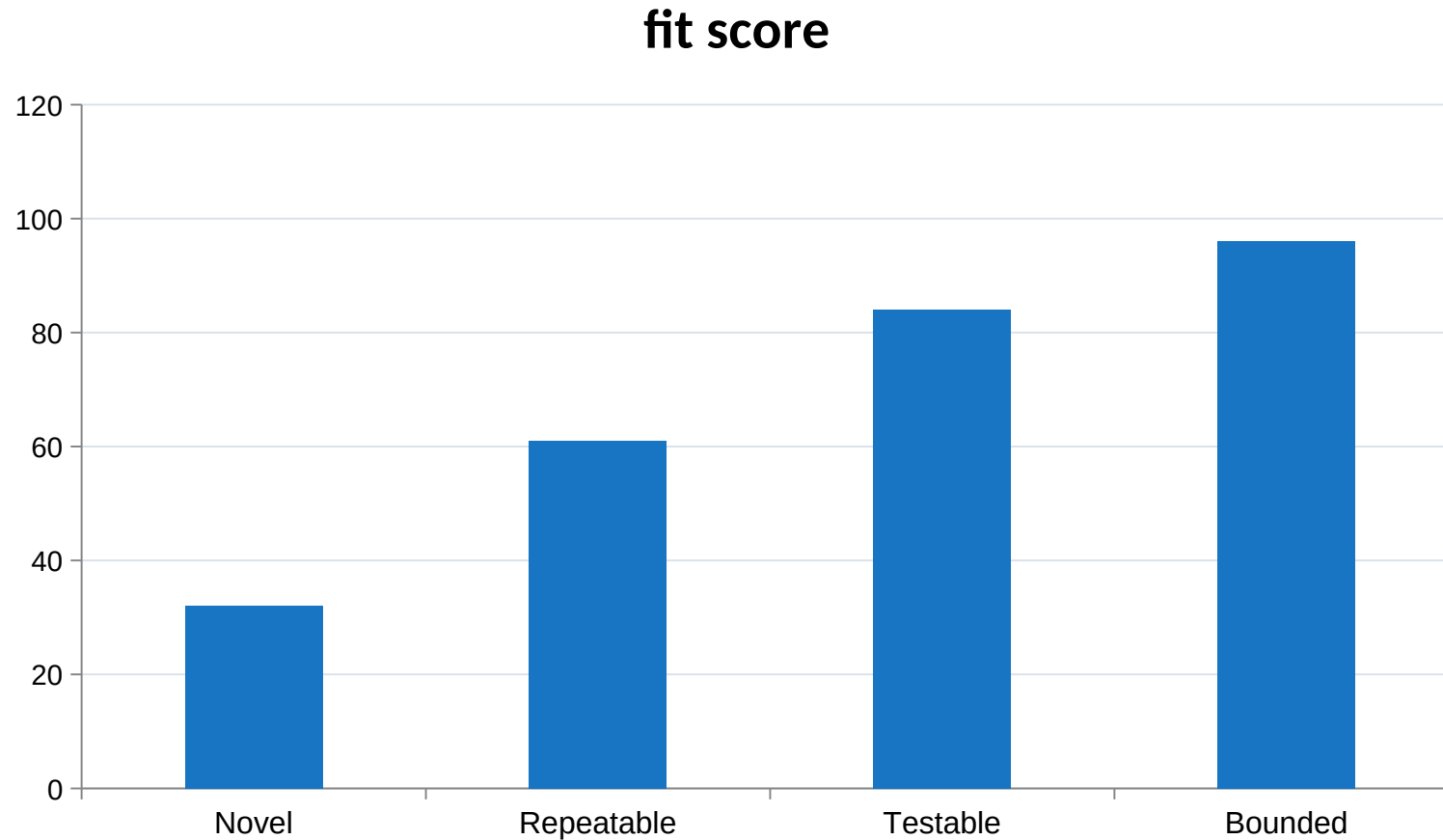
**A change is explainable when each identifier resolves to retrievable evidence.**

# Limitation taxonomy: ambiguity, context, capability and consequence

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# Applicability rises with repeatability and verifiability

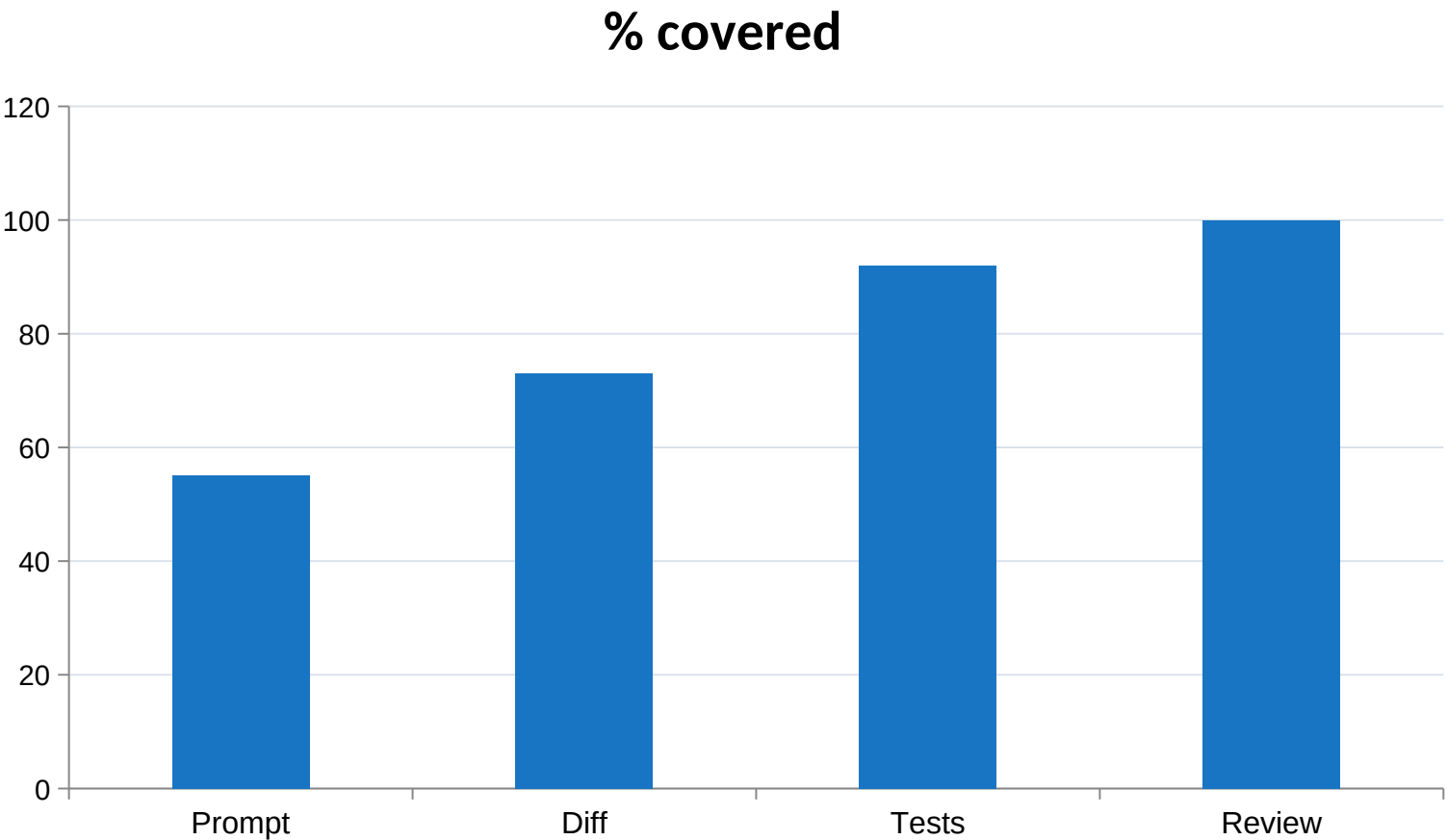


## INSIGHT

**Codex fits best when the task is repeatable, bounded and objectively verifiable.**

Illustrative synthetic values for mechanism explanation.

# Verification coverage closes the completion gap

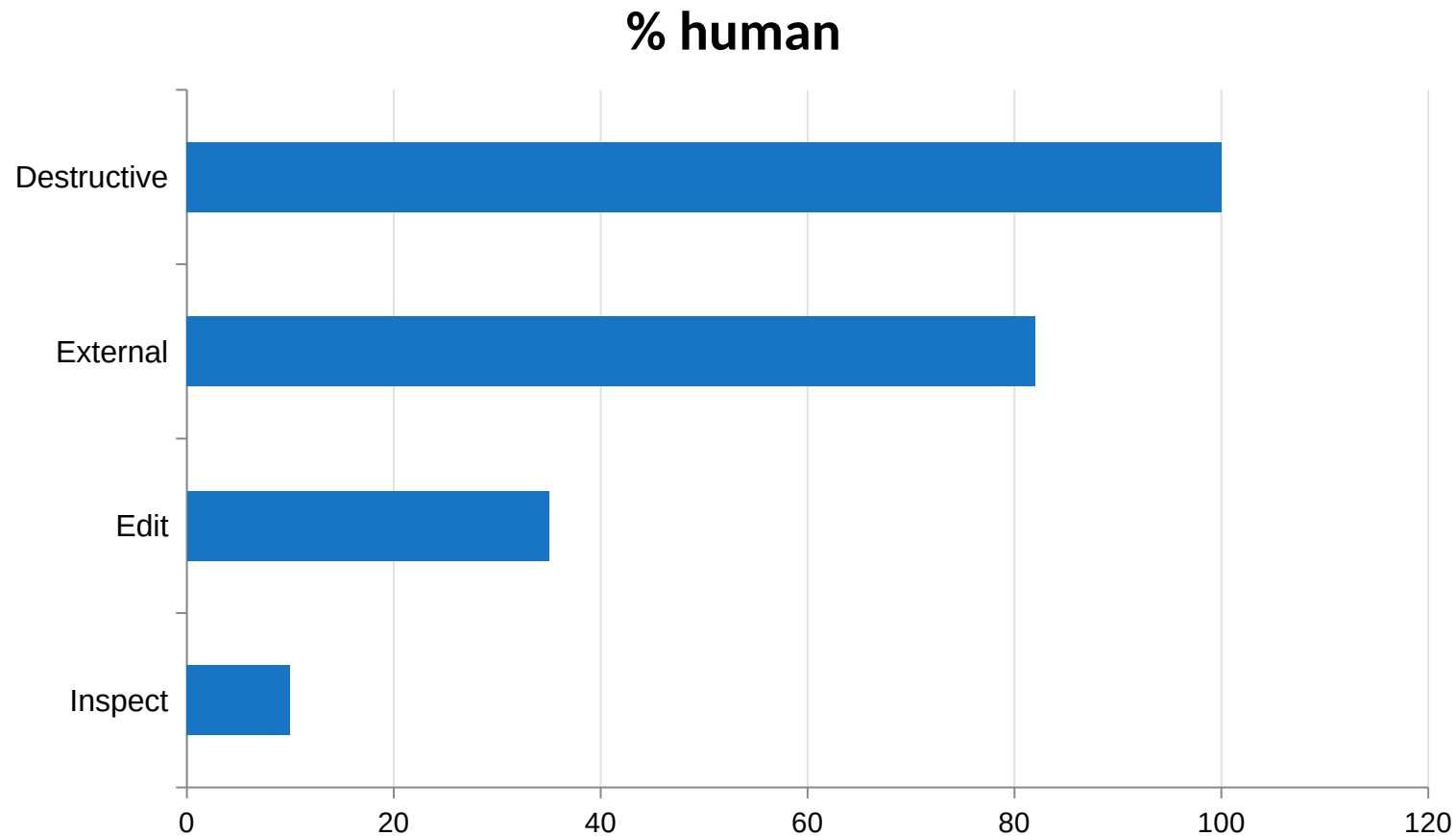


INSIGHT

**Completion is credible only when the request, changed state, checks and review remain connected.**

Illustrative synthetic values for mechanism explanation.

# Human ownership increases with consequence



## INSIGHT

**Human accountability rises as an action becomes external, destructive or otherwise consequential.**

Illustrative synthetic values for mechanism explanation.

# Lab 01 · Repository Reconnaissance and Baseline

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## SCENARIO

**Inspect a TaskBoard repository, identify its instruction and test boundaries, and record a measurable baseline before any edit.**

## EXPECTED OUTPUT

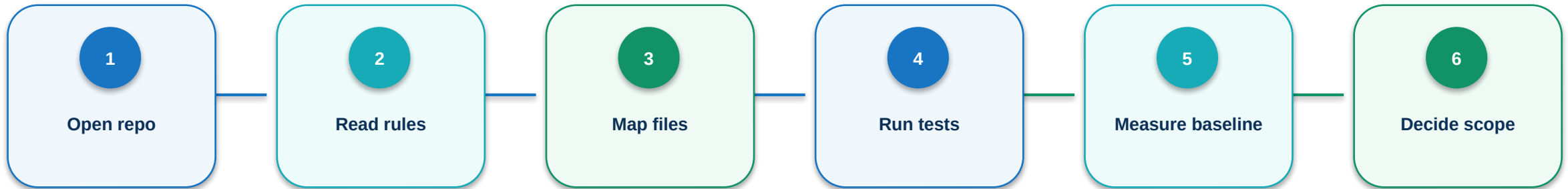
**Baseline report and verified repository map**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-01-repository-reconnaissance/**

## Lab 01 · Codex workflow contract

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**Each node consumes named keys, emits a status, and leaves retrievable execution evidence.**

## Lab 01 · evidence and failure gates

### PASS EVIDENCE

The report names the applicable instruction chain, test command, baseline result and scoped target files.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Lab 02 · Prompt-to-Plan Feature Delivery

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### SCENARIO

**Turn an outcome contract into a bounded plan, implement one TaskBoard feature and verify the changed behavior.**

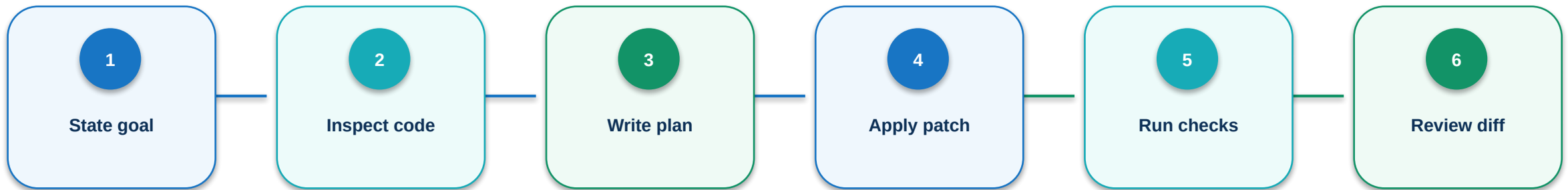
### EXPECTED OUTPUT

**Minimal patch, passing tests and decision summary**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-02-prompt-plan-feature/**

## Lab 02 · Codex workflow contract



**Consequential actions require a current diff, passing checks and a named decision.**

### HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

## Lab 02 · evidence and failure gates

### PASS EVIDENCE

The diff is scoped, the new behavior is tested, all supplied tests pass and the final summary cites evidence.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

# Topic 1 technical checkpoint

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1

**Which state field prevents a silent downstream assumption?**

2

**What evidence proves the mechanism completed successfully?**

3

**Which failure must stop rather than retry?**

4

**Who owns the consequential decision?**

# Topic 1 transfer challenge

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1

**Change one input constraint and predict the affected nodes.**

2

**Name the identifier that preserves provenance across the change.**

3

**Define one machine threshold and one human decision.**

4

**Describe the minimum evidence required to resume the run.**

# 2

## TOPIC 2

# Harness Design, Context, Sandbox and Process Efficiency

Design a bounded Codex harness that turns project context, permissions and Git isolation into an efficient process.

1

CONTROL

2

EVIDENCE

3

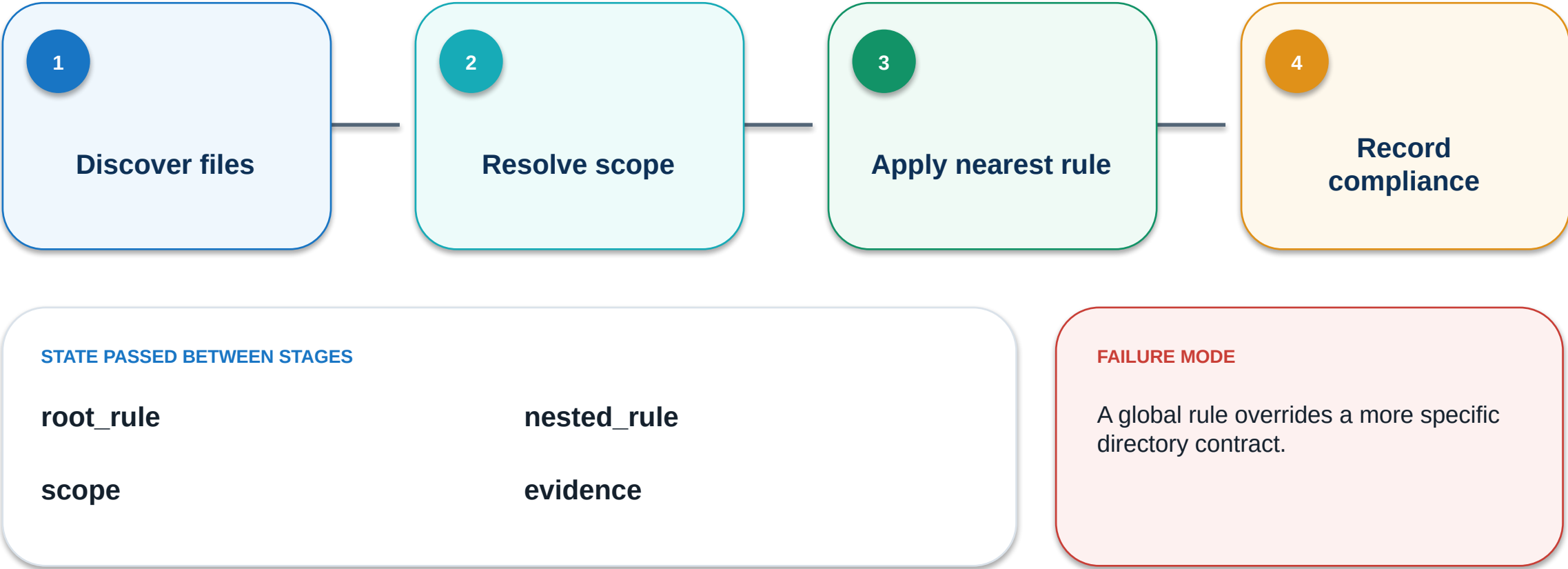
AUTOMATION

4

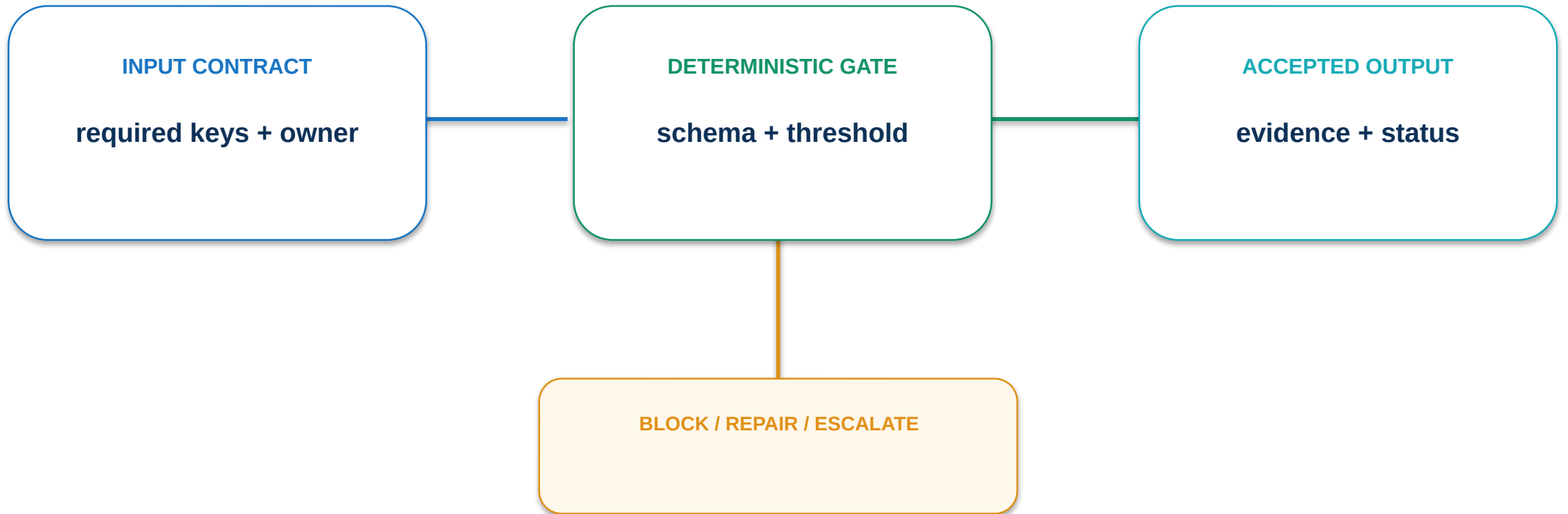
DECISION

Every mechanism ends in observable evidence—not an agent's assertion.

# 01 · AGENTS.md instruction hierarchy: execution path



# 01 · AGENTS.md instruction hierarchy: contract and control



Resolve instructions from repository root to target file.

## 01 · AGENTS.md instruction hierarchy: evidence and decision

### MEASURE

**Edits compliant with the nearest applicable instruction**

### ACCEPTANCE EVIDENCE

Instruction chain and compliant patch

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

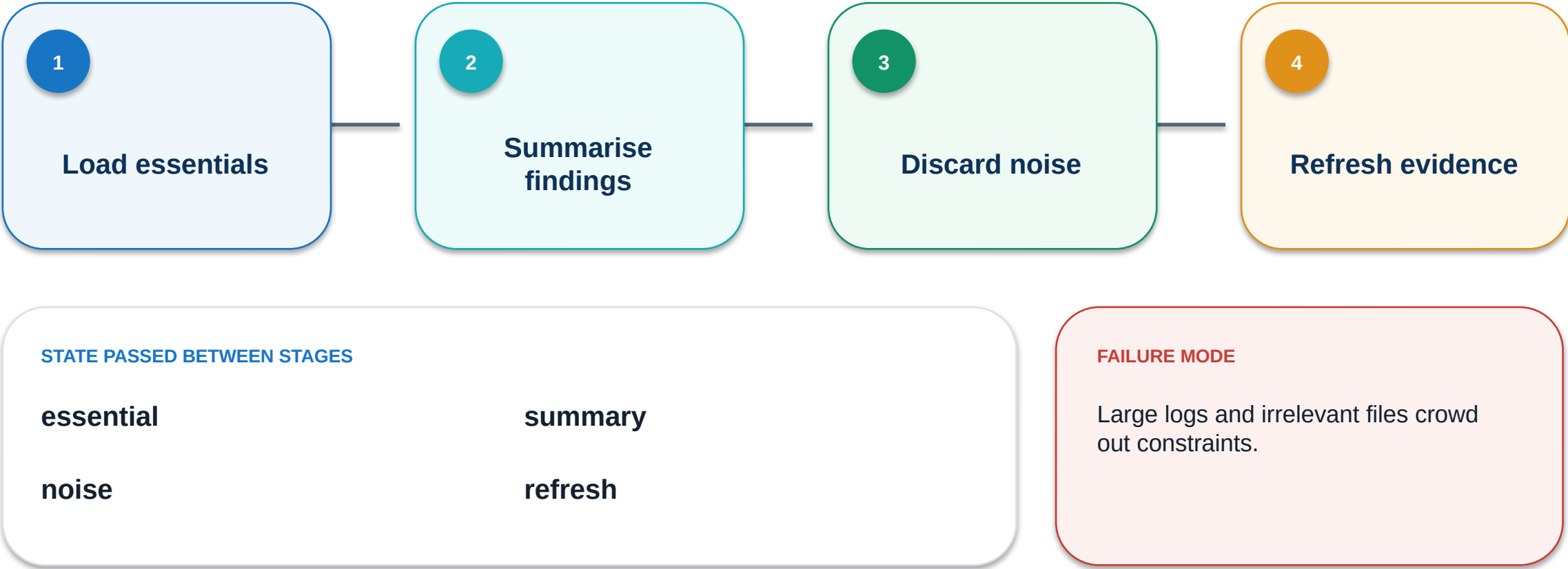
PASS / REPAIR

### OWNER

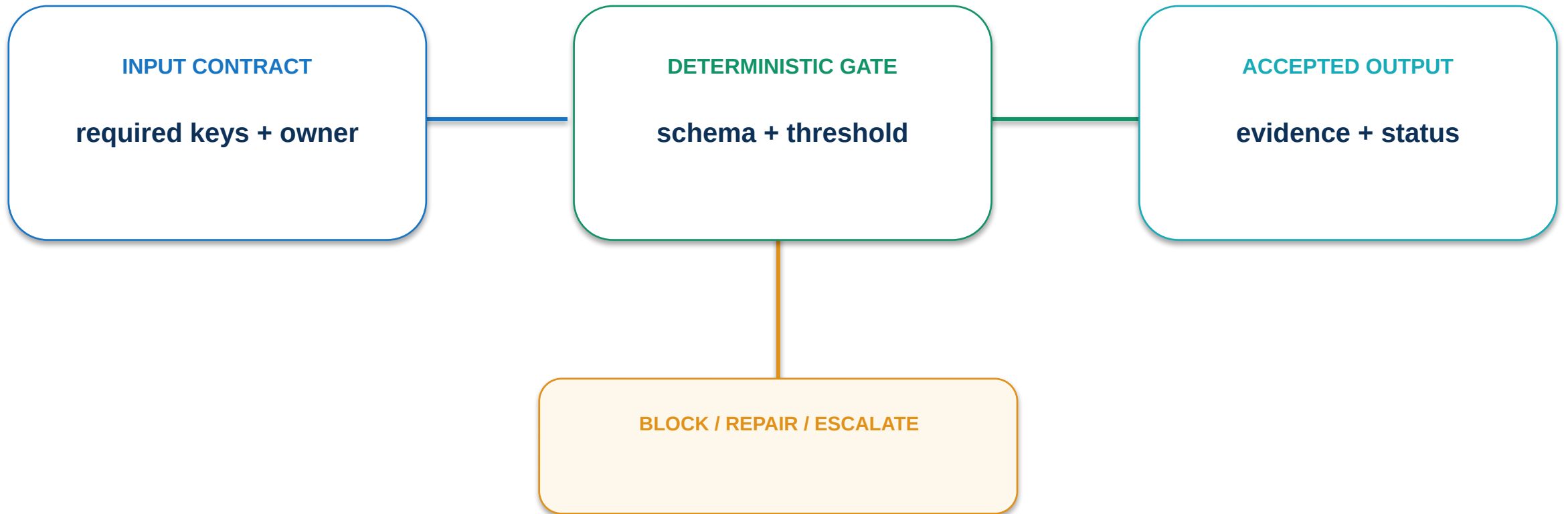
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 02 · Context budget engineering: execution path



## 02 · Context budget engineering: contract and control



Load progressively and summarise before expanding scope.

## 02 · Context budget engineering: evidence and decision

MEASURE

**Decision-relevant context / total loaded context**

ACCEPTANCE EVIDENCE

Compact context ledger

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

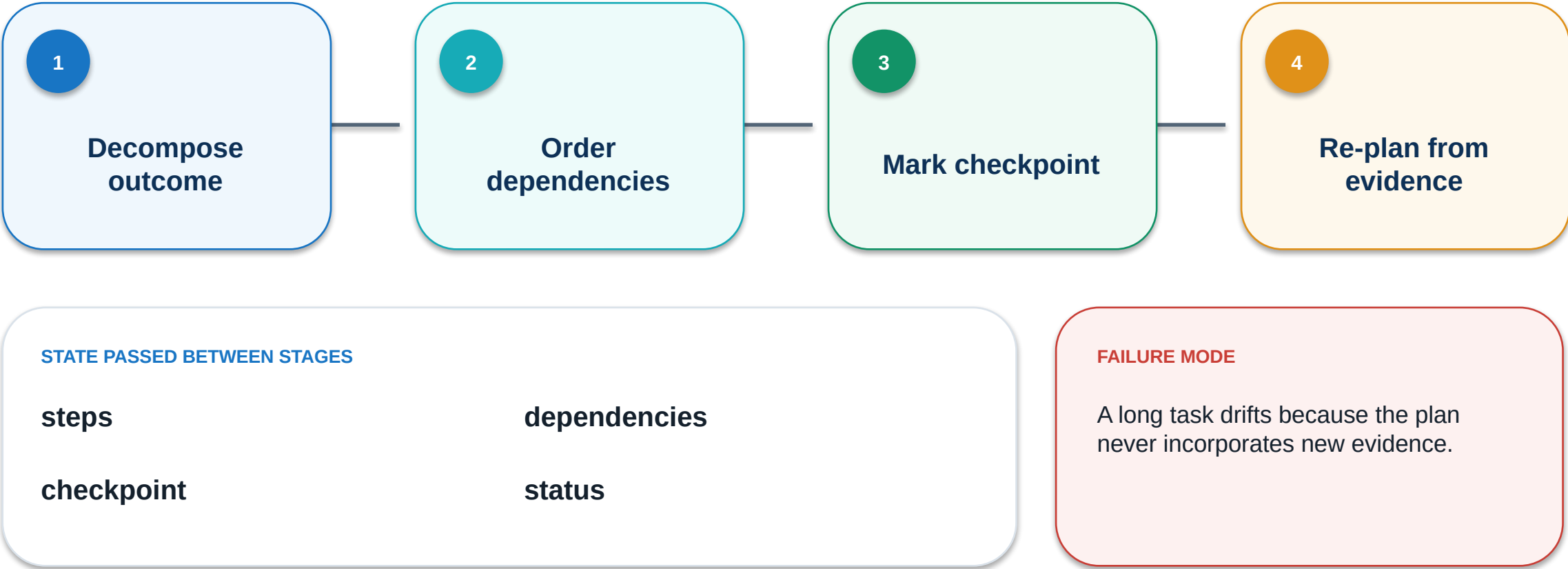
PASS / REPAIR

OWNER

named human

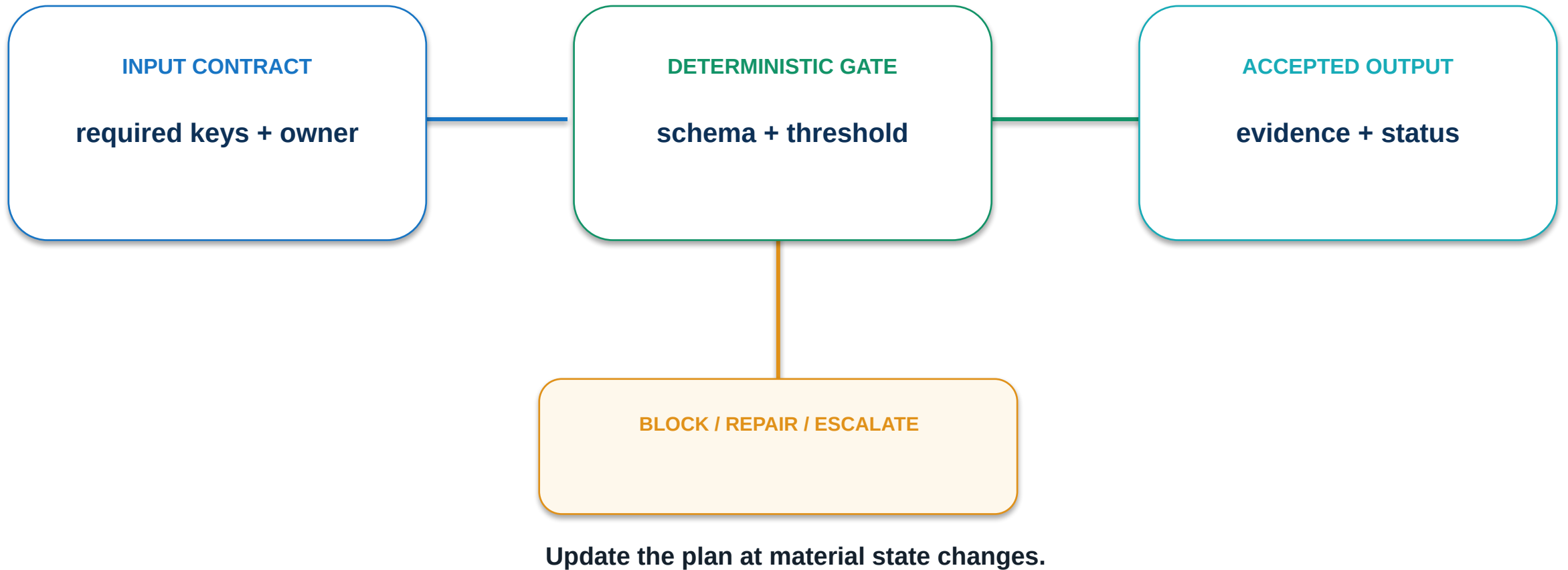
**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 03 · Planning and checkpoints: execution path



## 03 · Planning and checkpoints: contract and control

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## 03 · Planning and checkpoints: evidence and decision

### MEASURE

**Completed checkpoints with verification**

### ACCEPTANCE EVIDENCE

Plan history and checkpoint result

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

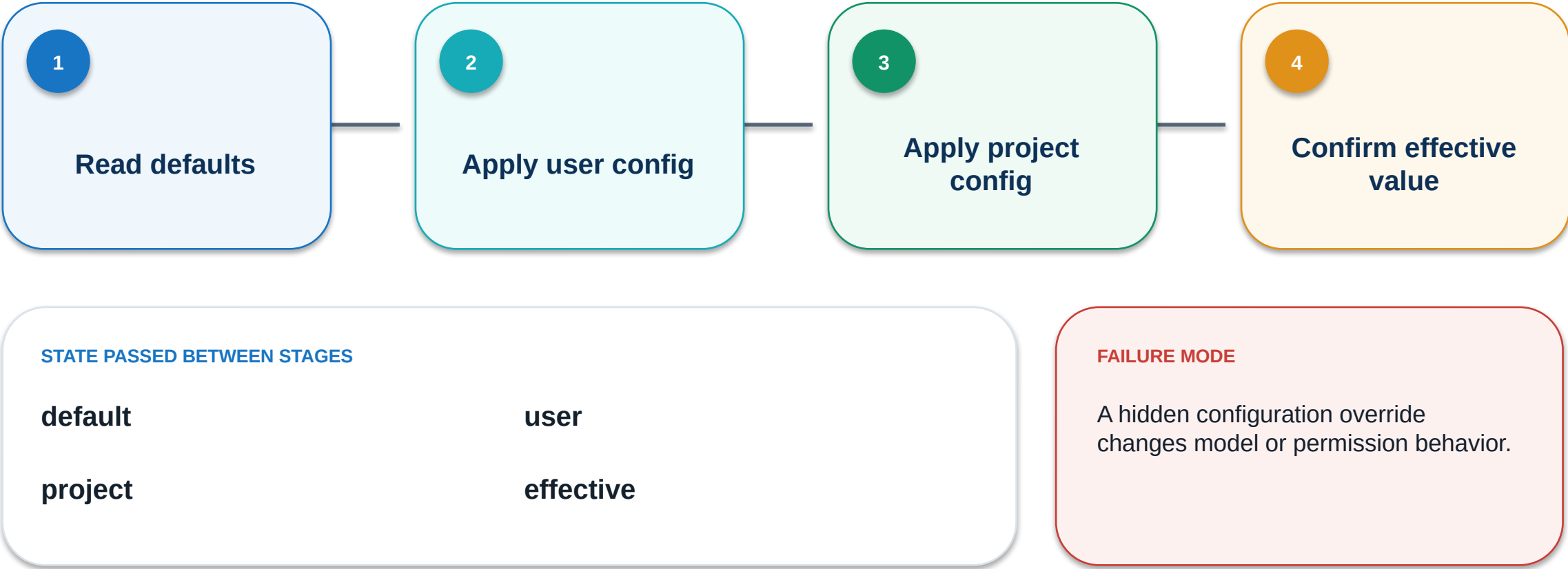
PASS / REPAIR

### OWNER

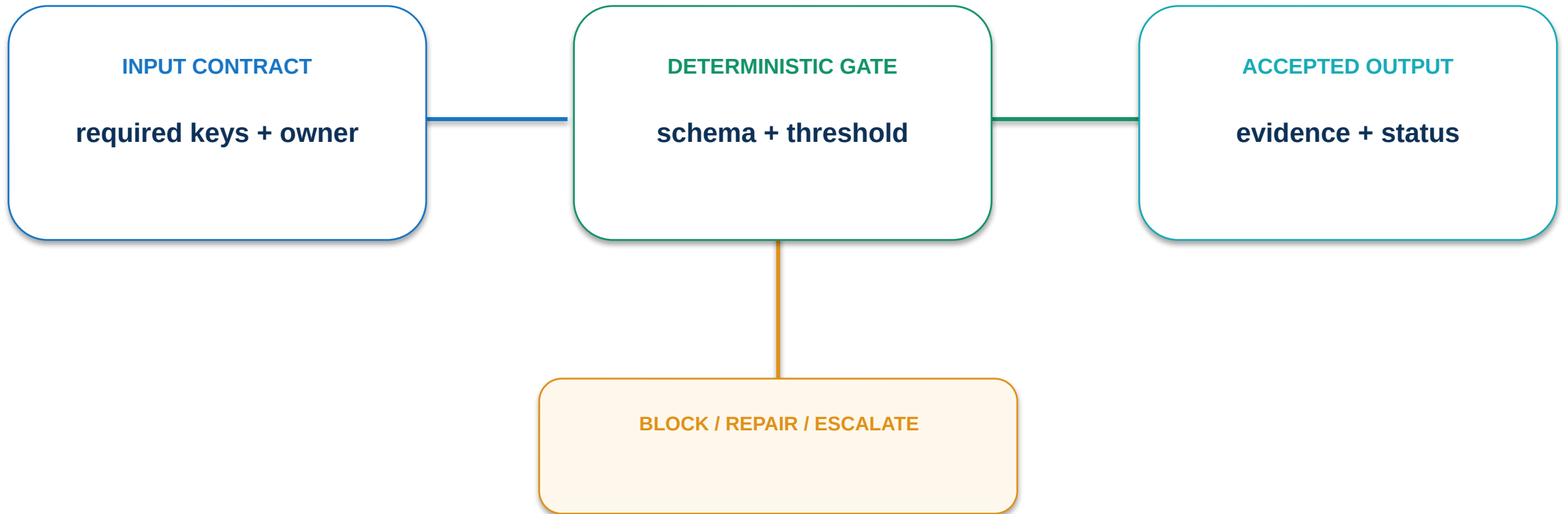
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 04 · Configuration layers: execution path



## 04 · Configuration layers: contract and control



Expose configuration precedence and validate the effective state.

## 04 · Configuration layers: evidence and decision

MEASURE

**Configuration values with known provenance**

ACCEPTANCE EVIDENCE

Resolved configuration table

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

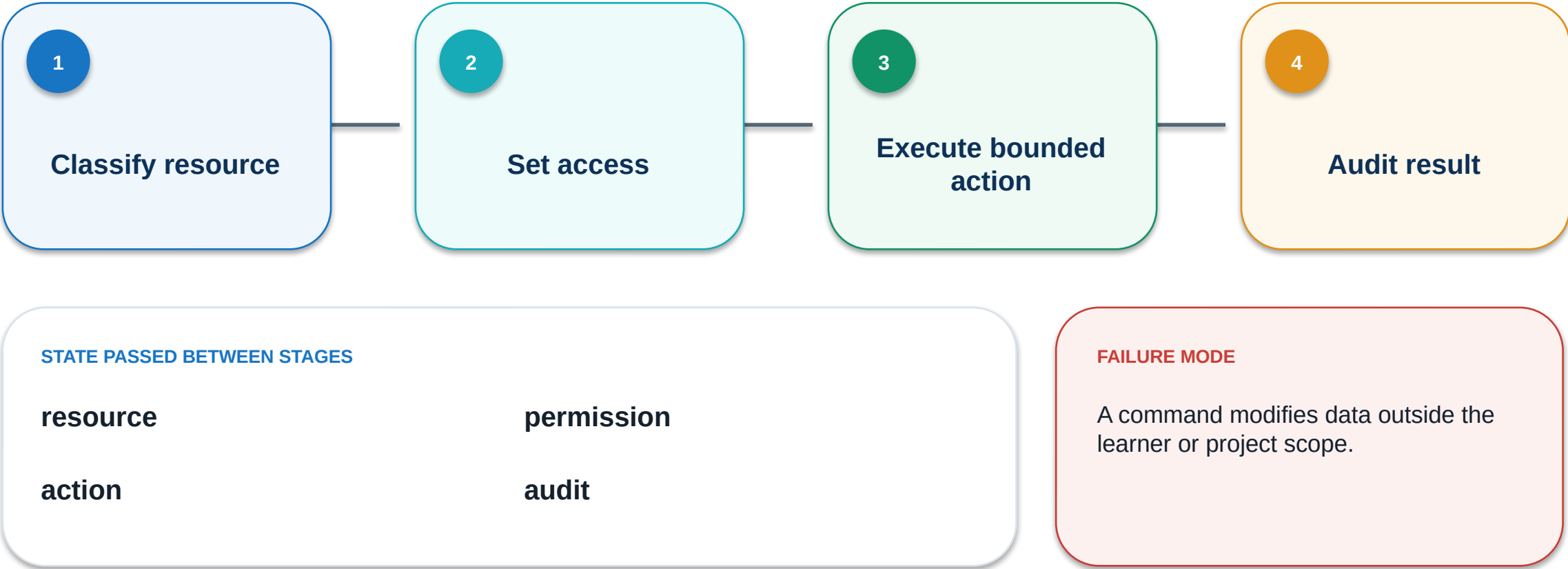
PASS / REPAIR

OWNER

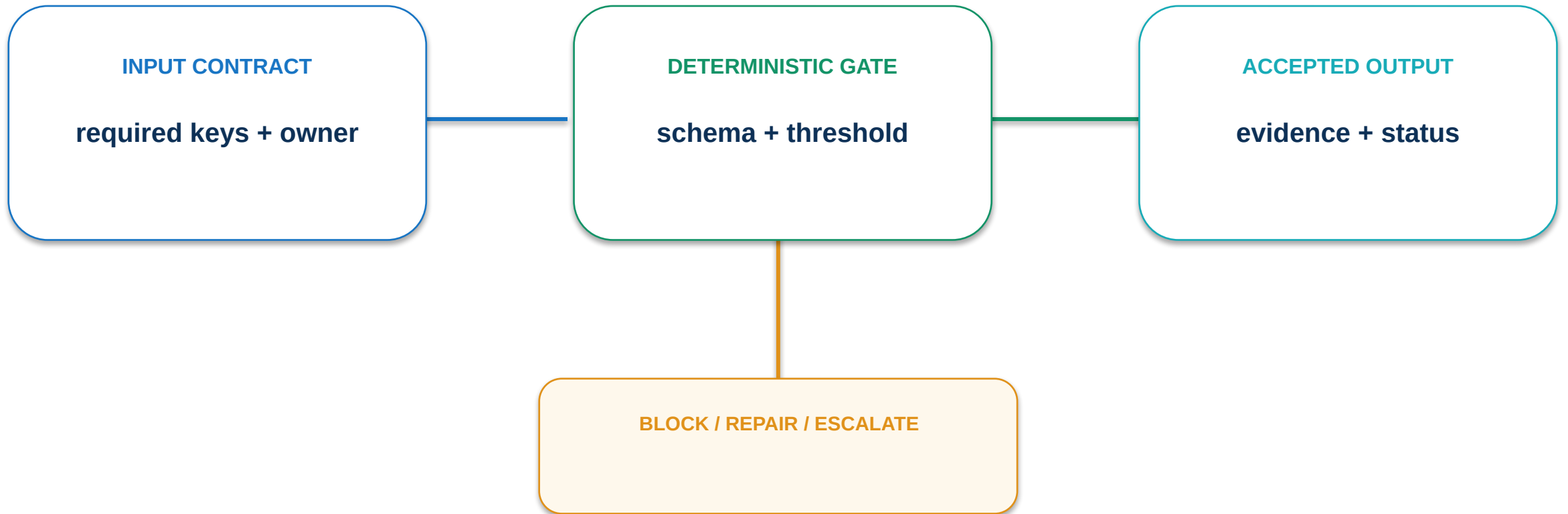
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 05 · Sandbox policy: execution path



## 05 · Sandbox policy: contract and control



Use the narrowest filesystem and network permissions that work.

## 05 · Sandbox policy: evidence and decision

### MEASURE

**Writes contained within the approved boundary**

### ACCEPTANCE EVIDENCE

Sandbox policy and audited path list

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

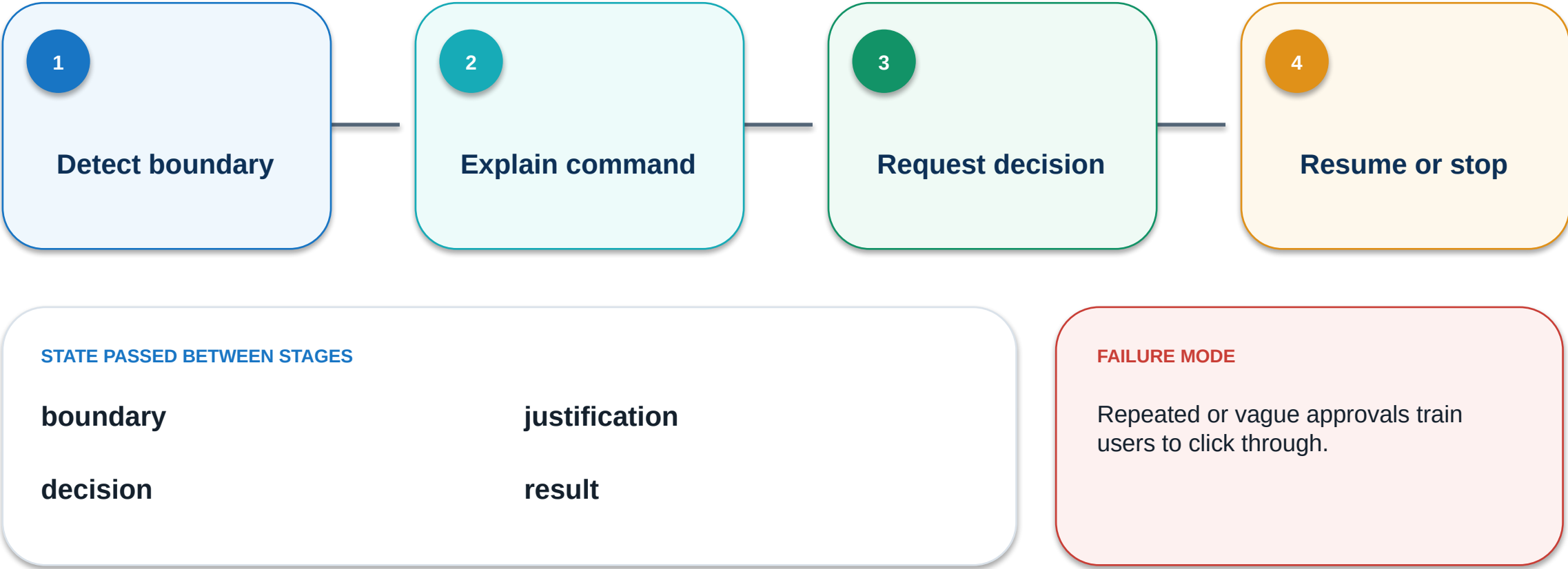
PASS / REPAIR

### OWNER

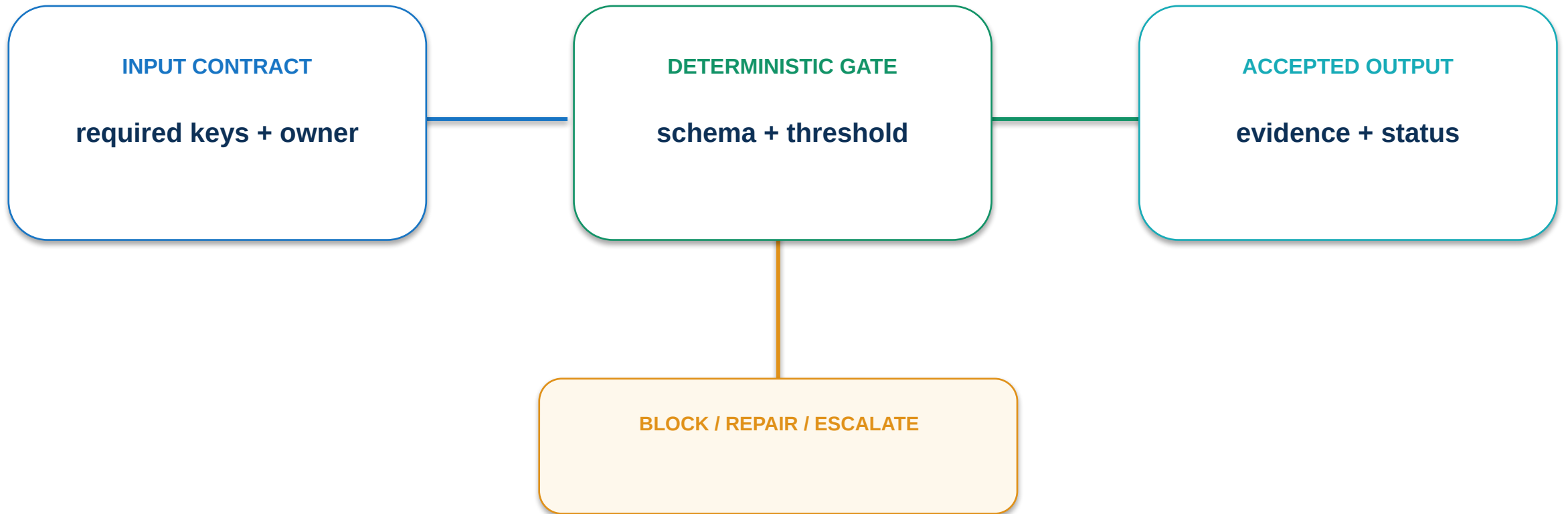
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 06 · Approval design: execution path



## 06 · Approval design: contract and control



Ask only at real boundaries with exact targets and effects.

## 06 · Approval design: evidence and decision

### MEASURE

**Approval prompts that identify exact consequence**

### ACCEPTANCE EVIDENCE

Approval decision tied to command and output

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

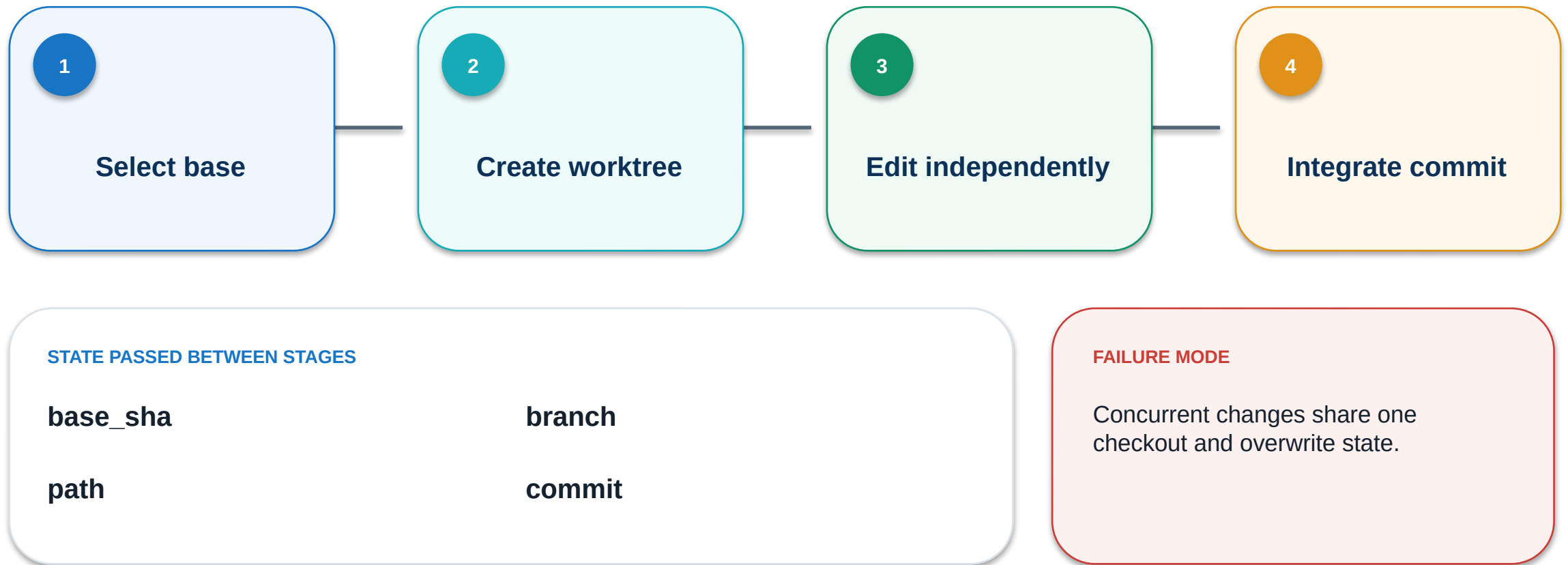
PASS / REPAIR

### OWNER

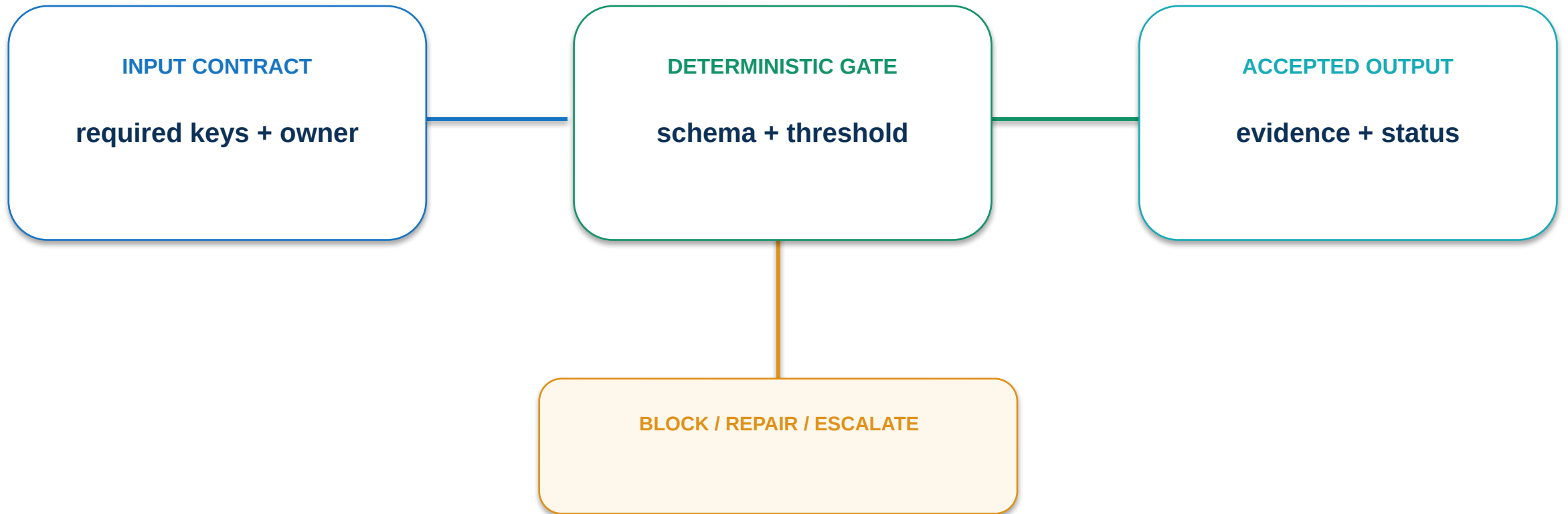
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 07 · Git worktree isolation: execution path



## 07 · Git worktree isolation: contract and control



Allocate one worktree and branch per independent mutation stream.

## 07 · Git worktree isolation: evidence and decision

### MEASURE

**Parallel tasks integrated without working-tree collision**

### ACCEPTANCE EVIDENCE

Distinct paths, branches and commit SHAs

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

PASS / REPAIR

### OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 08 · Patch discipline: execution path

PATCH REVIEW SURFACE

- 1

INSPECT

Read target and working state
- 2

PATCH

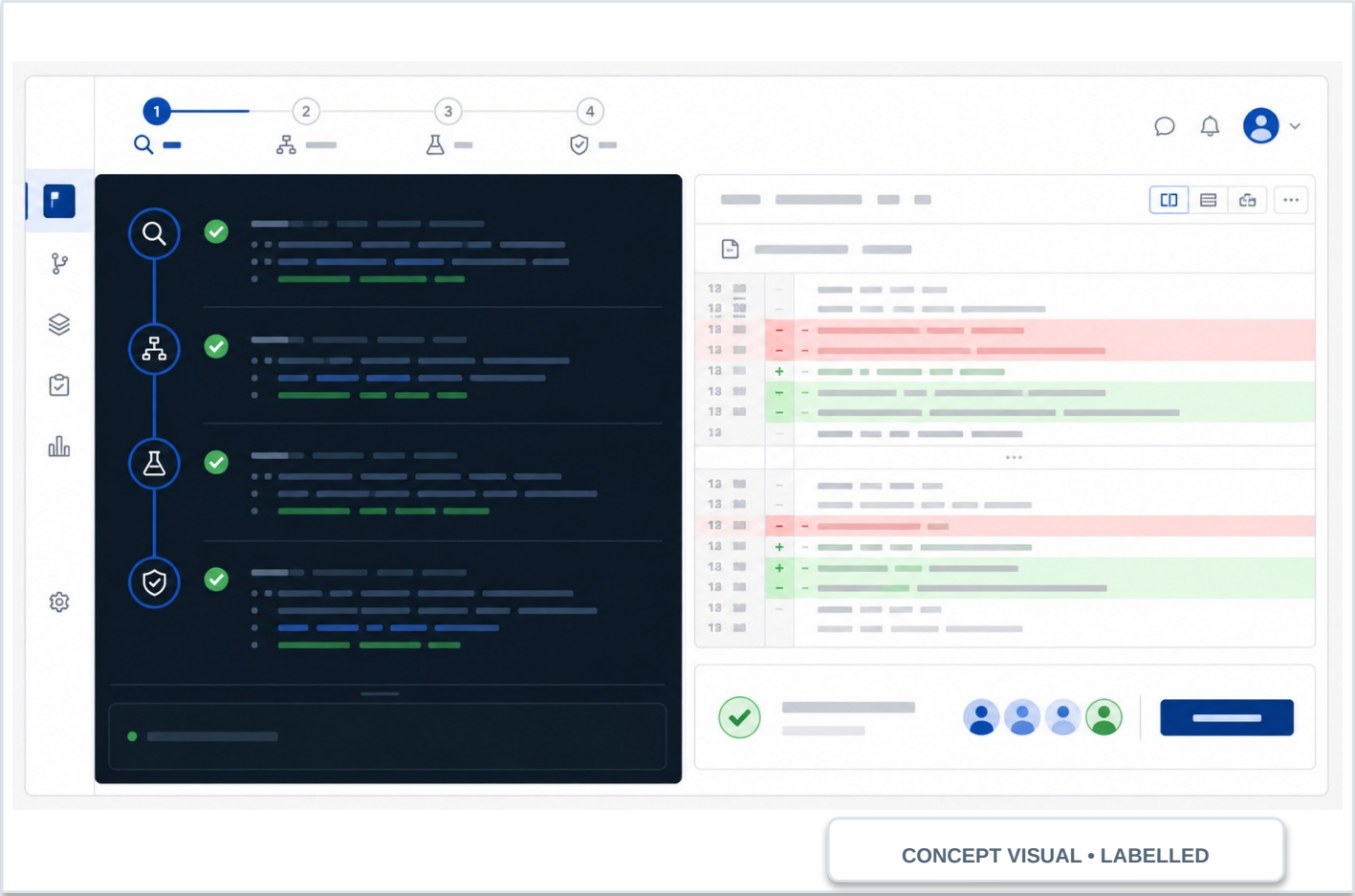
Change the smallest causal layer
- 3

DIFF

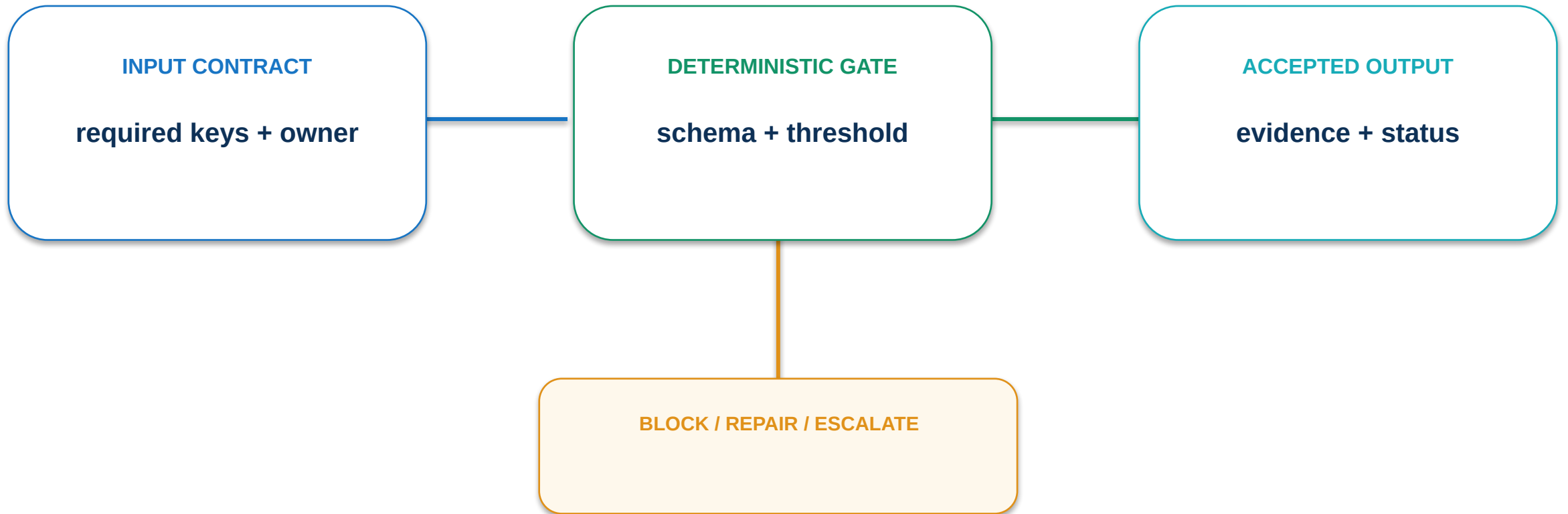
Review removed and added lines
- 4

VERIFY

Run focused and regression checks



## 08 · Patch discipline: contract and control



Prefer minimal, reviewable patches and preserve unrelated state.

## 08 · Patch discipline: evidence and decision

### MEASURE

**Changed lines that directly serve acceptance criteria**

### ACCEPTANCE EVIDENCE

Scoped diff and focused test output

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

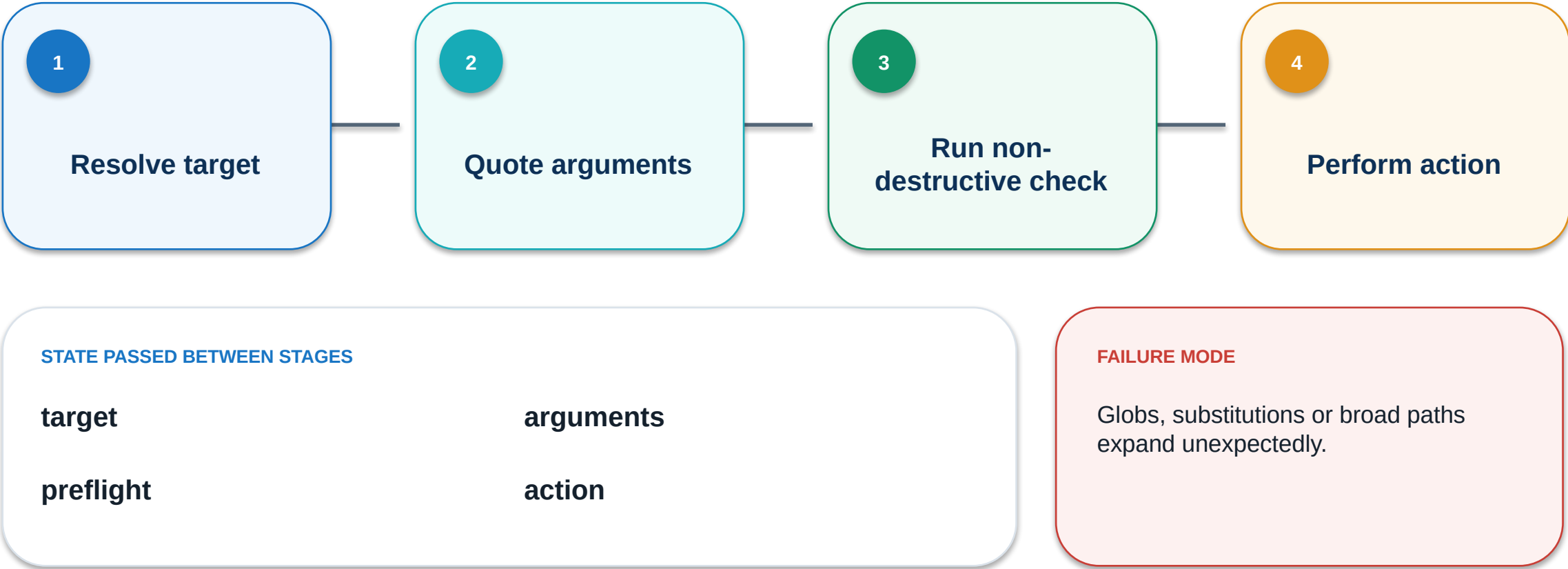
PASS / REPAIR

### OWNER

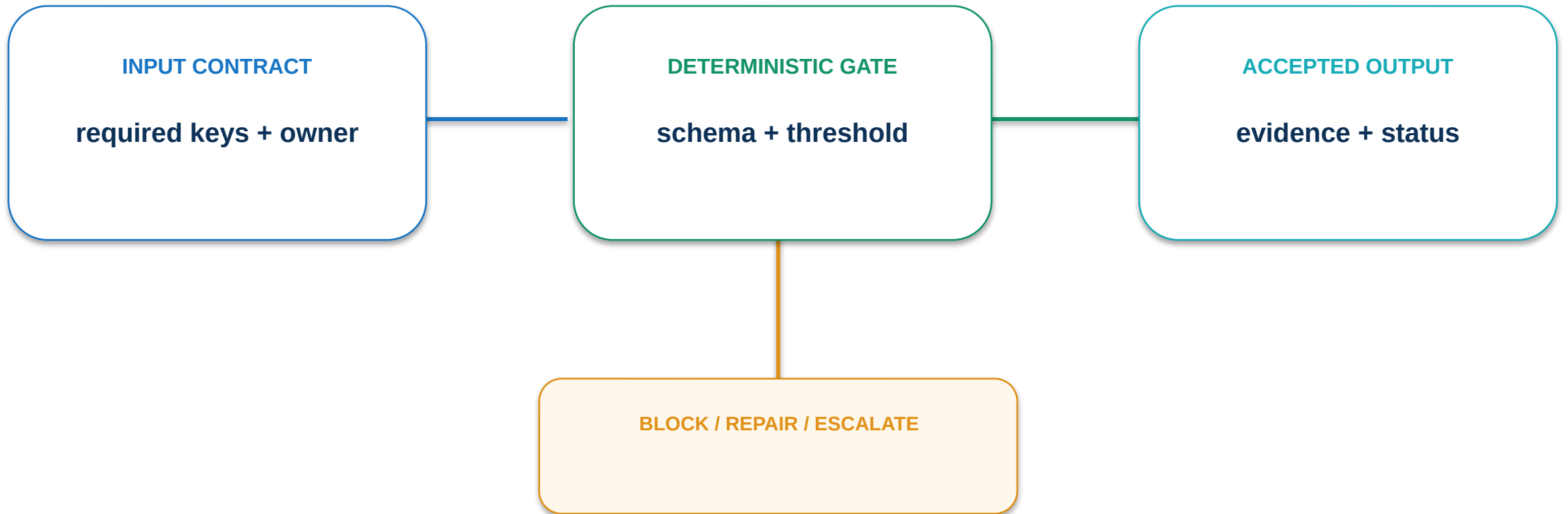
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 09 · Command safety: execution path



## 09 · Command safety: contract and control



Resolve explicit targets and prefer recoverable operations.

## 09 · Command safety: evidence and decision

### MEASURE

**Potentially destructive commands preceded by exact target checks**

### ACCEPTANCE EVIDENCE

Preflight listing and action log

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

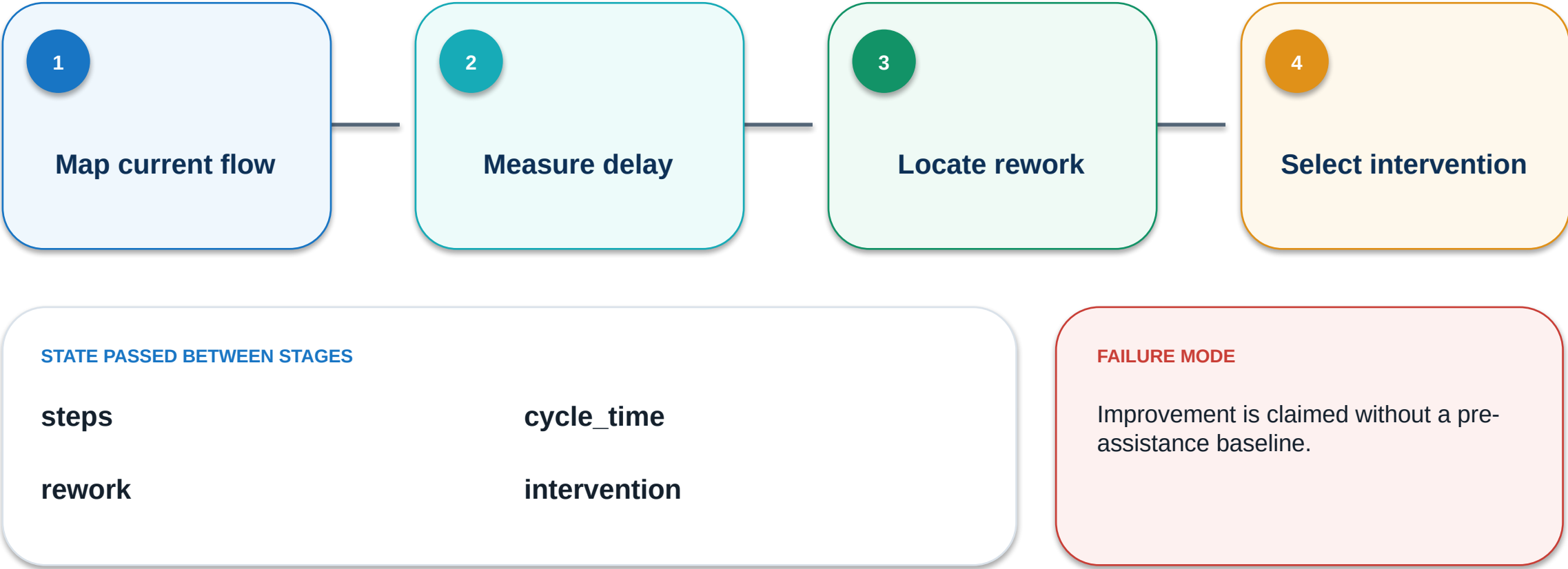
PASS / REPAIR

### OWNER

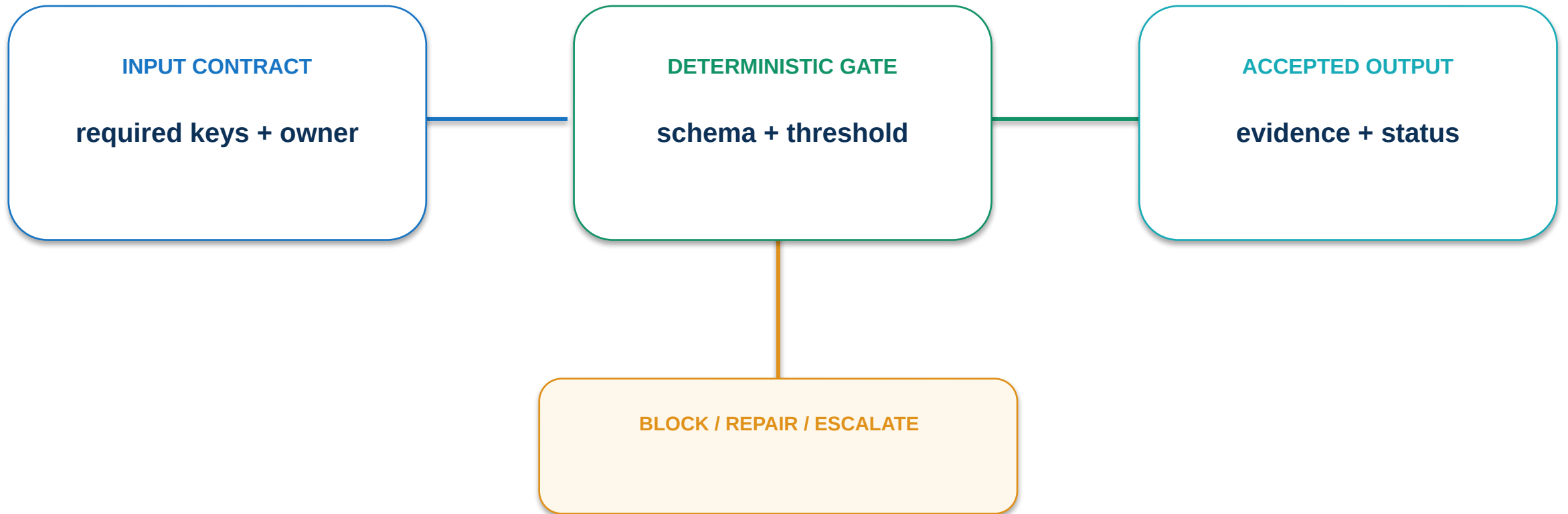
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 10 · Process baseline: execution path



## 10 · Process baseline: contract and control



Measure the existing process before introducing Codex.

## 10 · Process baseline: evidence and decision

### MEASURE

**Baseline cycle time and defect rate captured**

### ACCEPTANCE EVIDENCE

Baseline map and observed metrics

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

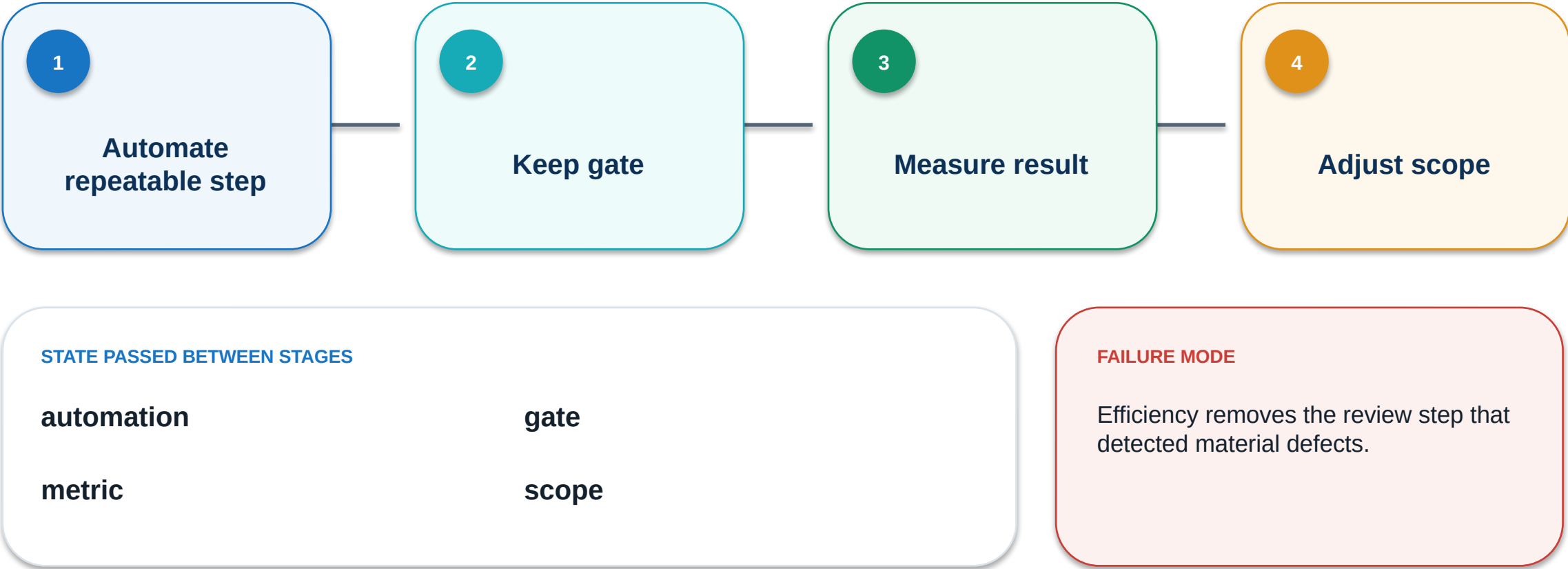
PASS / REPAIR

### OWNER

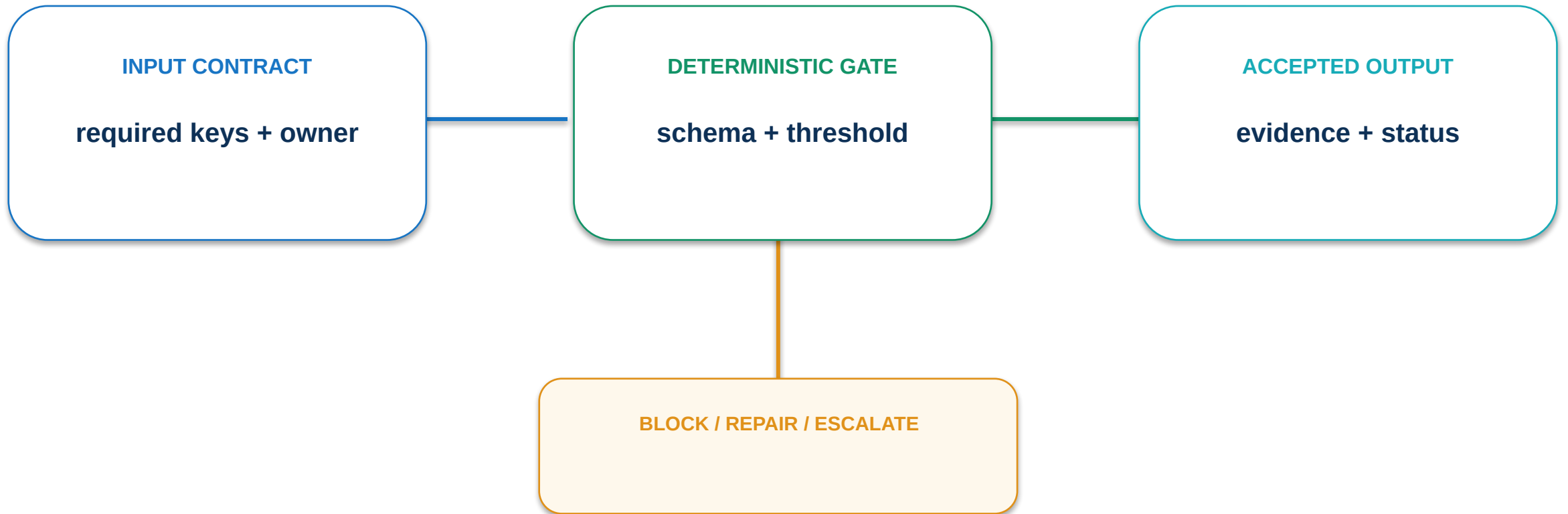
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 11 · Efficiency without control loss: execution path



## 11 · Efficiency without control loss: contract and control



Automate preparation and checking while retaining consequential gates.

## 11 · Efficiency without control loss: evidence and decision

### MEASURE

**Cycle-time gain with equal or better verification coverage**

### ACCEPTANCE EVIDENCE

Before/after measures and unchanged control map

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

PASS / REPAIR

### OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# Harness boundary: instructions, sandbox and approval

---

1

## INSTRUCTIONS

**Resolve nearest  
repository rule**

2

## SANDBOX

**Contain files  
and network**

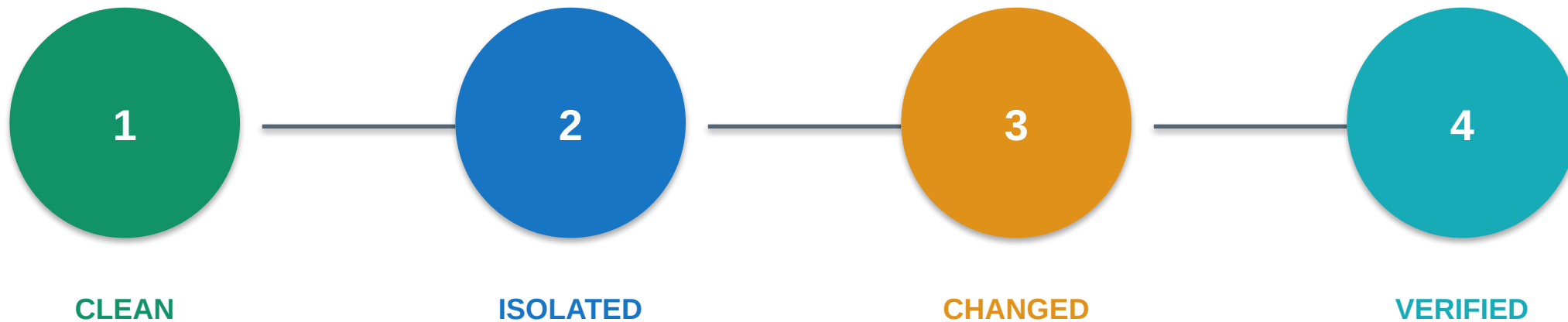
3

## APPROVAL

**Name target  
and consequence**

# Working-state model: CLEAN, ISOLATED, CHANGED and VERIFIED

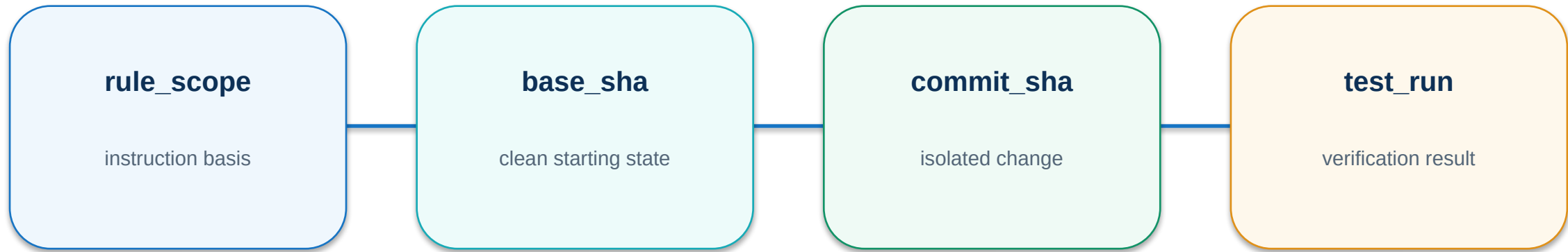
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**VERIFIED** means the isolated change passed focused and regression checks.

# Repository evidence chain: instruction → base SHA → patch → checks

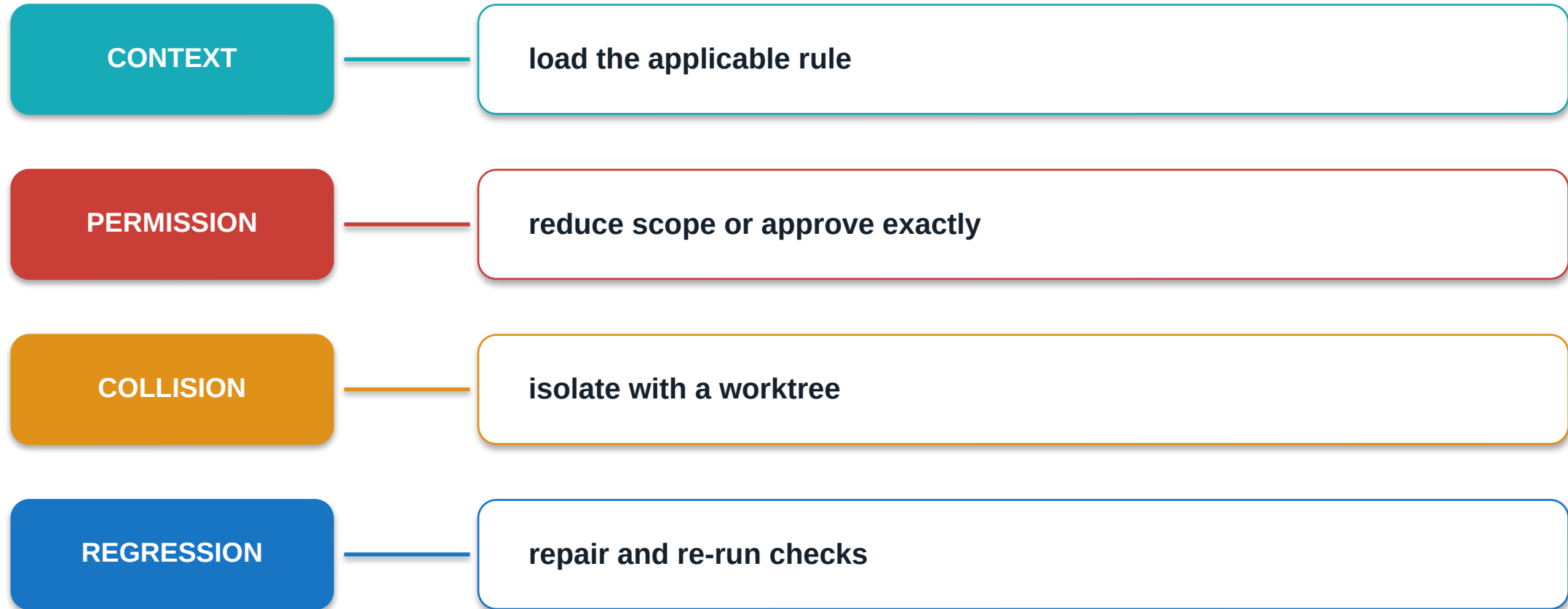
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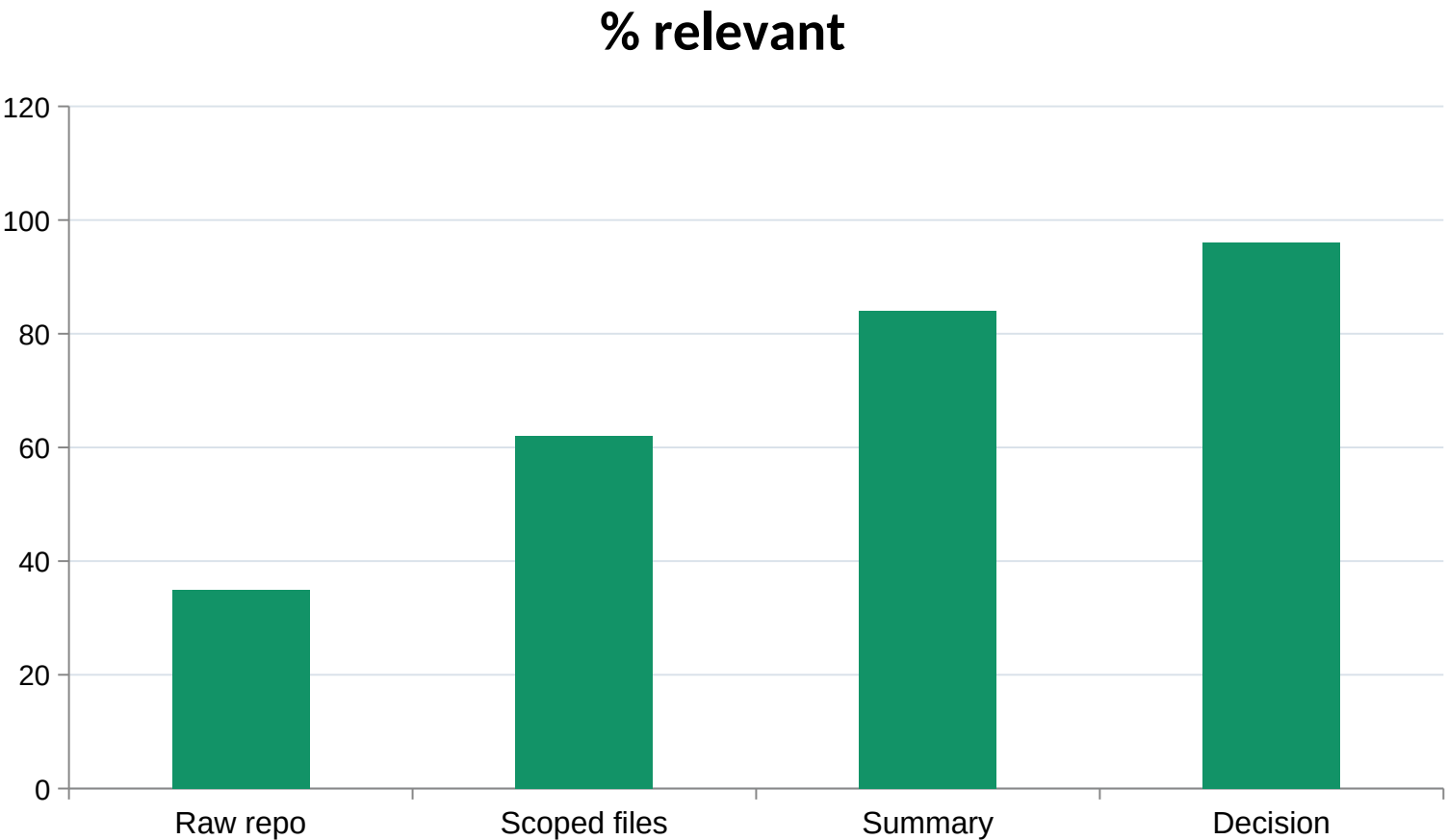
**Repository evidence preserves instruction, isolation, change and verification.**

# Control taxonomy: context, permission, collision and regression

---



# Bounded context improves useful signal

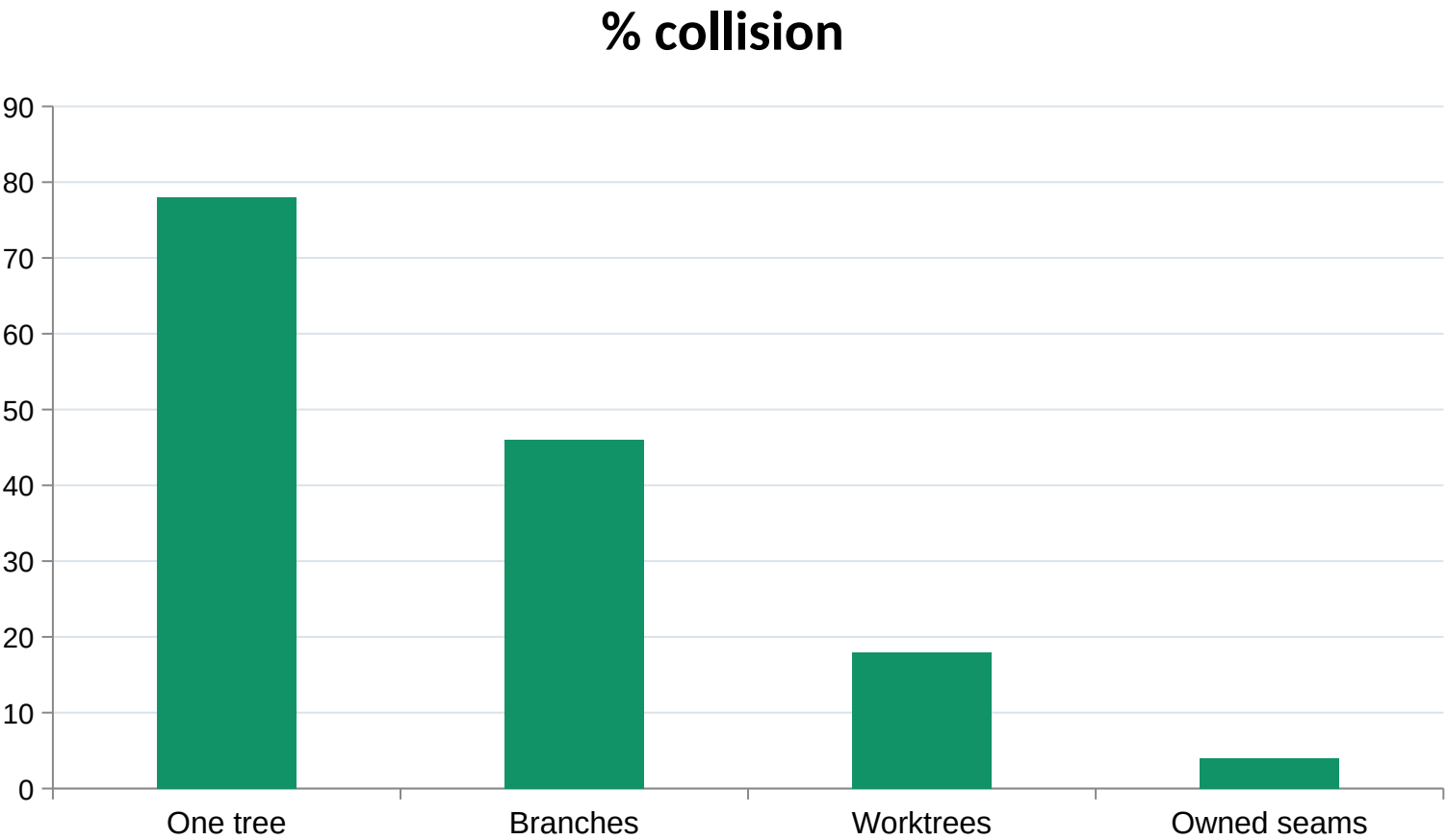


INSIGHT

**Context quality rises when inspection narrows evidence for the next decision.**

Illustrative synthetic values for mechanism explanation.

# Isolation reduces parallel-edit collision

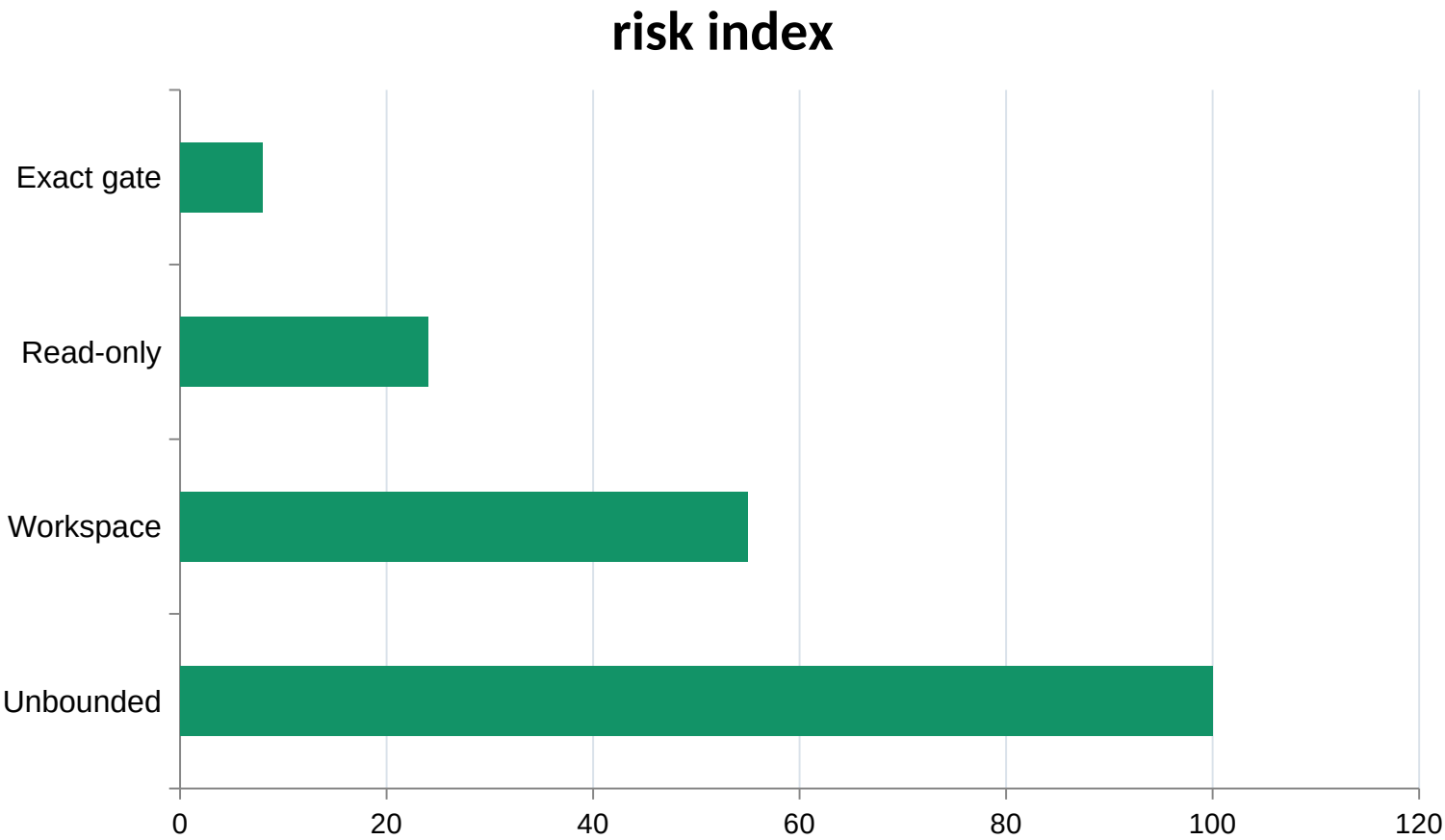


INSIGHT

**Worktrees help, but exclusive file ownership provides the strongest collision control.**

Illustrative synthetic values for mechanism explanation.

# Least privilege contains action exposure



INSIGHT

**The narrowest workable permission posture reduces accidental exposure.**

Illustrative synthetic values for mechanism explanation.

## Lab 03 · Scoped AGENTS.md Hierarchy

---

### SCENARIO

**Design root and nested instructions, then verify that a task applies the nearest rule without conflicting with repository policy.**

### EXPECTED OUTPUT

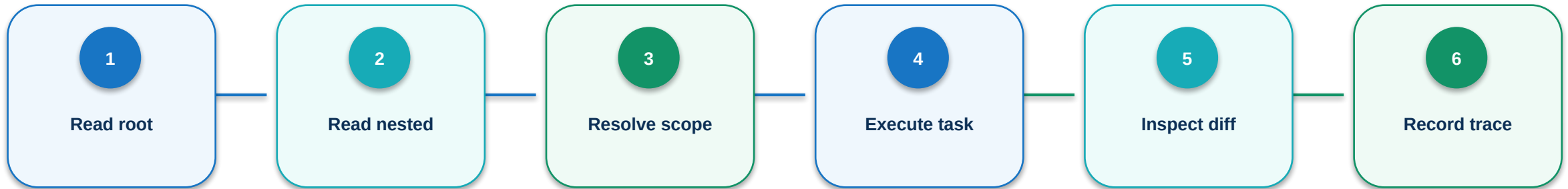
**Two-level instruction hierarchy and compliance trace**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-03-agents-hierarchy/**

## Lab 03 · Codex workflow contract

---



**Each node consumes named keys, emits a status, and leaves retrievable execution evidence.**

## Lab 03 · evidence and failure gates

### PASS EVIDENCE

The trace distinguishes root and nested rules and the patch complies with both in the correct precedence.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Lab 04 · Sandbox and Approval Experiment

---

### SCENARIO

**Compare read-only, workspace-write and approval boundaries using safe commands and an explicit consequence matrix.**

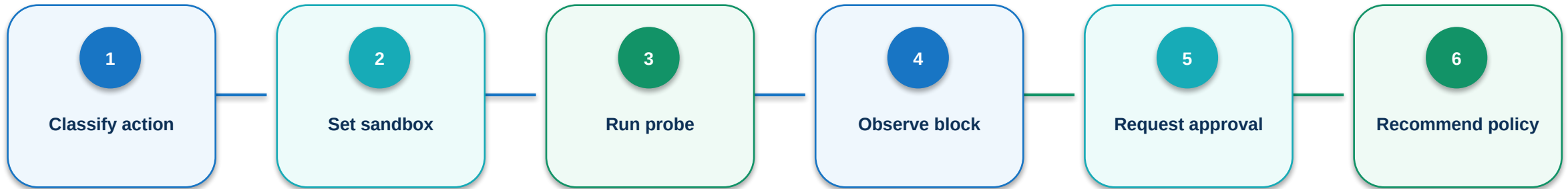
### EXPECTED OUTPUT

**Permission experiment log and recommended policy**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-04-sandbox-approval/**

# Lab 04 · Codex workflow contract



**Consequential actions require a current diff, passing checks and a named decision.**

**HUMAN GATE**

Approval resumes only when the reviewed diff and current state still match.

## Lab 04 · evidence and failure gates

### PASS EVIDENCE

**The log contains safe probes for three permission cases and a least-privilege recommendation supported by results.**

### FAIL CLOSED WHEN

**a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.**

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Lab 05 · Parallel Worktrees without Collision

---

### SCENARIO

**Split two independent TaskBoard changes into isolated worktrees and integrate them in a controlled order.**

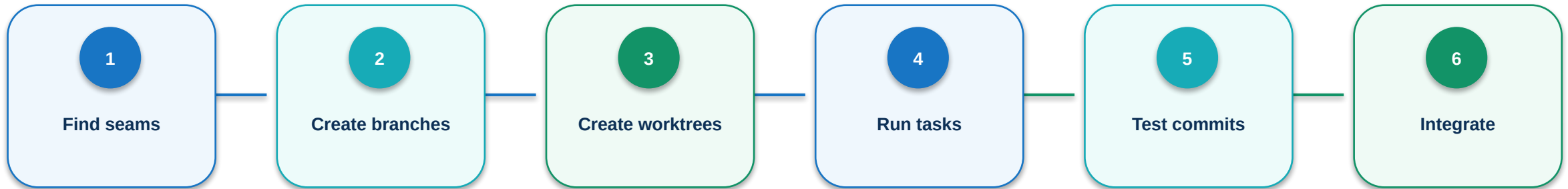
### EXPECTED OUTPUT

**Isolation map, two commits and integration evidence**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-05-parallel-worktrees/**

# Lab 05 · Codex workflow contract



**Consequential actions require a current diff, passing checks and a named decision.**

**HUMAN GATE**

Approval resumes only when the reviewed diff and current state still match.

## Lab 05 · evidence and failure gates

### PASS EVIDENCE

The worktrees use distinct paths and branches, produce separate commits, and integrate with the full test suite passing.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Topic 2 technical checkpoint

---

1

**Which state field prevents a silent downstream assumption?**

2

**What evidence proves the mechanism completed successfully?**

3

**Which failure must stop rather than retry?**

4

**Who owns the consequential decision?**

## Topic 2 transfer challenge

---

1

**Change one input constraint and predict the affected nodes.**

2

**Name the identifier that preserves provenance across the change.**

3

**Define one machine threshold and one human decision.**

4

**Describe the minimum evidence required to resume the run.**

# 3

## TOPIC 3

# Tools, Skills, MCP, Subagents and Cloud Automation

Apply the Codex extension surface to implement repeatable, observable and portable agent workflows.

1

CONTROL

2

EVIDENCE

3

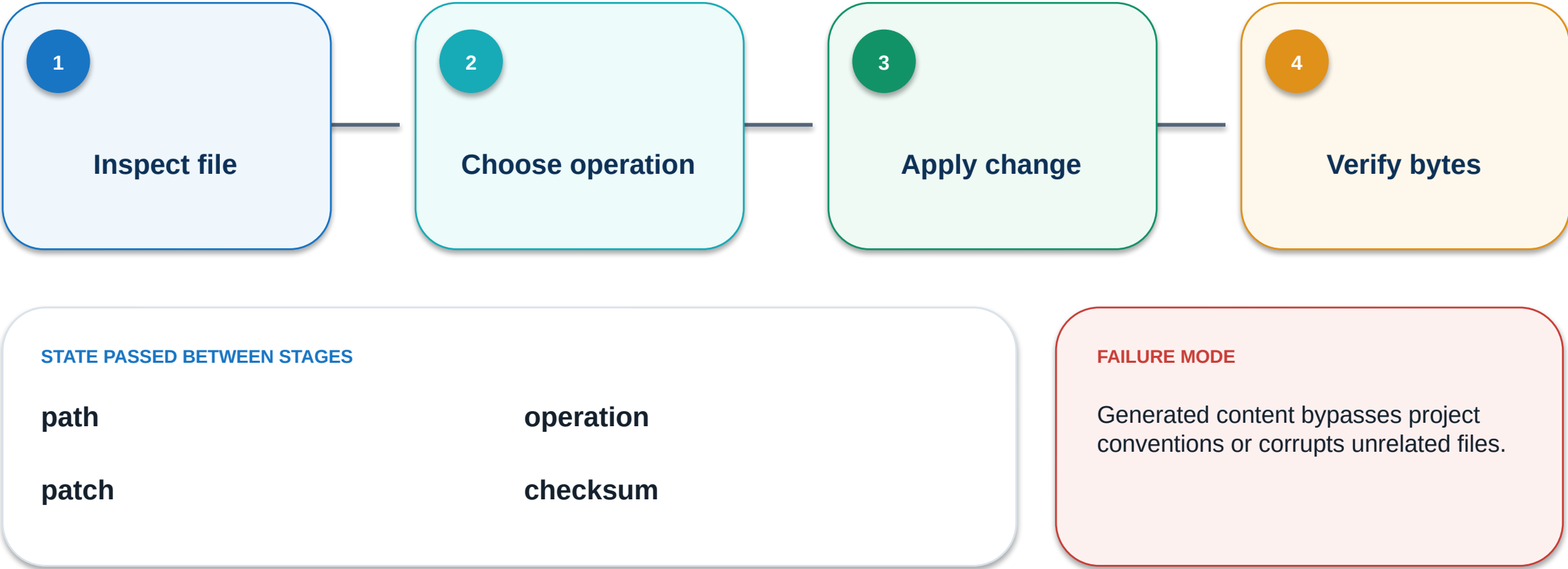
AUTOMATION

4

DECISION

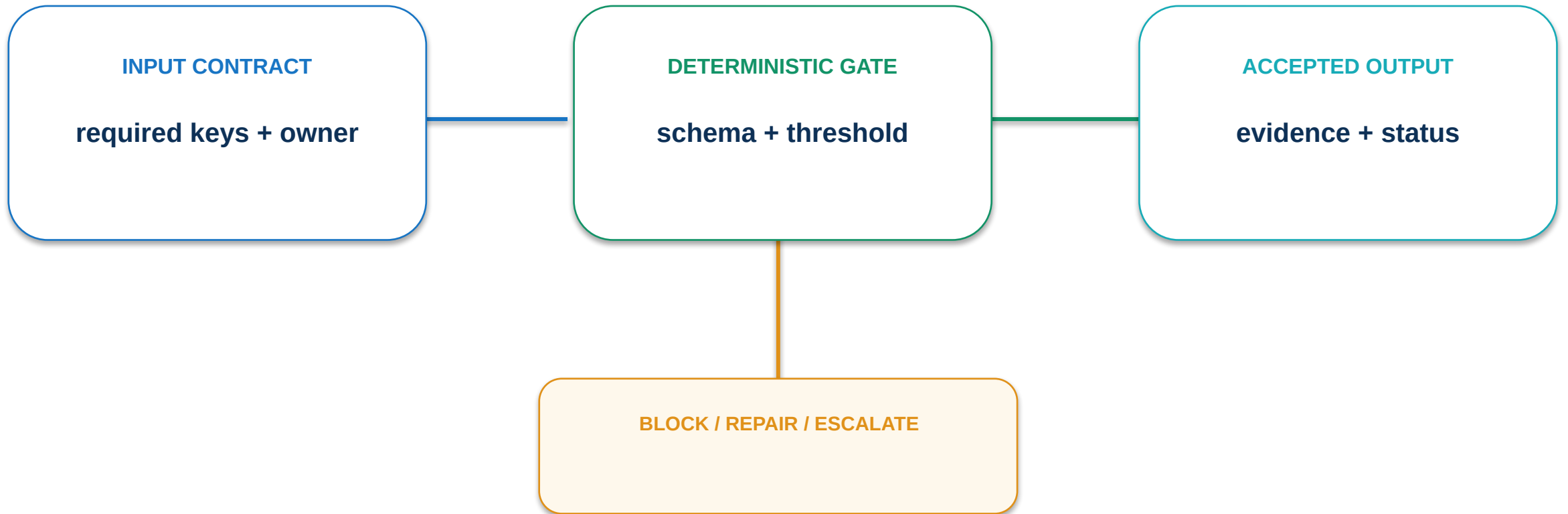
Every mechanism ends in observable evidence—not an agent's assertion.

# 01 · Terminal and file tools: execution path



## 01 · Terminal and file tools: contract and control

---



Use targeted search, patch and verification tools.

## 01 · Terminal and file tools: evidence and decision

### MEASURE

**File operations with explicit before-and-after evidence**

### ACCEPTANCE EVIDENCE

Diff plus test or checksum

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

PASS / REPAIR

### OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 02 · Browser-based verification: execution path

BROWSER EVIDENCE SURFACE

- 1

PATH

Exercise the real user journey
- 2

VIEWPORT

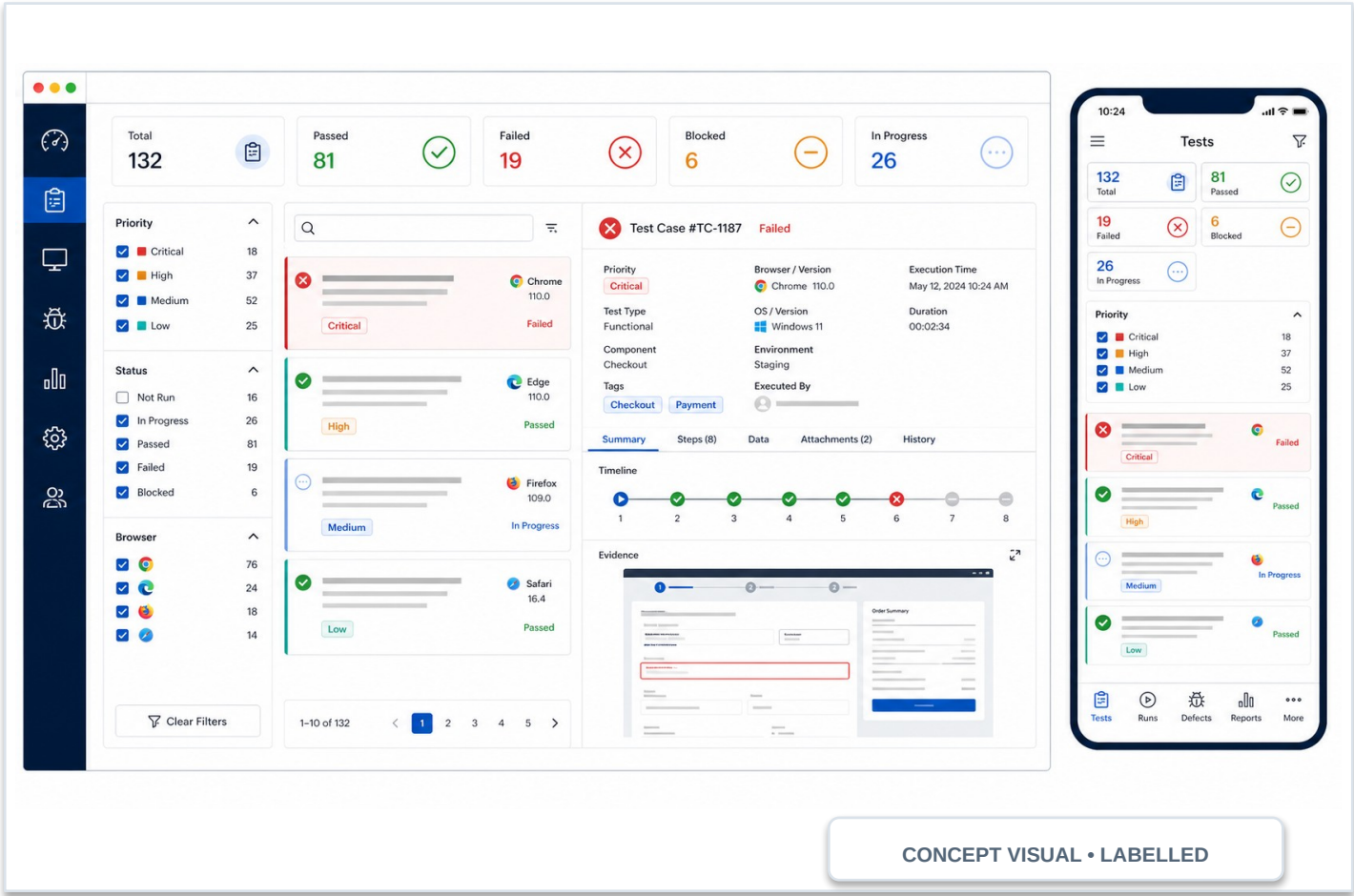
Check desktop and mobile state
- 3

DEFECT

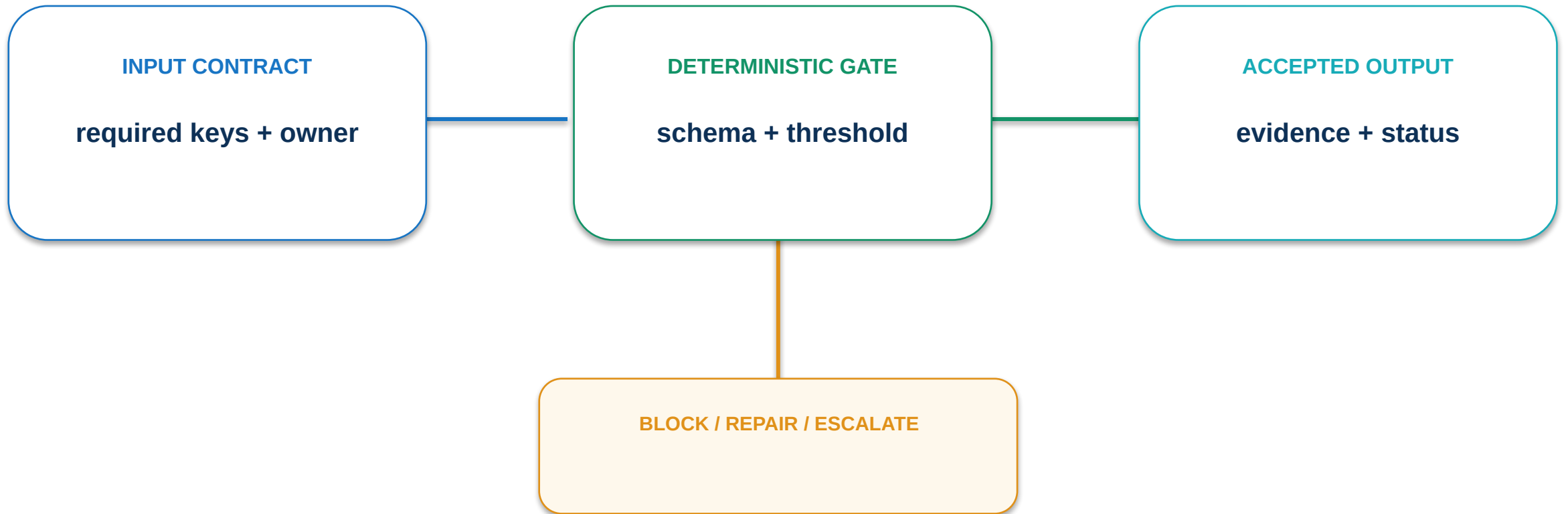
Capture reproducible failure
- 4

RE-TEST

Prove the scoped repair



## 02 · Browser-based verification: contract and control



Pair implementation tests with real-browser evidence.

## 02 · Browser-based verification: evidence and decision

MEASURE

**Critical user paths verified at target viewport**

ACCEPTANCE EVIDENCE

Browser trace and screenshot

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

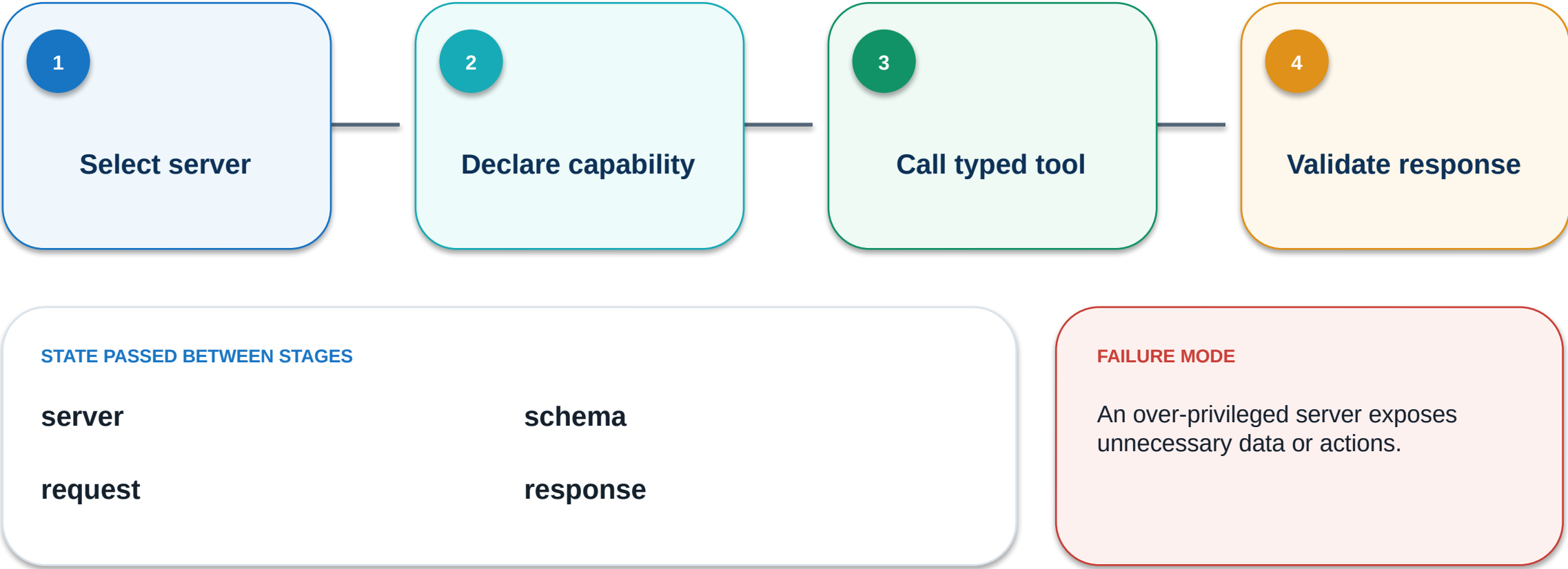
PASS / REPAIR

OWNER

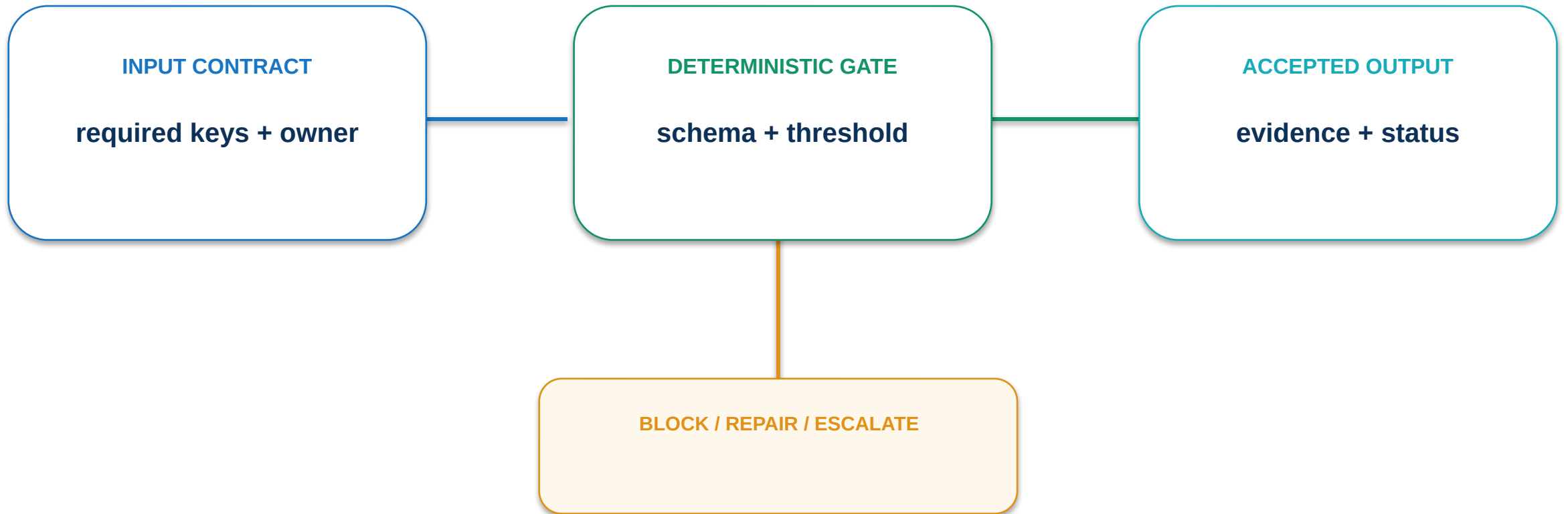
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 03 · Model Context Protocol: execution path



## 03 · Model Context Protocol: contract and control



Install only required servers and constrain their permissions.

## 03 · Model Context Protocol: evidence and decision

MEASURE

**MCP calls returning schema-valid results**

ACCEPTANCE EVIDENCE

Server inventory, call trace and validation

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

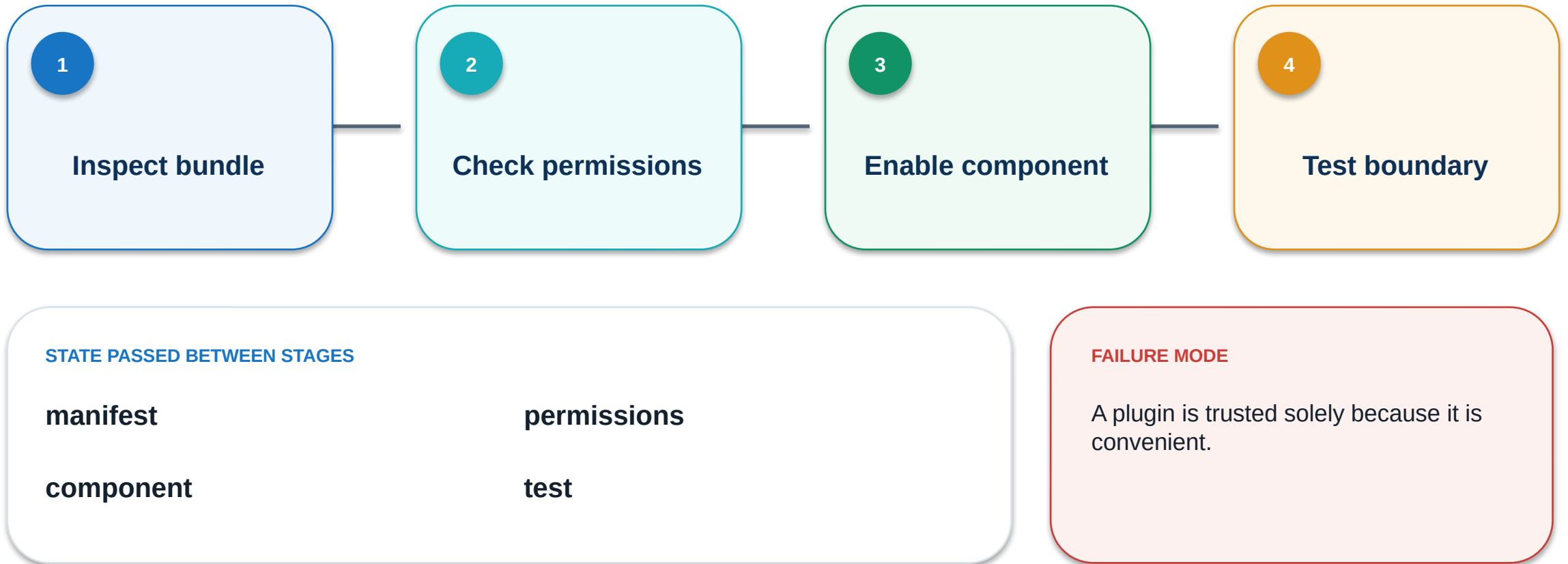
PASS / REPAIR

OWNER

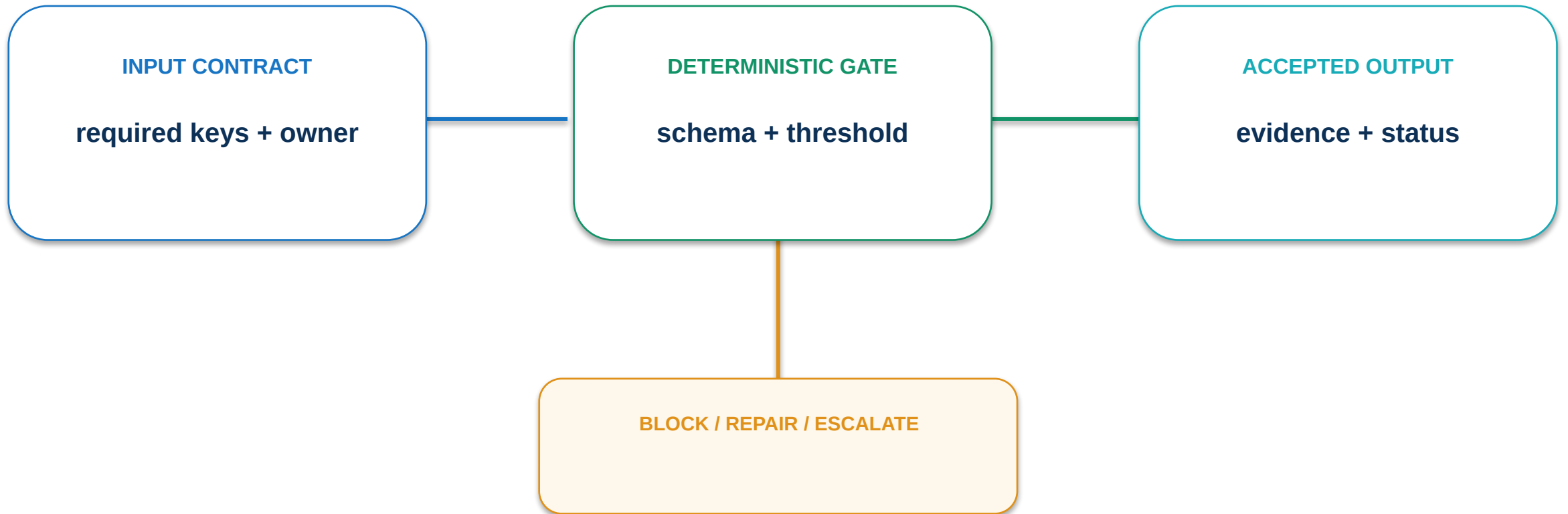
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 04 · Plugins and packaged capability: execution path



## 04 · Plugins and packaged capability: contract and control



**Review source, permissions and dependencies before enabling.**

## 04 · Plugins and packaged capability: evidence and decision

### MEASURE

**Enabled capabilities with reviewed dependencies**

### ACCEPTANCE EVIDENCE

Manifest review and bounded test

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

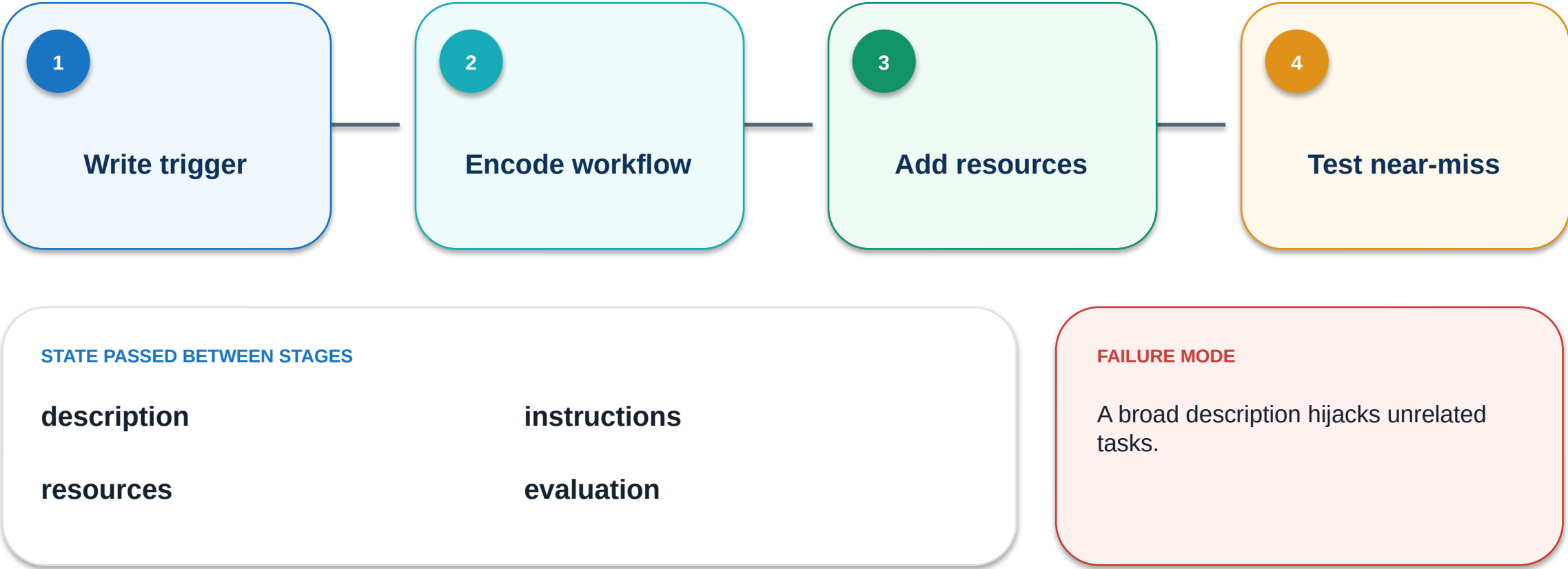
PASS / REPAIR

### OWNER

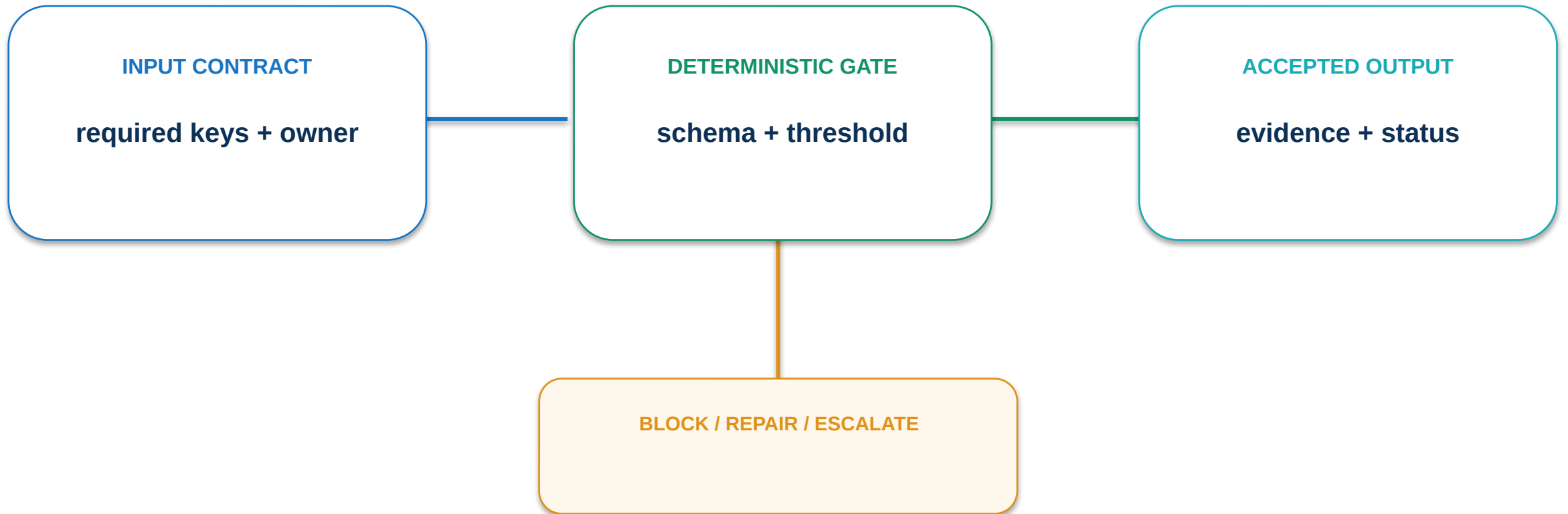
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 05 · Reusable skills: execution path



## 05 · Reusable skills: contract and control



Use narrow triggers and verify both trigger and non-trigger cases.

## 05 · Reusable skills: evidence and decision

### MEASURE

**Trigger precision across positive and near-miss prompts**

### ACCEPTANCE EVIDENCE

Skill package plus evaluation record

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

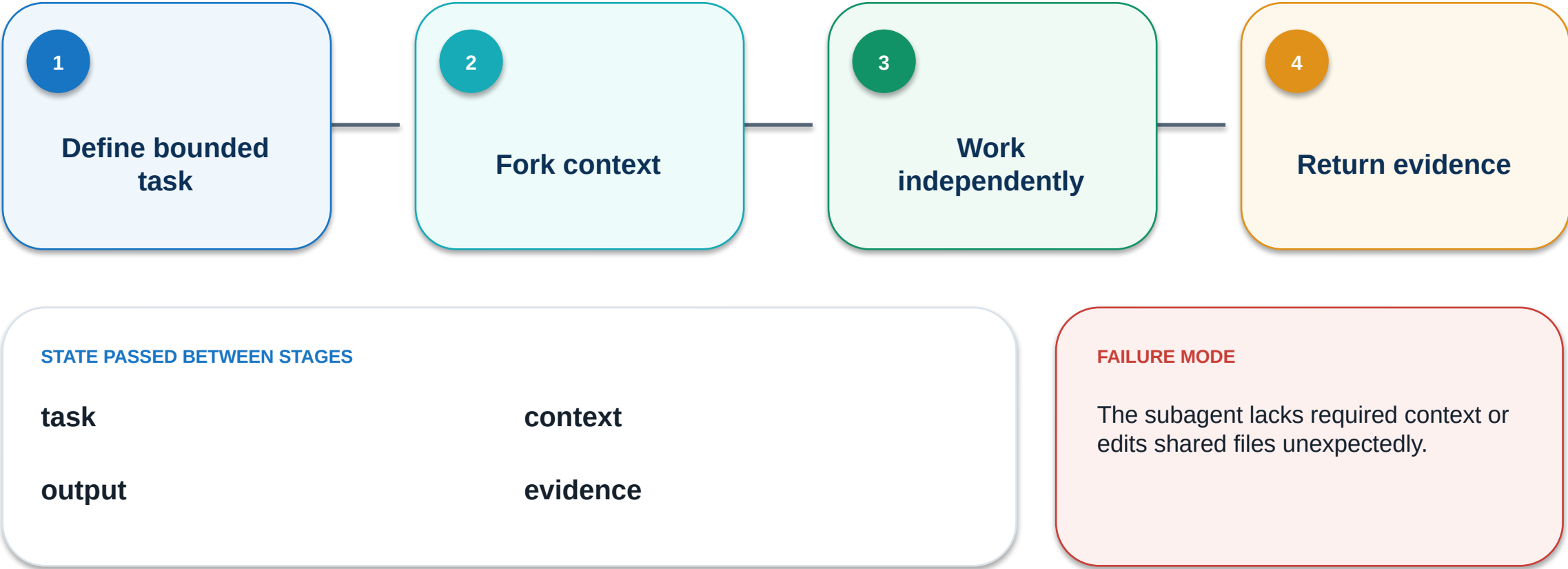
PASS / REPAIR

### OWNER

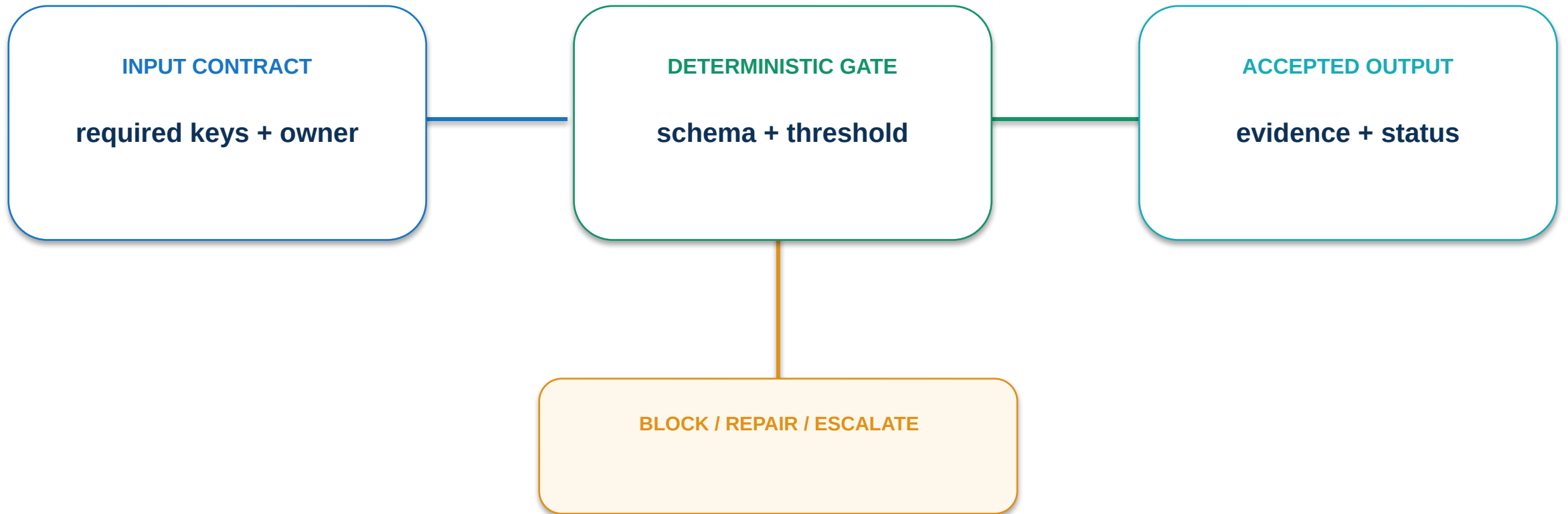
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 06 · Subagent delegation: execution path



## 06 · Subagent delegation: contract and control



Delegate concrete, dependency-light work with explicit deliverables.

## 06 · Subagent delegation: evidence and decision

MEASURE

**Delegated tasks completed without scope overlap**

ACCEPTANCE EVIDENCE

Task brief and returned evidence

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

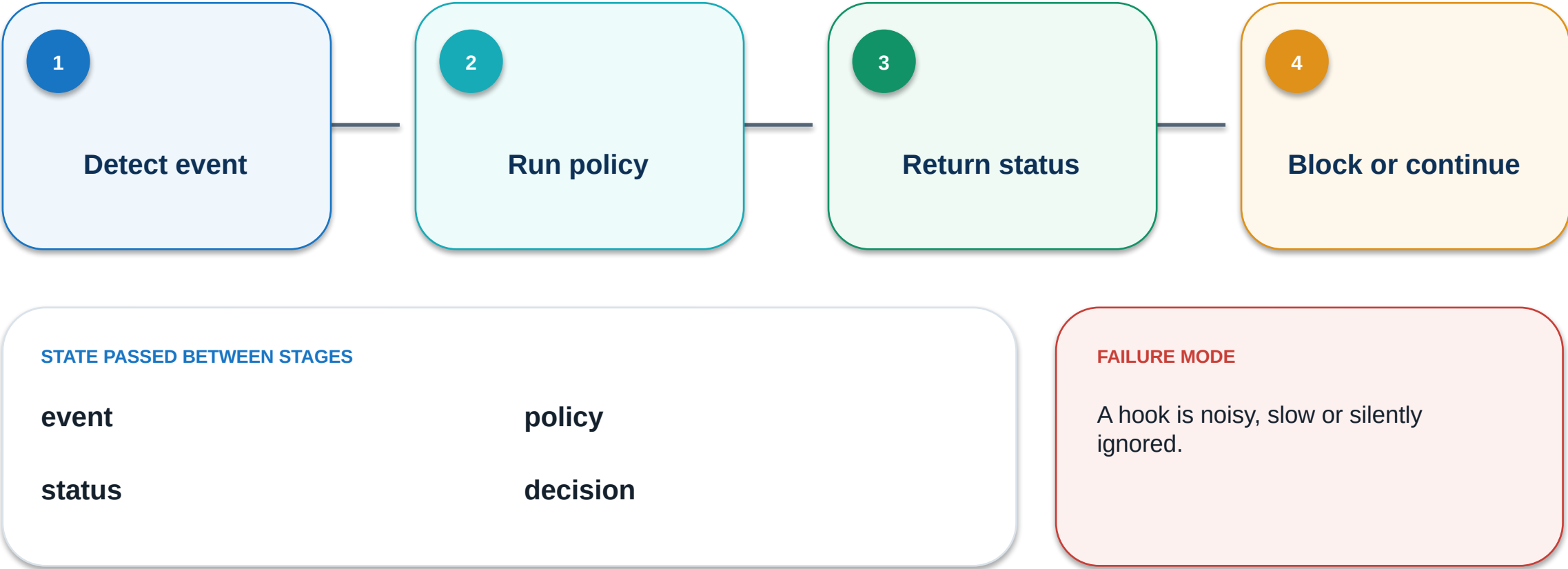
PASS / REPAIR

OWNER

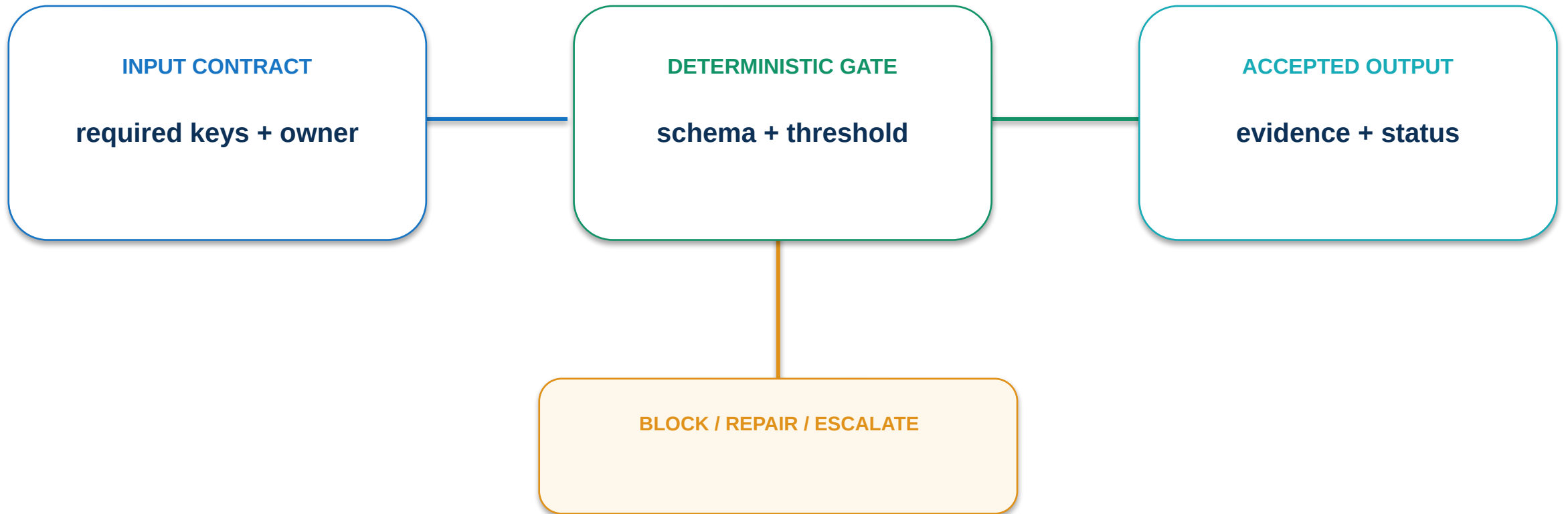
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 07 · Hooks and lifecycle checks: execution path



## 07 · Hooks and lifecycle checks: contract and control



Keep hooks deterministic, fast and fail-closed for material rules.

## 07 · Hooks and lifecycle checks: evidence and decision

MEASURE

**Policy violations blocked before completion**

ACCEPTANCE EVIDENCE

Hook output and blocked test case

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

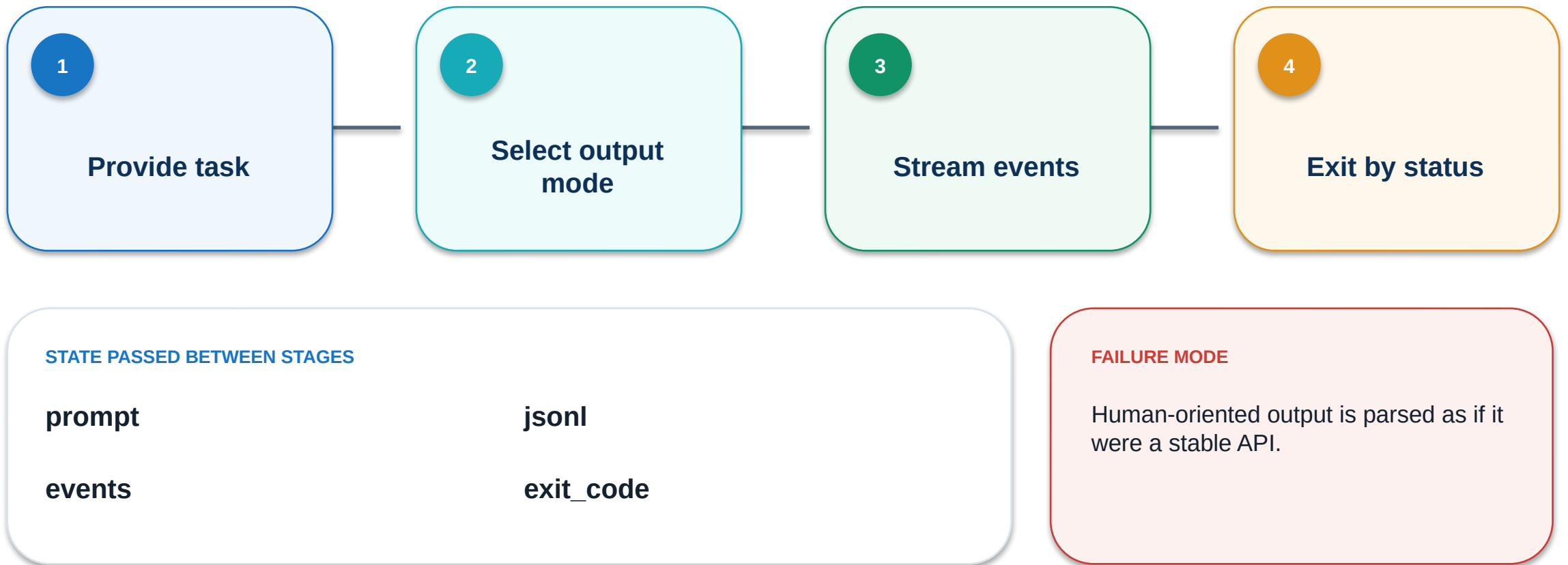
PASS / REPAIR

OWNER

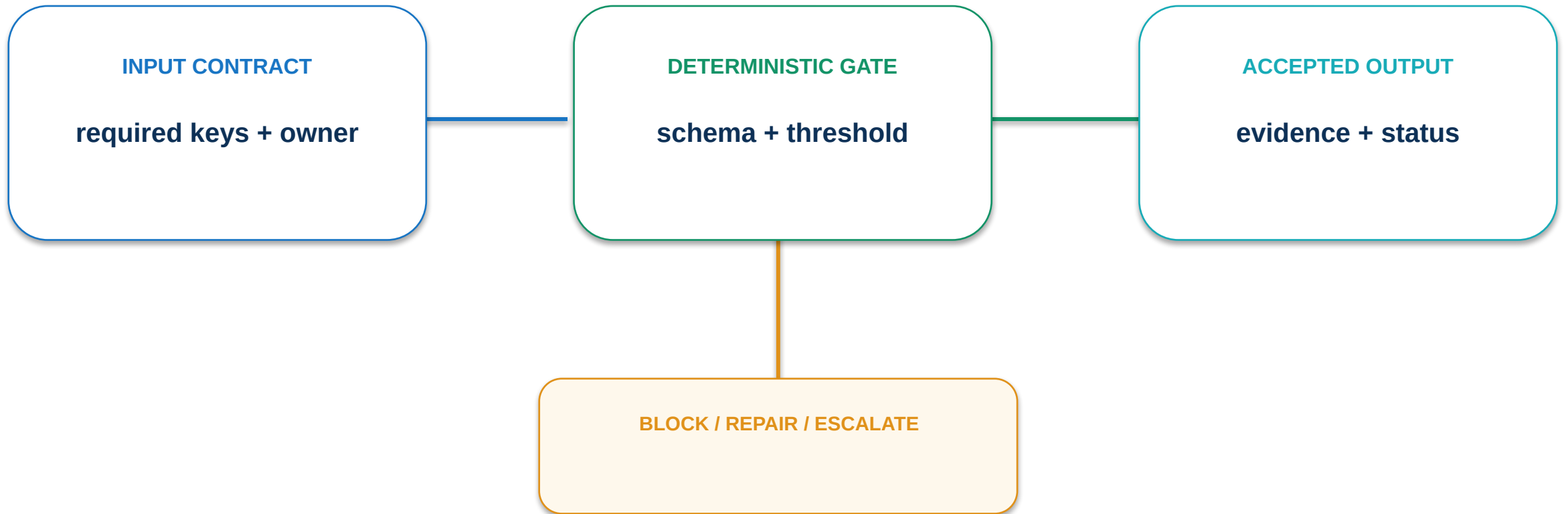
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 08 · Headless codex exec: execution path



## 08 · Headless codex exec: contract and control



Use structured JSON events and explicit finish conditions.

## 08 · Headless codex exec: evidence and decision

### MEASURE

**Machine-readable runs with reliable non-zero failure status**

### ACCEPTANCE EVIDENCE

JSONL event log and process exit code

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

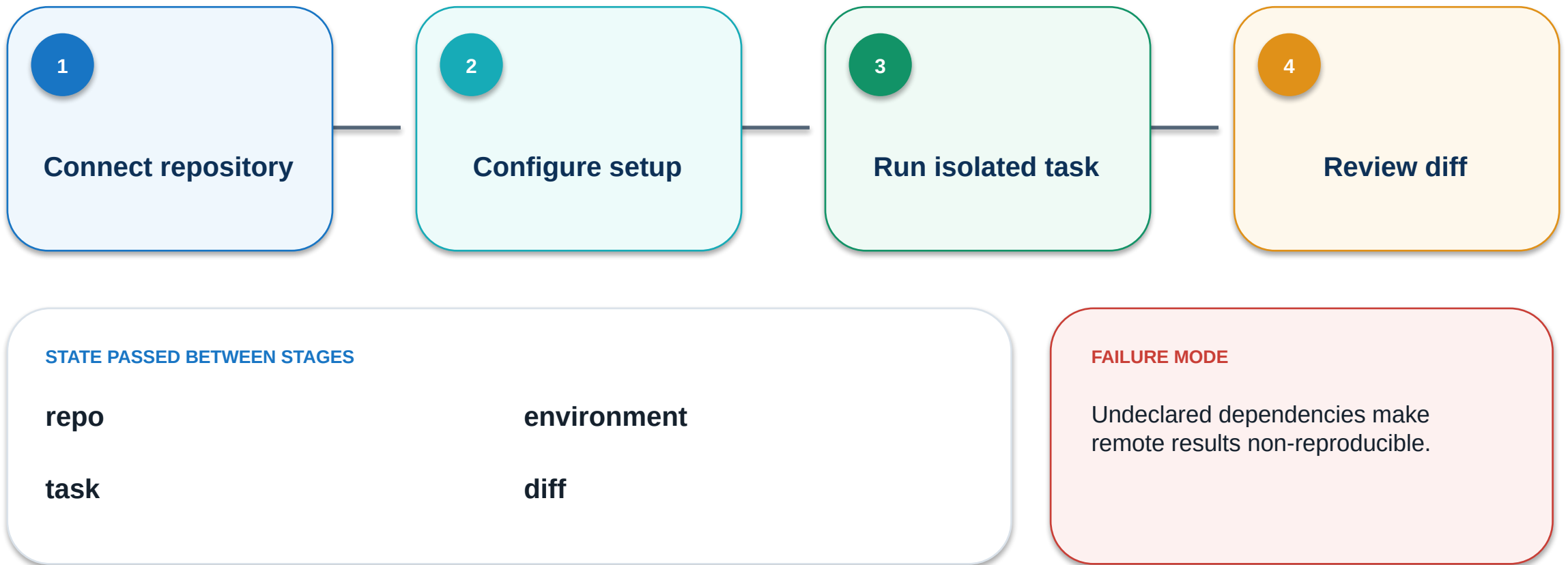
PASS / REPAIR

### OWNER

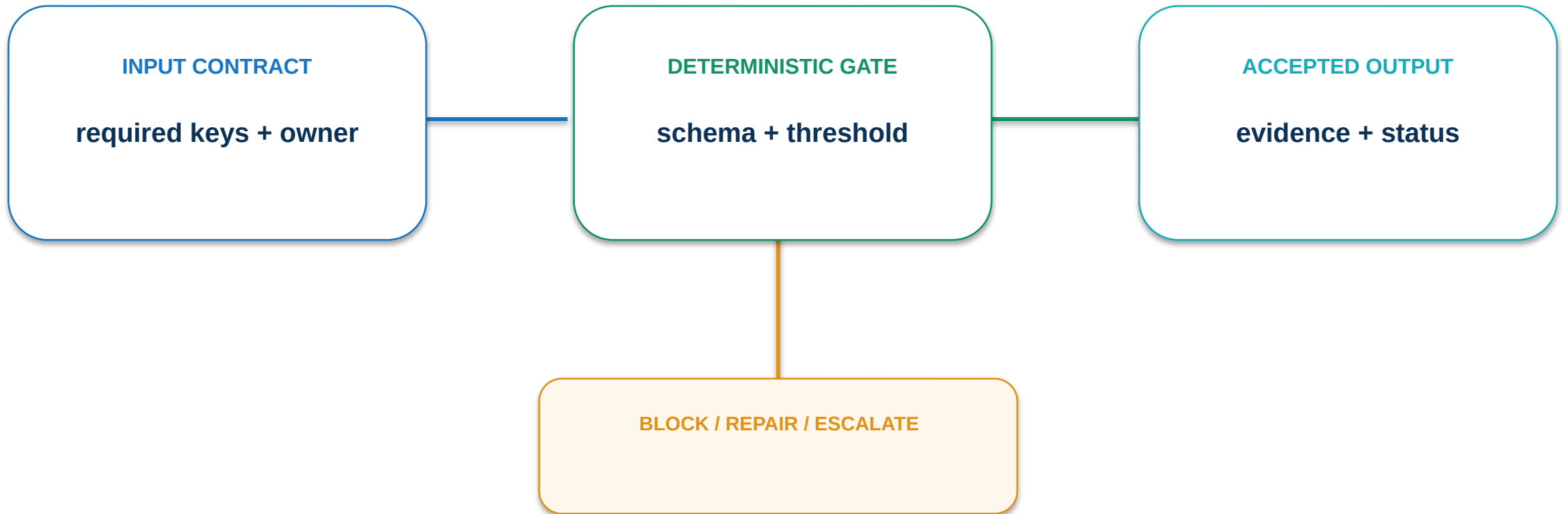
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

## 09 · Cloud environments: execution path



## 09 · Cloud environments: contract and control



Pin setup steps, environment variables and safe secrets.

## 09 · Cloud environments: evidence and decision

### MEASURE

**Cloud tasks reproducible from declared setup**

### ACCEPTANCE EVIDENCE

Environment config and cloud task summary

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

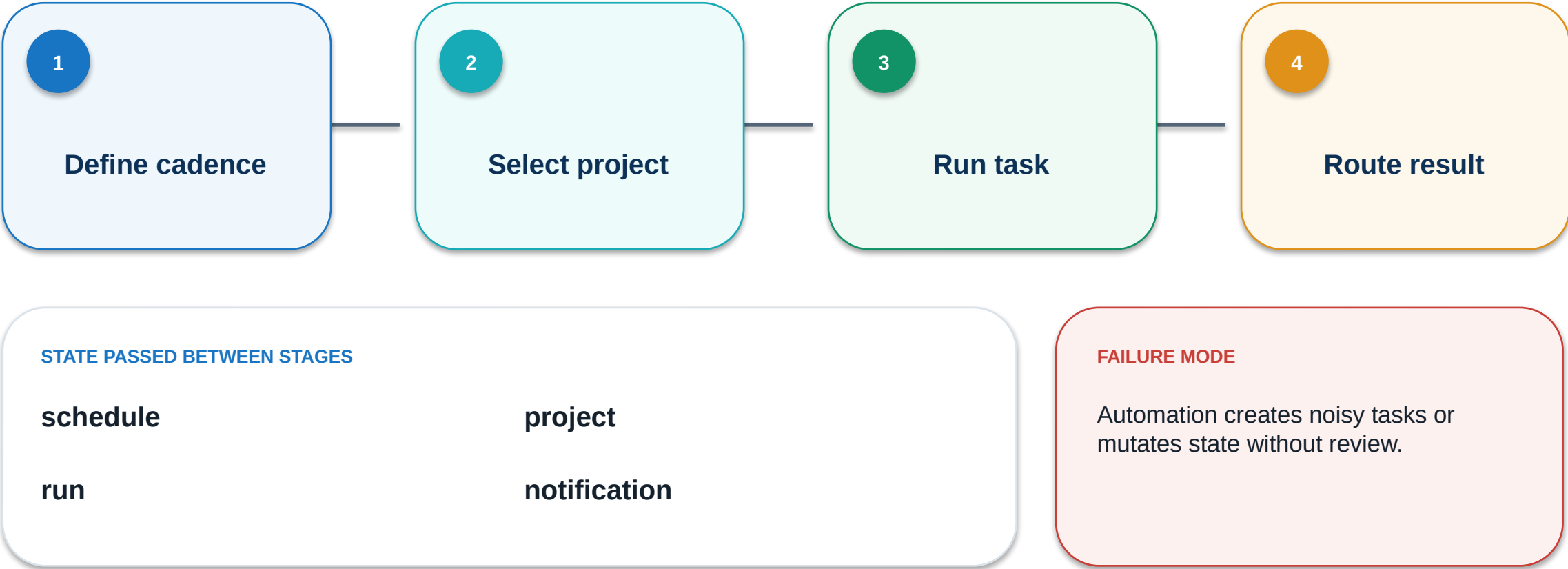
PASS / REPAIR

### OWNER

named human

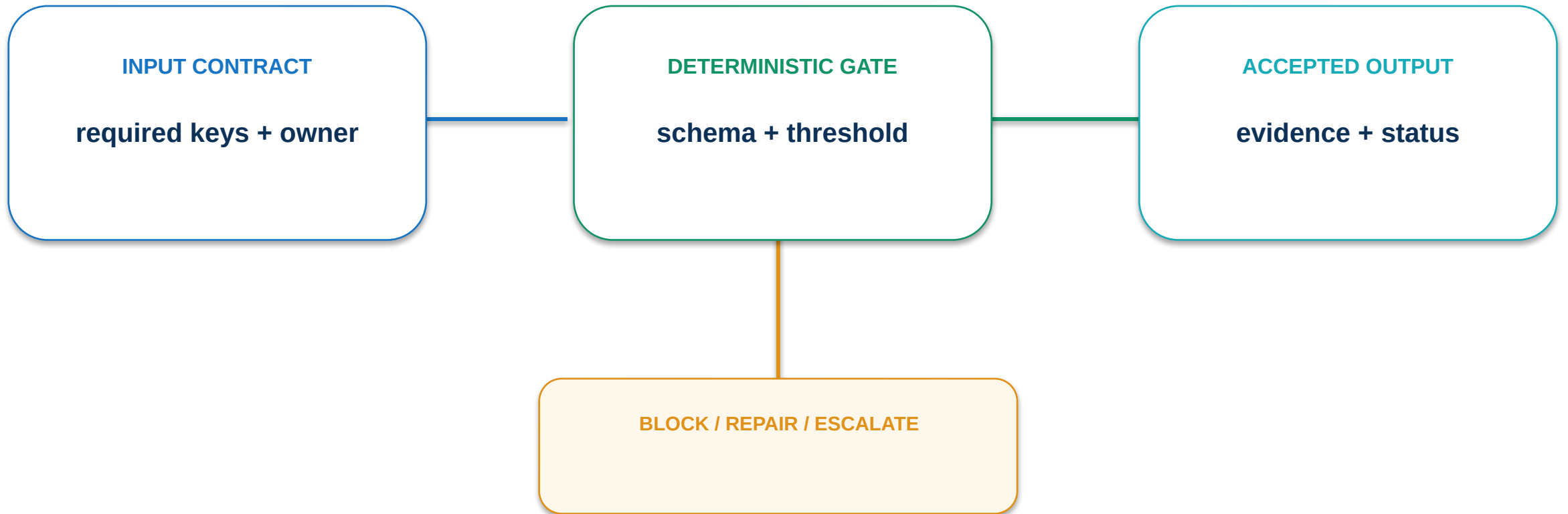
**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 10 · Background and scheduled work: execution path



## 10 · Background and scheduled work: contract and control

---



Use bounded prompts, idempotency and appropriate notification policy.

## 10 · Background and scheduled work: evidence and decision

### MEASURE

**Scheduled runs producing actionable, non-duplicate results**

### ACCEPTANCE EVIDENCE

Run history and action decision

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

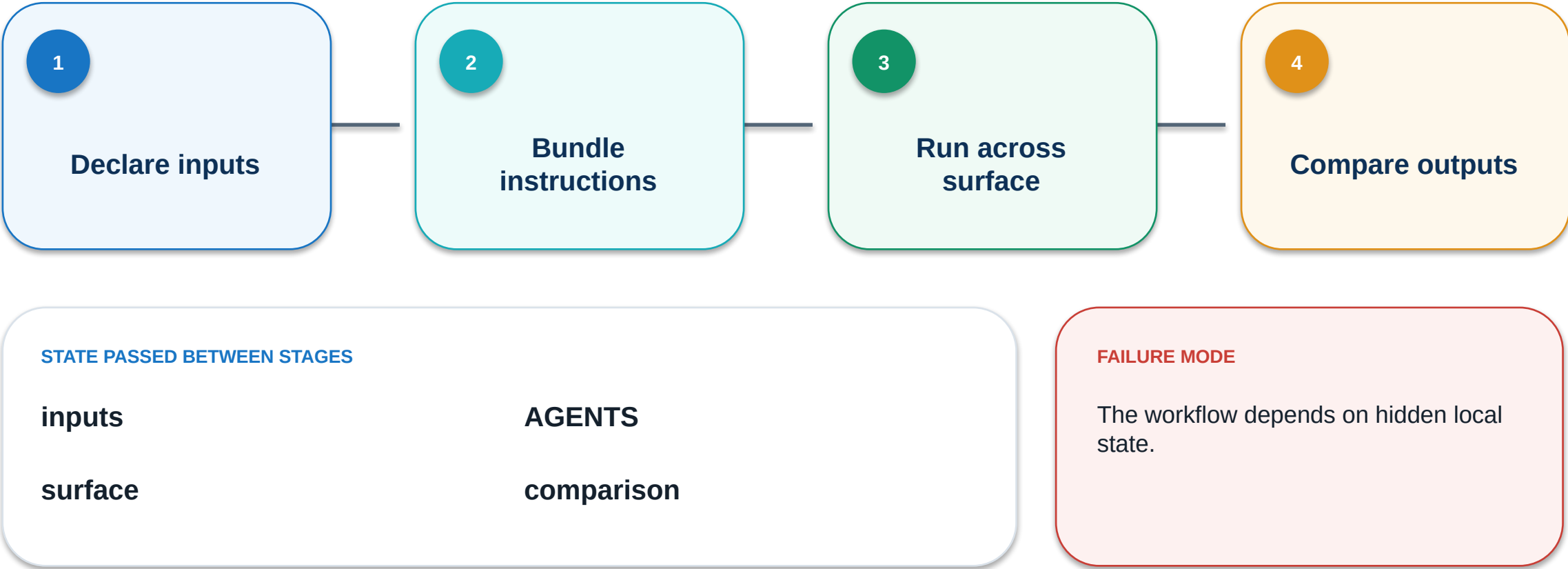
PASS / REPAIR

### OWNER

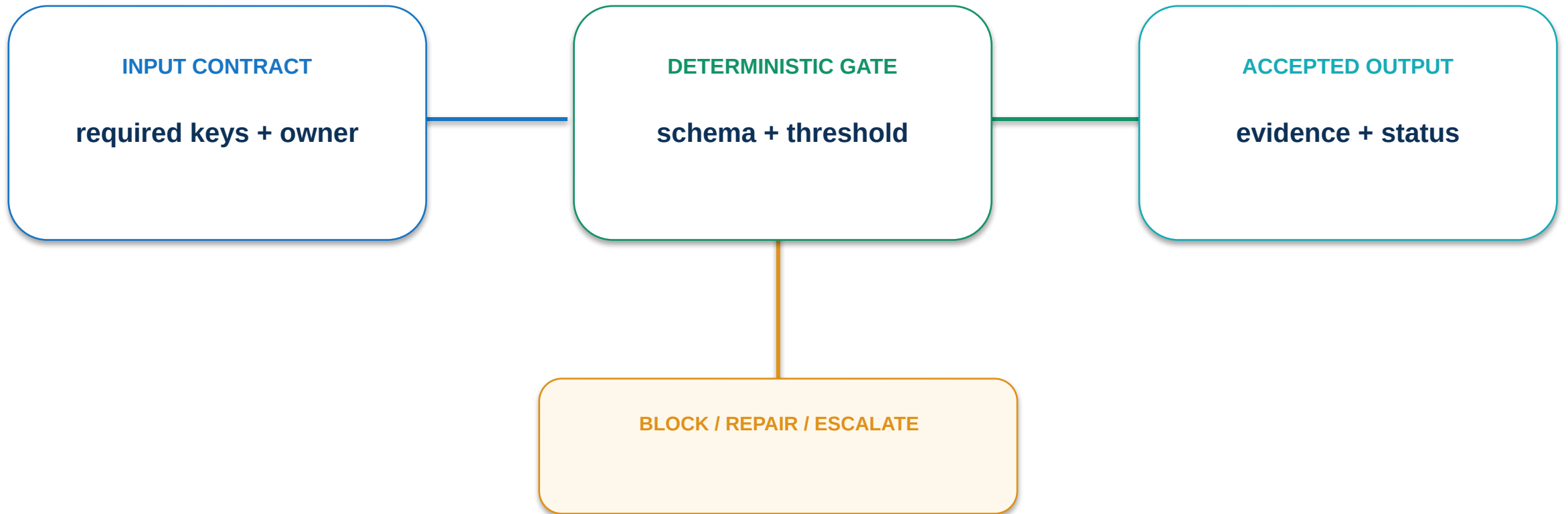
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 11 · Portable agent workflow: execution path



## 11 · Portable agent workflow: contract and control



Move requirements into versioned repository artifacts.

## 11 · Portable agent workflow: evidence and decision

### MEASURE

**Equivalent acceptance results across supported surfaces**

### ACCEPTANCE EVIDENCE

Portable repository and cross-surface comparison

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

PASS / REPAIR

### OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

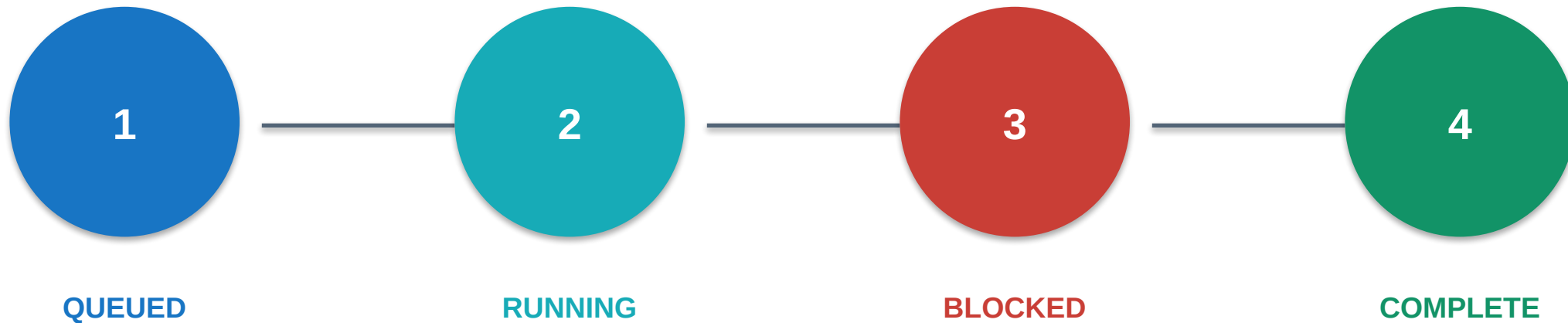
## Extension boundary: skill, tool server and human owner

---



# Automation state: QUEUED, RUNNING, BLOCKED and COMPLETE

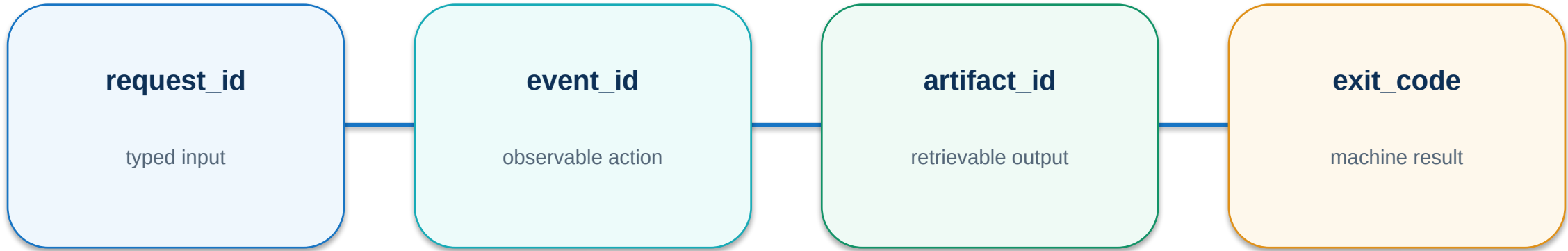
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**COMPLETE** requires a typed result, retrievable artifact and reliable exit status.

## Tool evidence chain: request → event → artifact → exit status

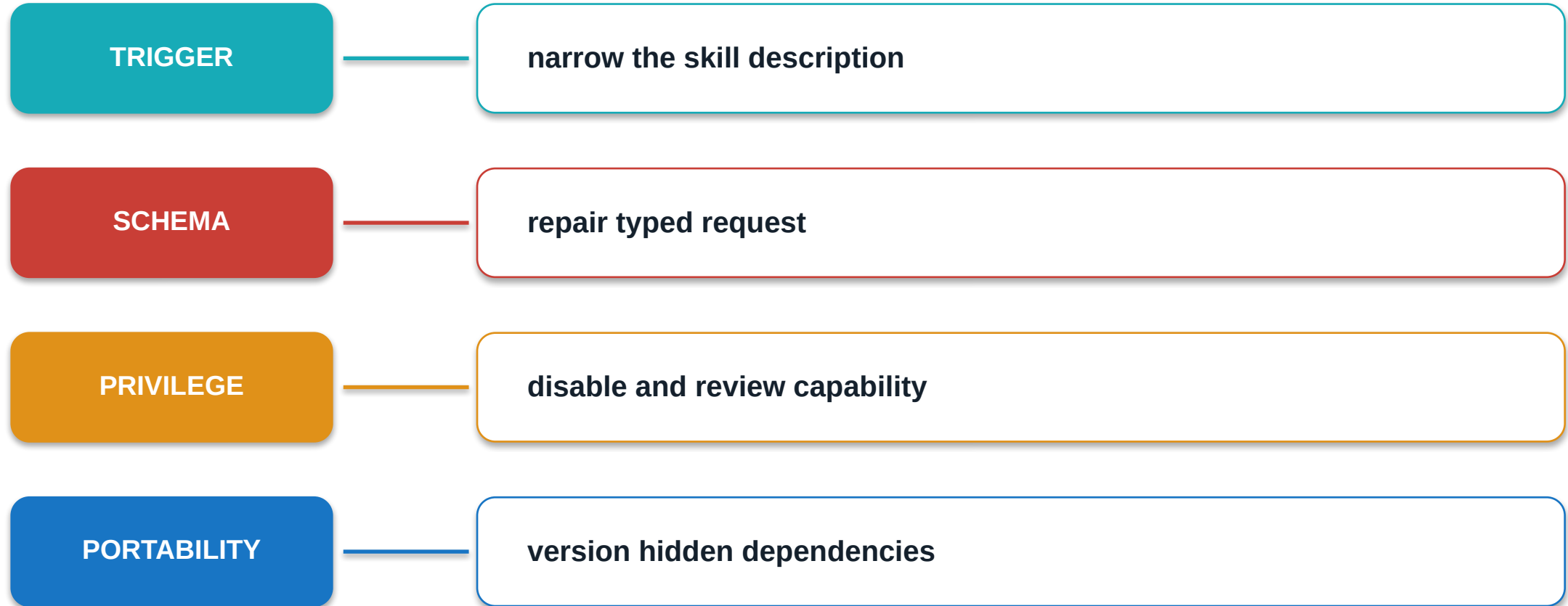
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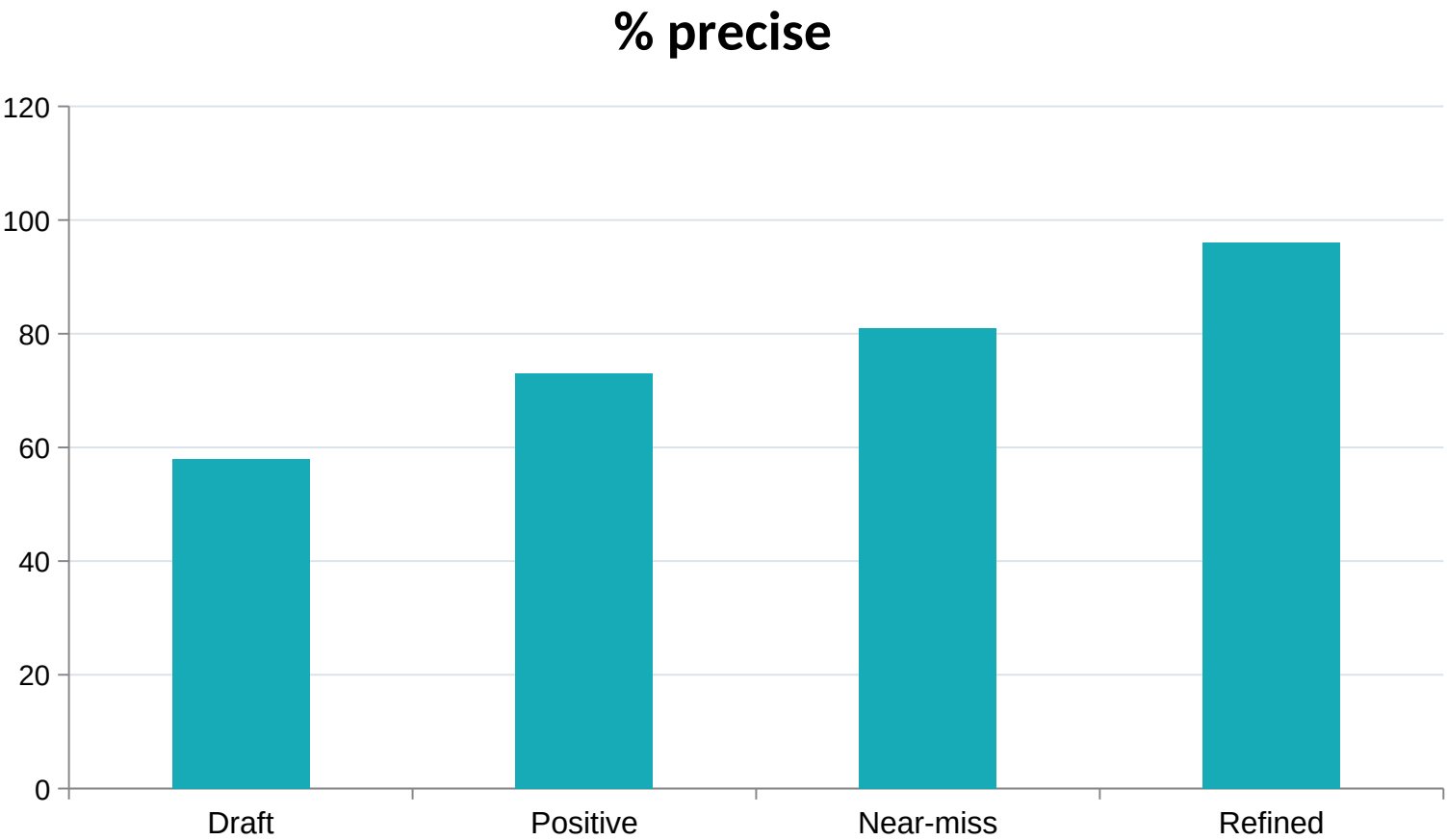
**Portable automation exposes typed inputs, events, artifacts and status.**

# Extension taxonomy: trigger, schema, privilege and portability

---



# Skill precision improves after near-miss tests

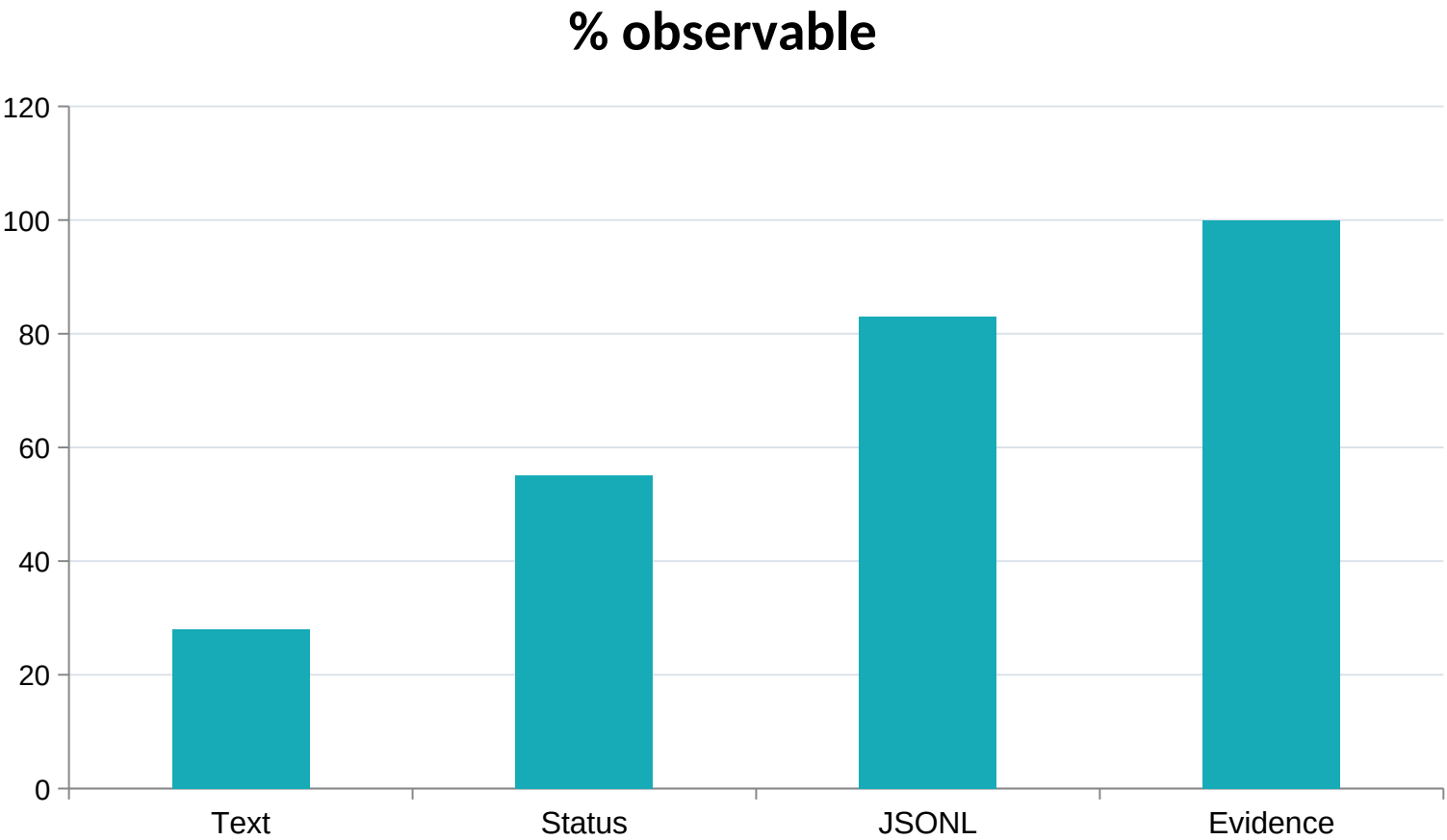


INSIGHT

**A reusable skill  
needs both trigger  
and near-miss  
evidence.**

Illustrative synthetic values for  
mechanism explanation.

# Structured events improve automation observability

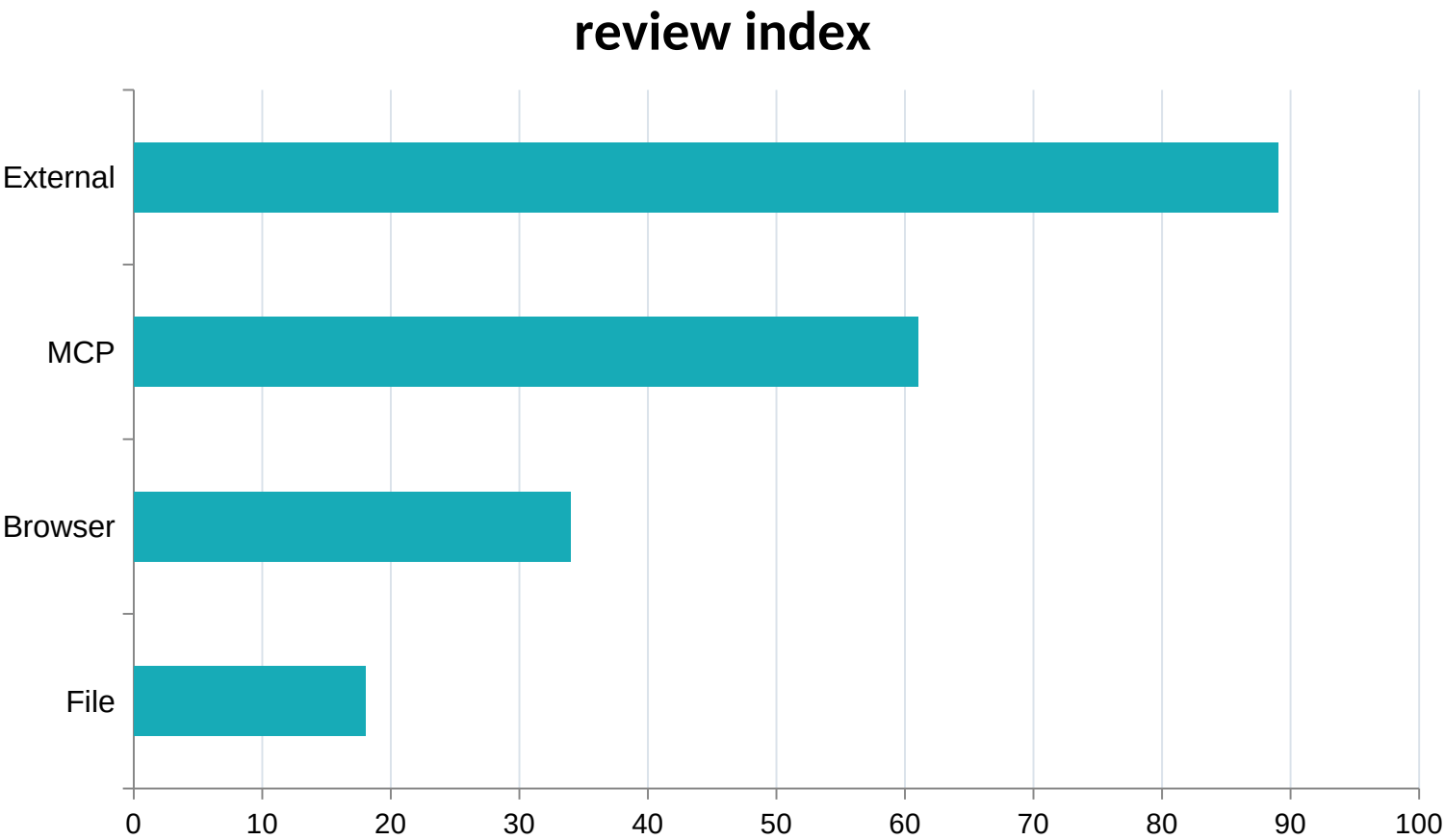


INSIGHT

**Machine-readable events turn a chat workflow into an observable automation contract.**

Illustrative synthetic values for mechanism explanation.

# Extension privilege expands review effort



INSIGHT

**More capable extensions require proportionally stronger permission and dependency review.**

Illustrative synthetic values for mechanism explanation.

## Lab 06 · Browser QA with Playwright

---

### SCENARIO

**Verify a local TaskBoard dashboard through a real browser and convert one visible defect into reproducible evidence.**

### EXPECTED OUTPUT

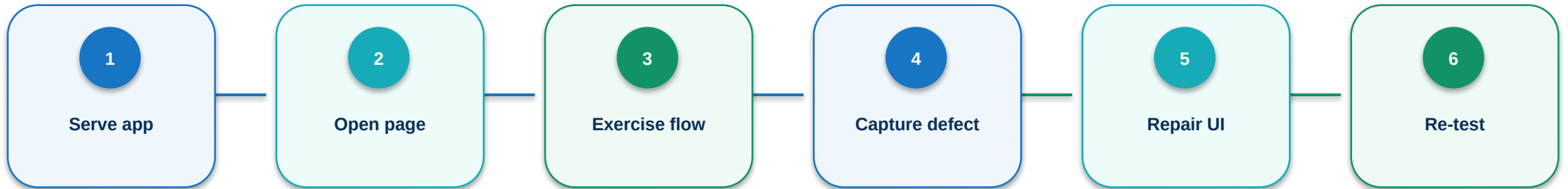
**Browser trace, screenshot and verified repair**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-06-browser-qa/**

## Lab 06 · Codex workflow contract

---



**Each node consumes named keys, emits a status, and leaves retrievable execution evidence.**

## Lab 06 · evidence and failure gates

### PASS EVIDENCE

The evidence includes pre-repair failure, scoped patch and passing desktop/mobile re-test.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Lab 07 · Build and Evaluate a Codex Skill

---

### SCENARIO

**Package a test-triage workflow as a reusable skill and measure trigger precision with positive and near-miss prompts.**

### EXPECTED OUTPUT

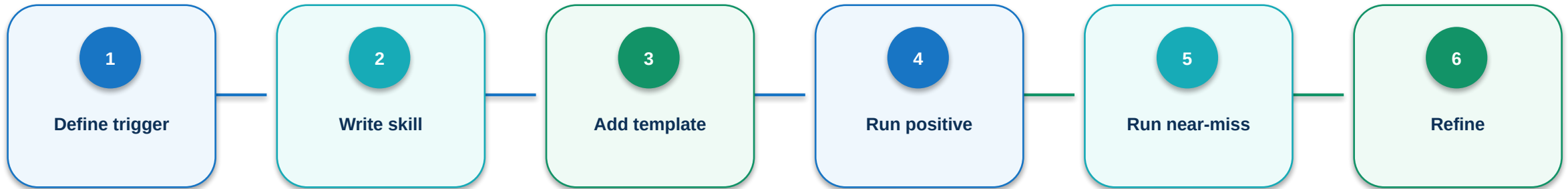
**Skill package and trigger evaluation report**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-07-reusable-skill/**

## Lab 07 · Codex workflow contract

---



**Each node consumes named keys, emits a status, and leaves retrievable execution evidence.**

## Lab 07 · evidence and failure gates

### PASS EVIDENCE

The skill passes structural checks and its evaluation shows correct trigger and near-miss behavior.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Lab 08 · Headless Codex Exec for CI

---

### SCENARIO

**Design a non-interactive Codex run that emits machine-readable events and fails safely when acceptance checks fail.**

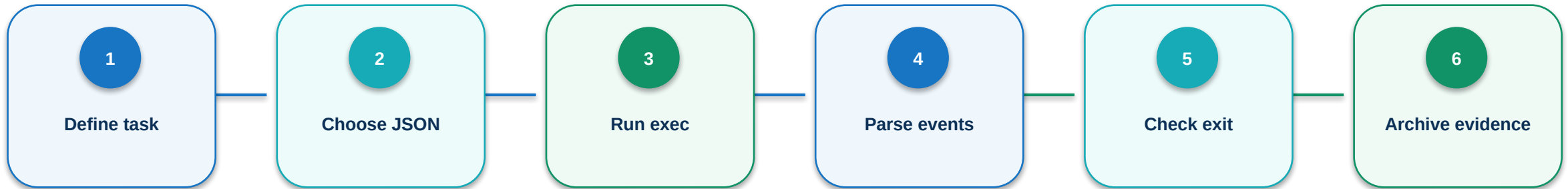
### EXPECTED OUTPUT

**CI-safe command wrapper and JSONL evidence**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-08-headless-exec/**

# Lab 08 · Codex workflow contract



**Consequential actions require a current diff, passing checks and a named decision.**



## Lab 08 · evidence and failure gates

### PASS EVIDENCE

The wrapper avoids embedded secrets, records structured events and distinguishes success from failure by exit status.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Topic 3 technical checkpoint

---

1

**Which state field prevents a silent downstream assumption?**

2

**What evidence proves the mechanism completed successfully?**

3

**Which failure must stop rather than retry?**

4

**Who owns the consequential decision?**

## Topic 3 transfer challenge

---

1

**Change one input constraint and predict the affected nodes.**

2

**Name the identifier that preserves provenance across the change.**

3

**Define one machine threshold and one human decision.**

4

**Describe the minimum evidence required to resume the run.**

# 4

## TOPIC 4

# Evaluation, Comparative Effectiveness and Governed Adoption

Evaluate Codex workflows with golden tasks, risk-adjusted measures and a defensible adoption recommendation.

1

CONTROL

2

EVIDENCE

3

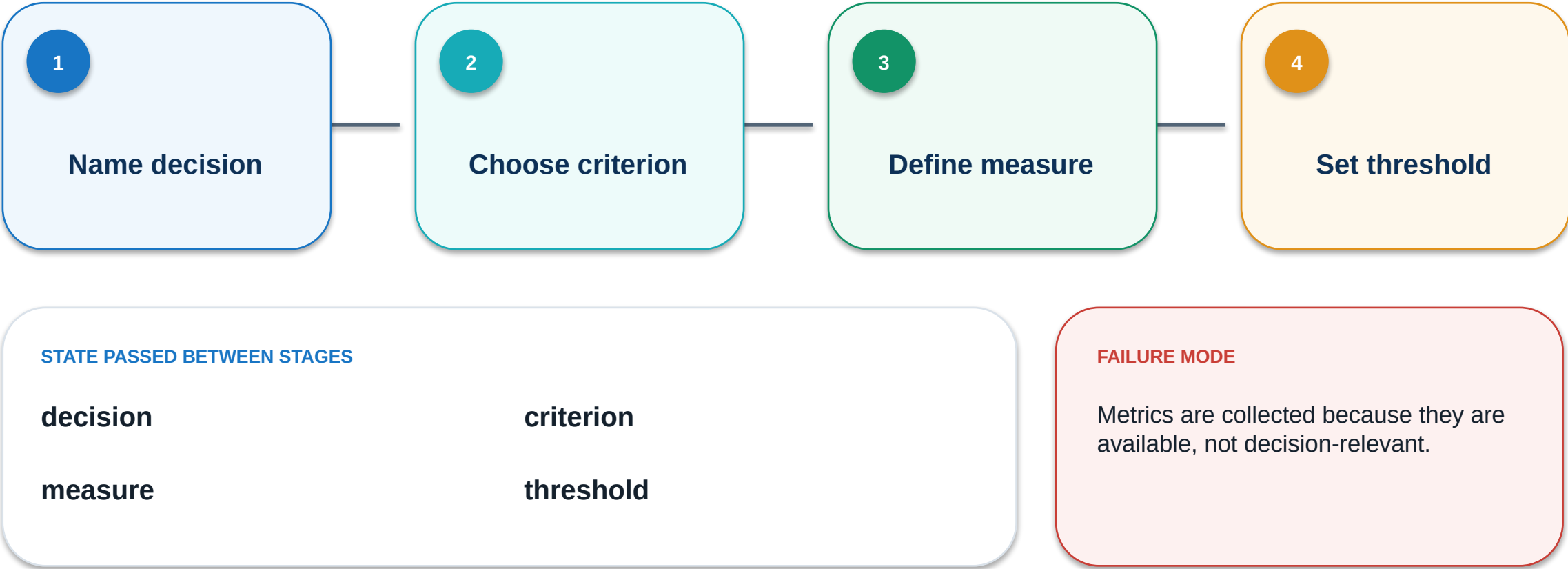
AUTOMATION

4

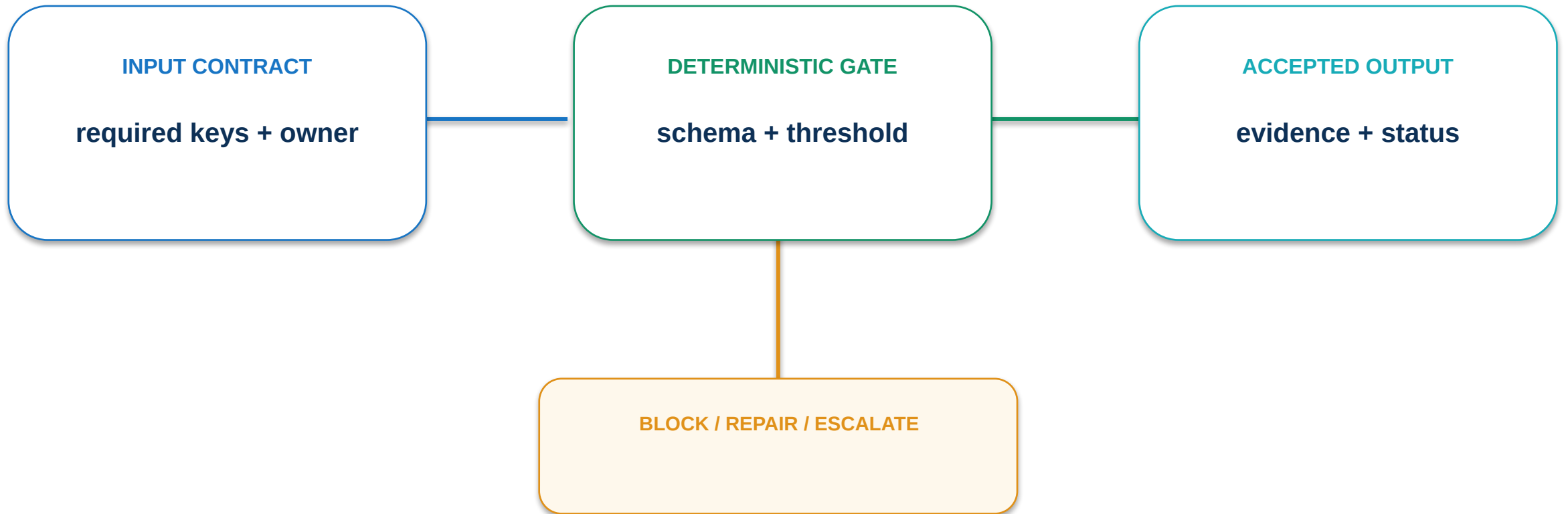
DECISION

Every mechanism ends in observable evidence—not an agent's assertion.

# 01 · Evaluation question design: execution path



## 01 · Evaluation question design: contract and control



Start from the adoption or remediation decision.

## 01 · Evaluation question design: evidence and decision

MEASURE

**Criteria with decision-linked thresholds**

ACCEPTANCE EVIDENCE

Evaluation charter

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 02 · Golden task suite: execution path

## GOLDEN-TASK DASHBOARD

- 1

CORRECT

Score accepted outcomes
- 2

RELIABLE

Repeat and measure variance
- 3

EFFICIENT

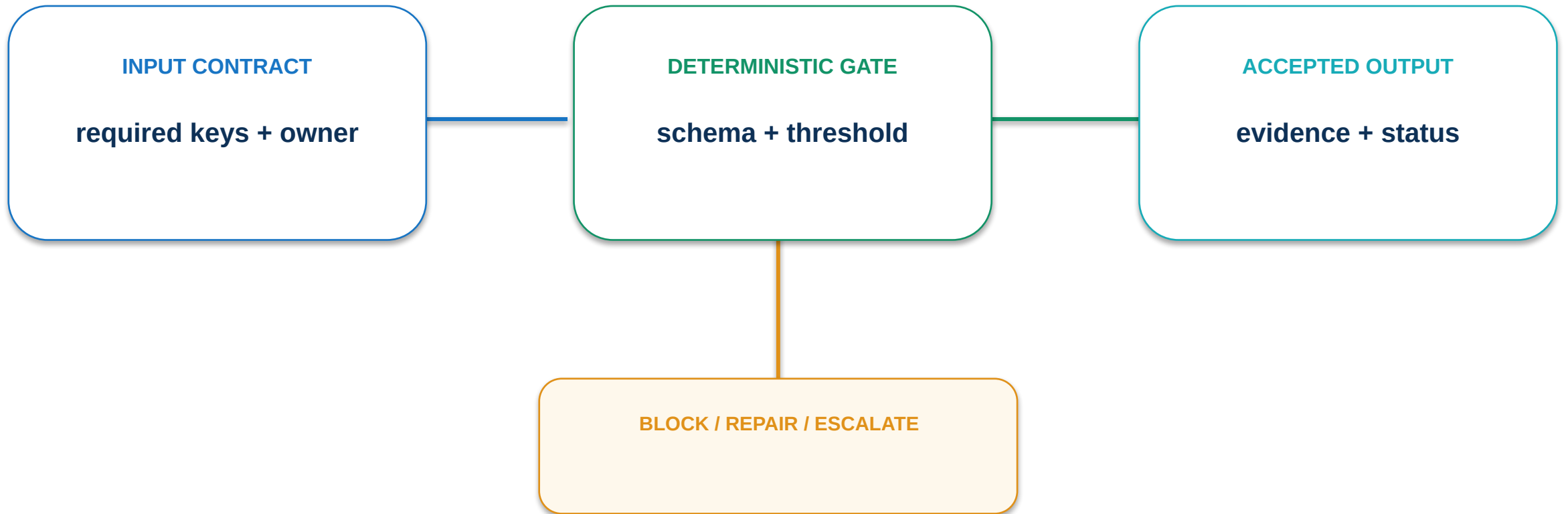
Track latency and full cost
- 4

GOVERN

Bind risk to a human posture



## 02 · Golden task suite: contract and control



Use a stratified task set including edge and failure cases.

## 02 · Golden task suite: evidence and decision

MEASURE

**Representative tasks passing repeatable checks**

ACCEPTANCE EVIDENCE

Versioned fixtures and run results

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

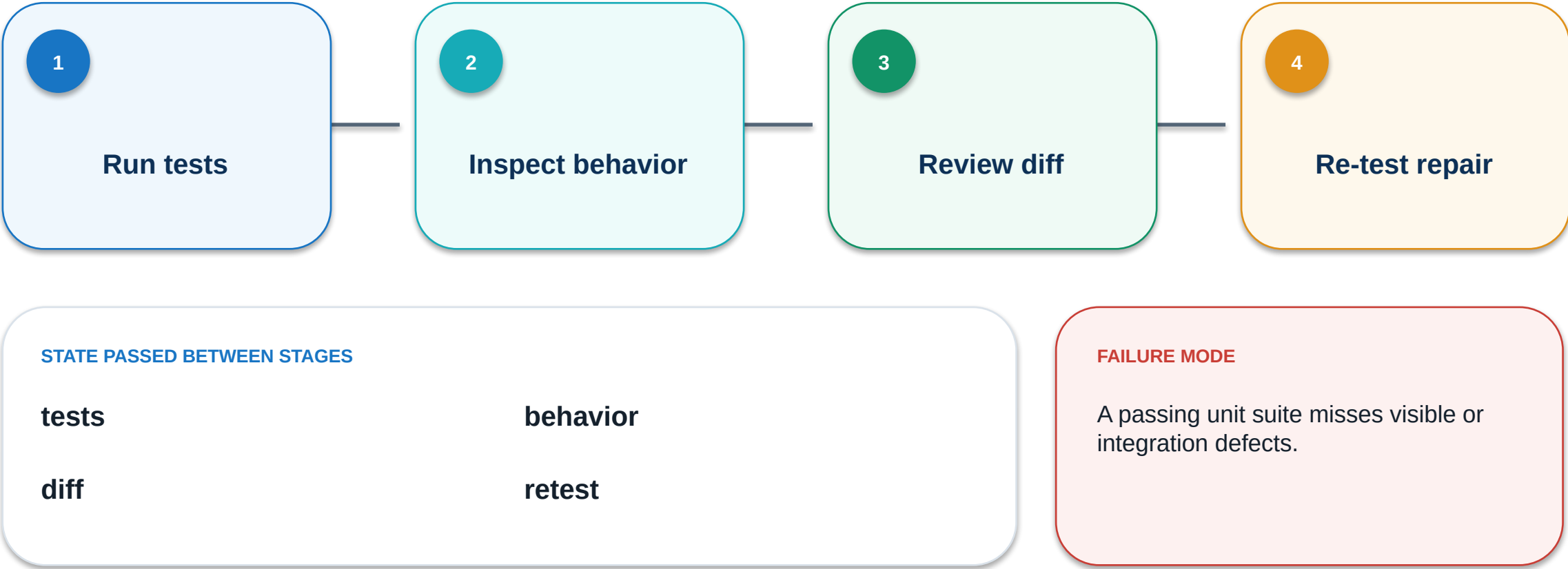
PASS / REPAIR

OWNER

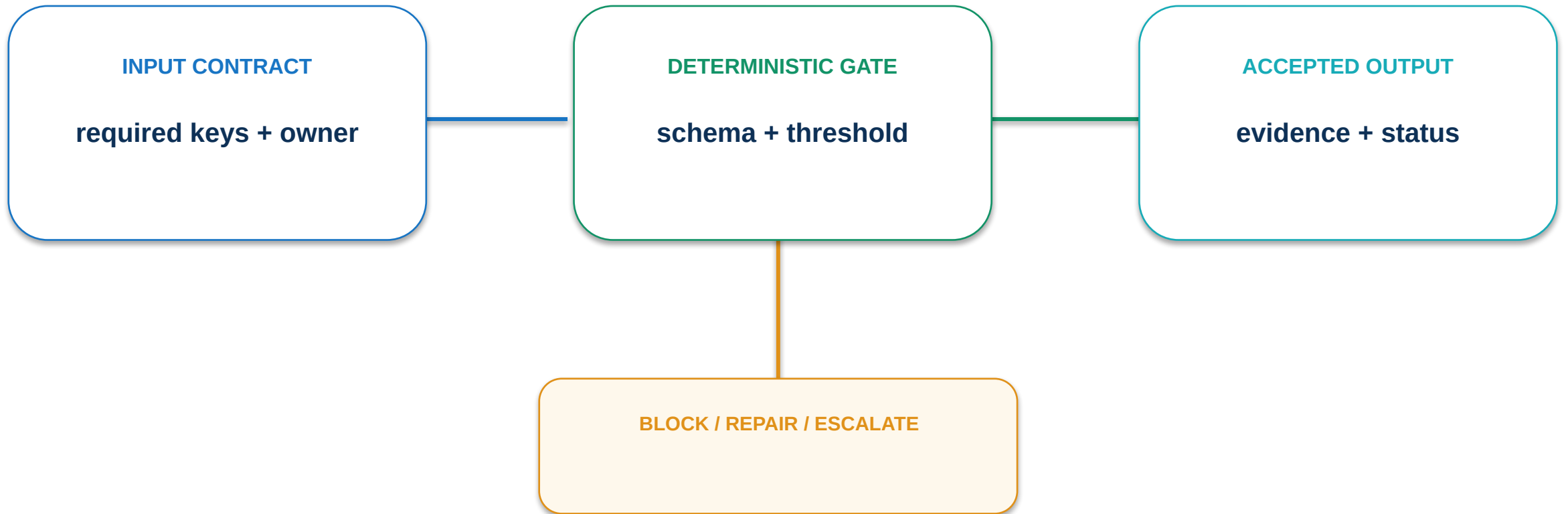
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 03 · Functional correctness: execution path



## 03 · Functional correctness: contract and control



Combine automated, browser and human review criteria.

## 03 · Functional correctness: evidence and decision

MEASURE

**Passed required checks / total required checks**

ACCEPTANCE EVIDENCE

Test report, observed behavior and diff

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

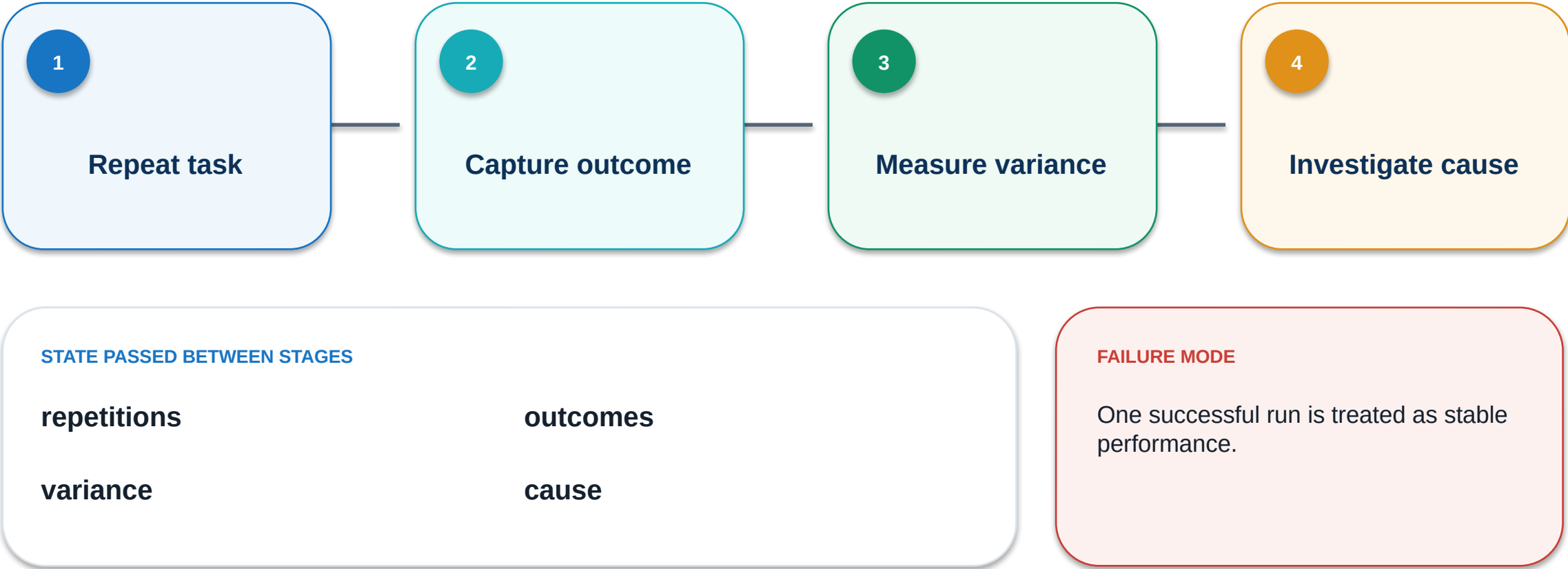
**PASS / REPAIR**

OWNER

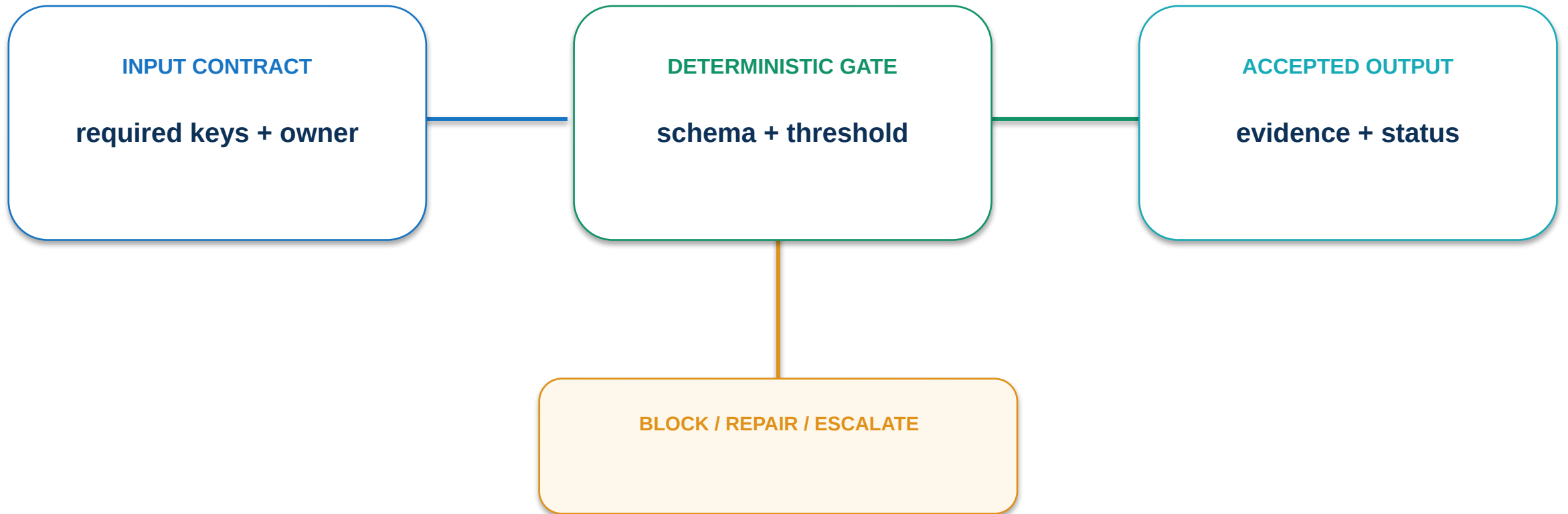
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 04 · Reliability and variance: execution path



## 04 · Reliability and variance: contract and control



Repeat representative tasks under controlled conditions.

## 04 · Reliability and variance: evidence and decision

### MEASURE

**Success rate and outcome variance over repeated runs**

### ACCEPTANCE EVIDENCE

Run series and variance summary

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

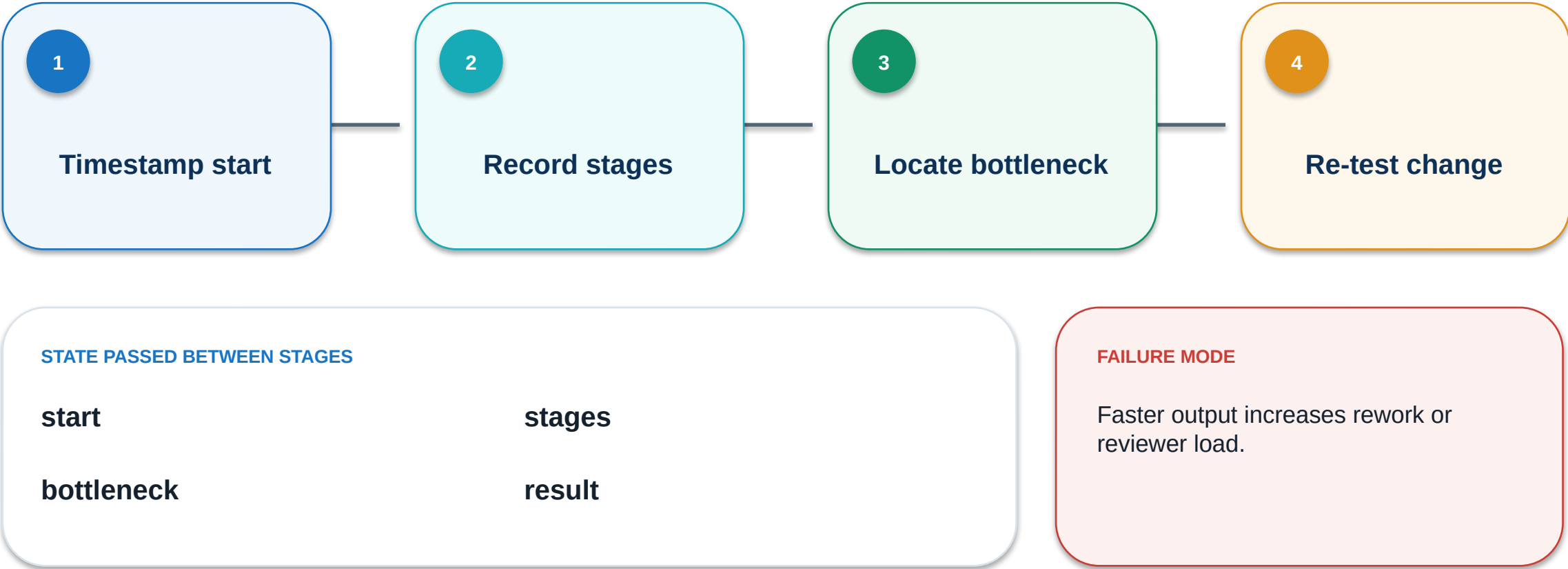
PASS / REPAIR

### OWNER

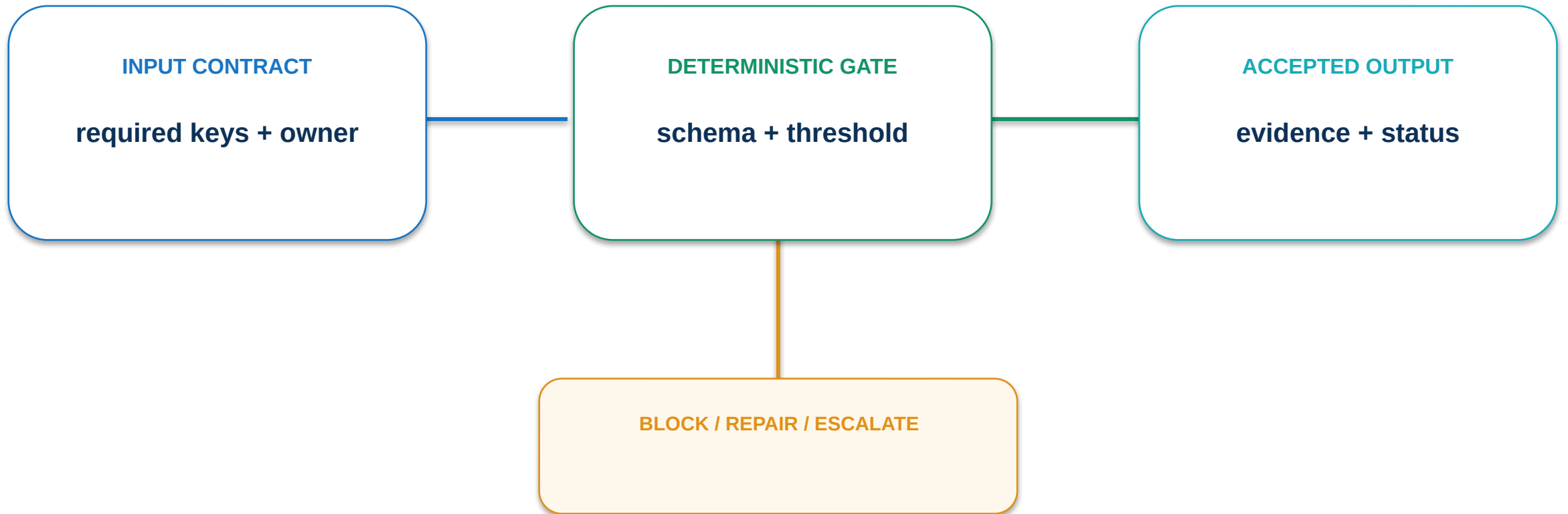
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 05 · Latency and throughput: execution path



## 05 · Latency and throughput: contract and control



Pair speed measures with defect and review effort.

## 05 · Latency and throughput: evidence and decision

### MEASURE

**Median completion time and useful tasks per hour**

### ACCEPTANCE EVIDENCE

Stage timings and quality-adjusted throughput

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

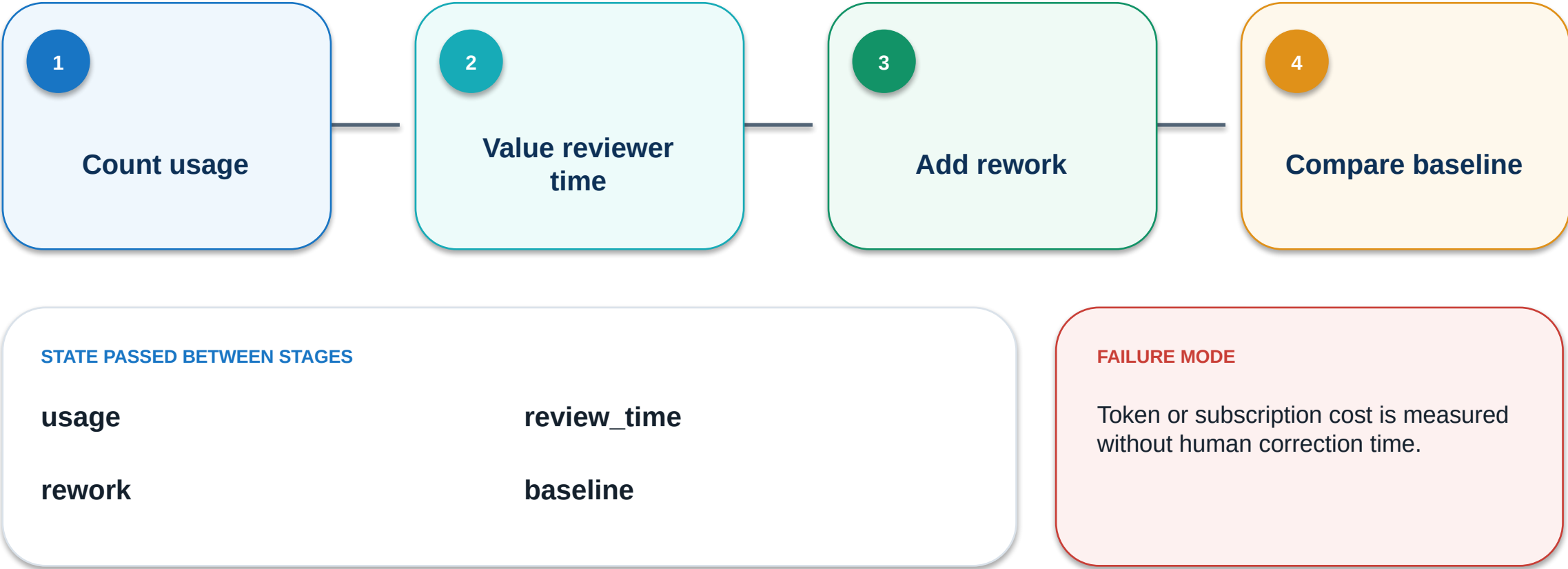
PASS / REPAIR

### OWNER

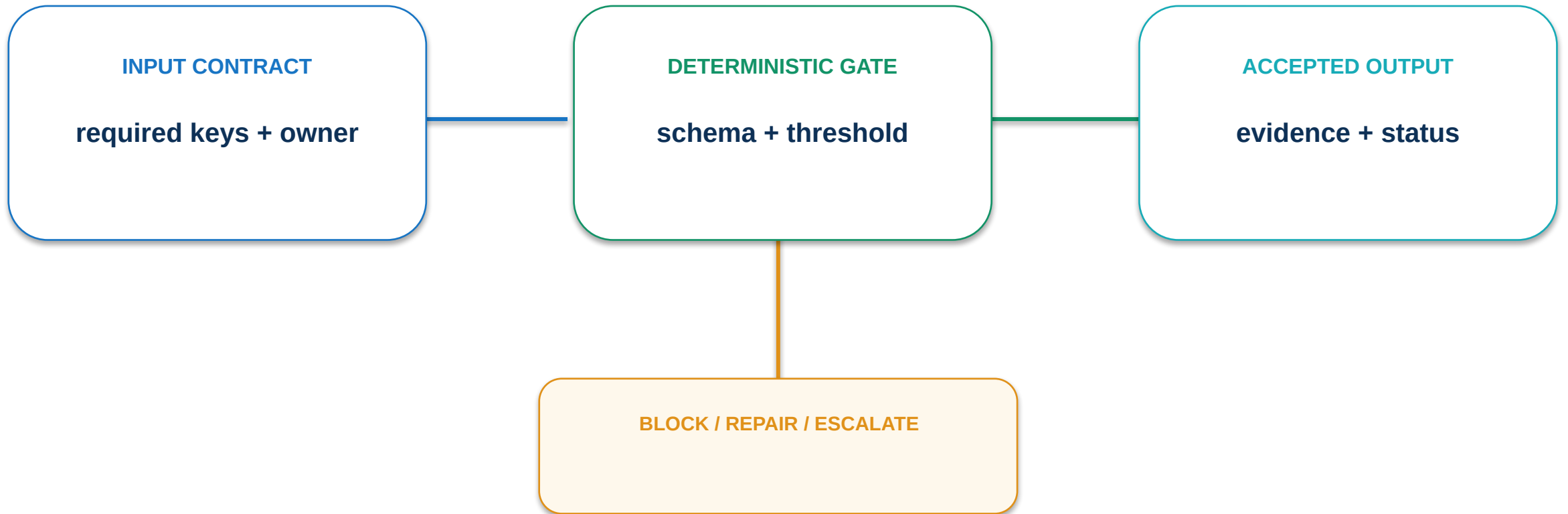
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 06 · Cost and resource efficiency: execution path



## 06 · Cost and resource efficiency: contract and control



Use fully loaded cost per accepted outcome.

## 06 · Cost and resource efficiency: evidence and decision

MEASURE

**Total cost per accepted task**

ACCEPTANCE EVIDENCE

Cost model and sensitivity table

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

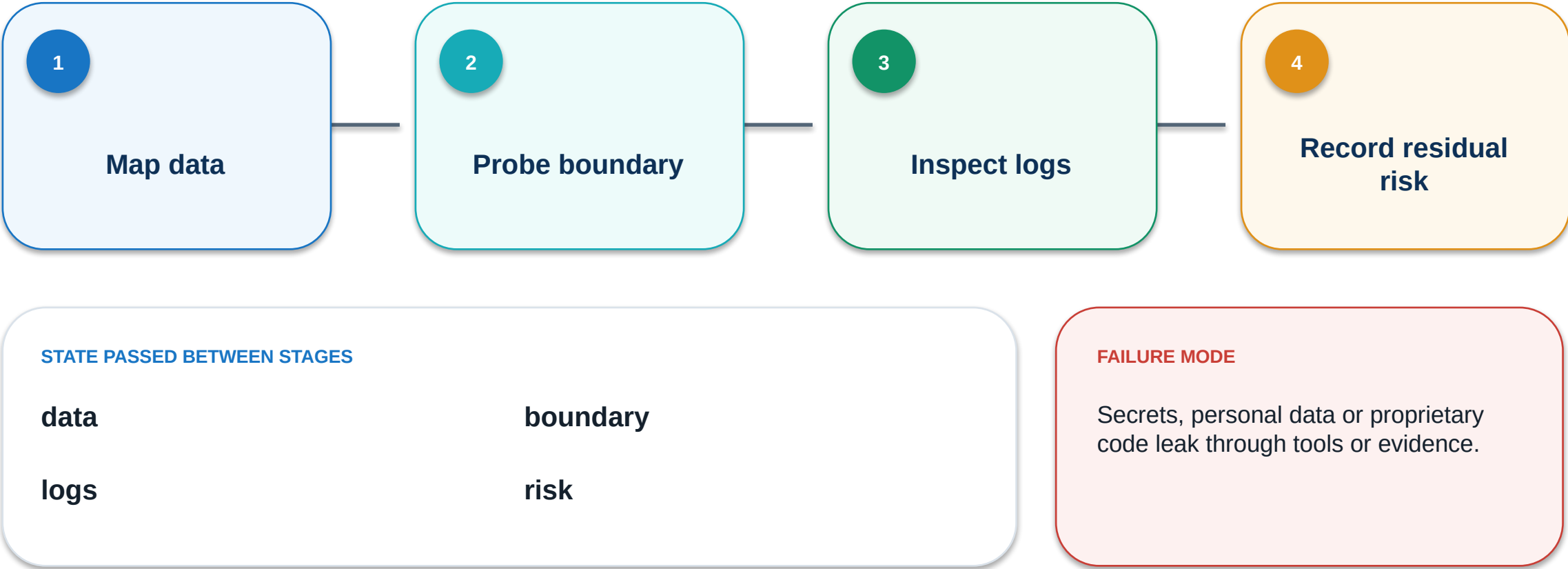
PASS / REPAIR

OWNER

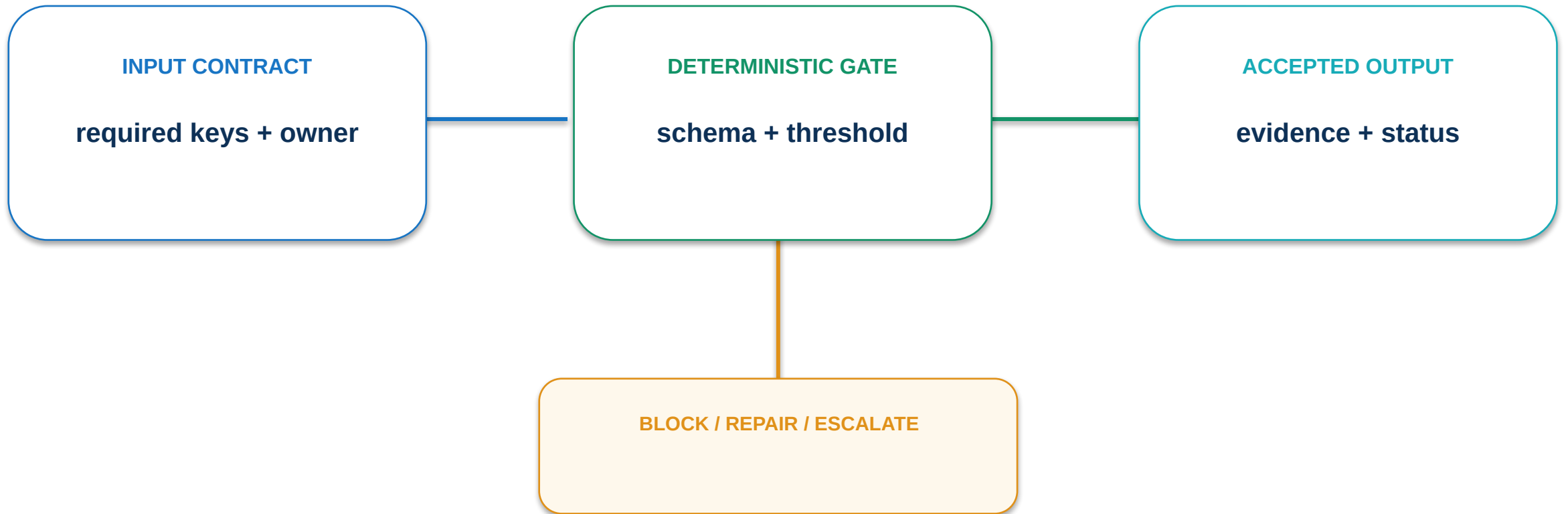
named human

Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.

# 07 · Security and privacy evaluation: execution path



## 07 · Security and privacy evaluation: contract and control



Use least privilege, redaction and explicit retention rules.

## 07 · Security and privacy evaluation: evidence and decision

### MEASURE

**Material risks with tested controls and owners**

### ACCEPTANCE EVIDENCE

Threat model, probe results and risk owner

### OBSERVED

**machine value**

### THRESHOLD

**approved rule**

### DECISION

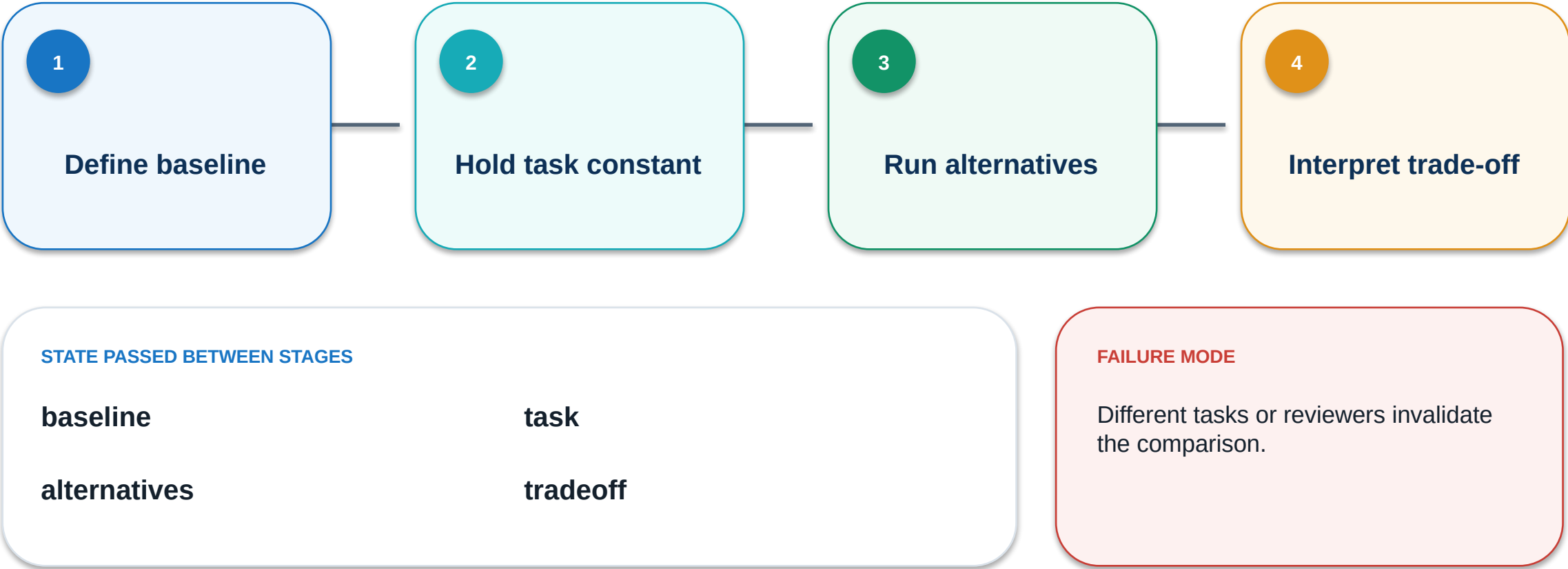
**PASS / REPAIR**

### OWNER

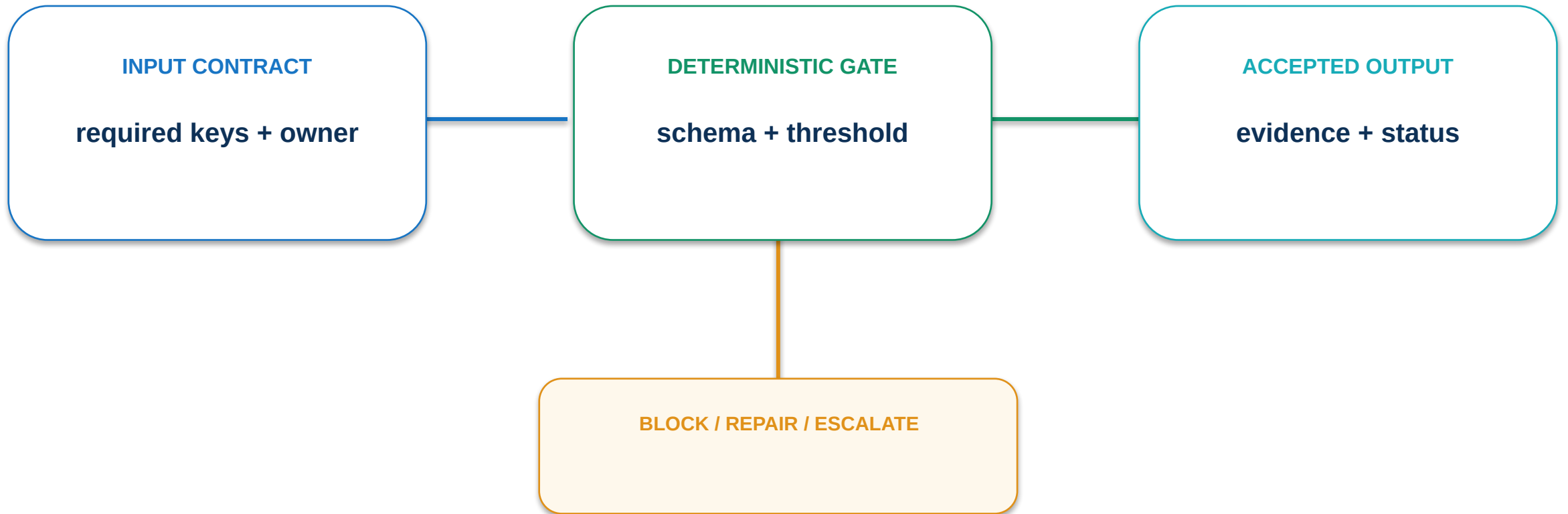
**named human**

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 08 · Comparative experiment: execution path



## 08 · Comparative experiment: contract and control



Use controlled paired trials and document confounders.

## 08 · Comparative experiment: evidence and decision

### MEASURE

**Paired results across identical tasks and criteria**

### ACCEPTANCE EVIDENCE

Comparison table and caveat register

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

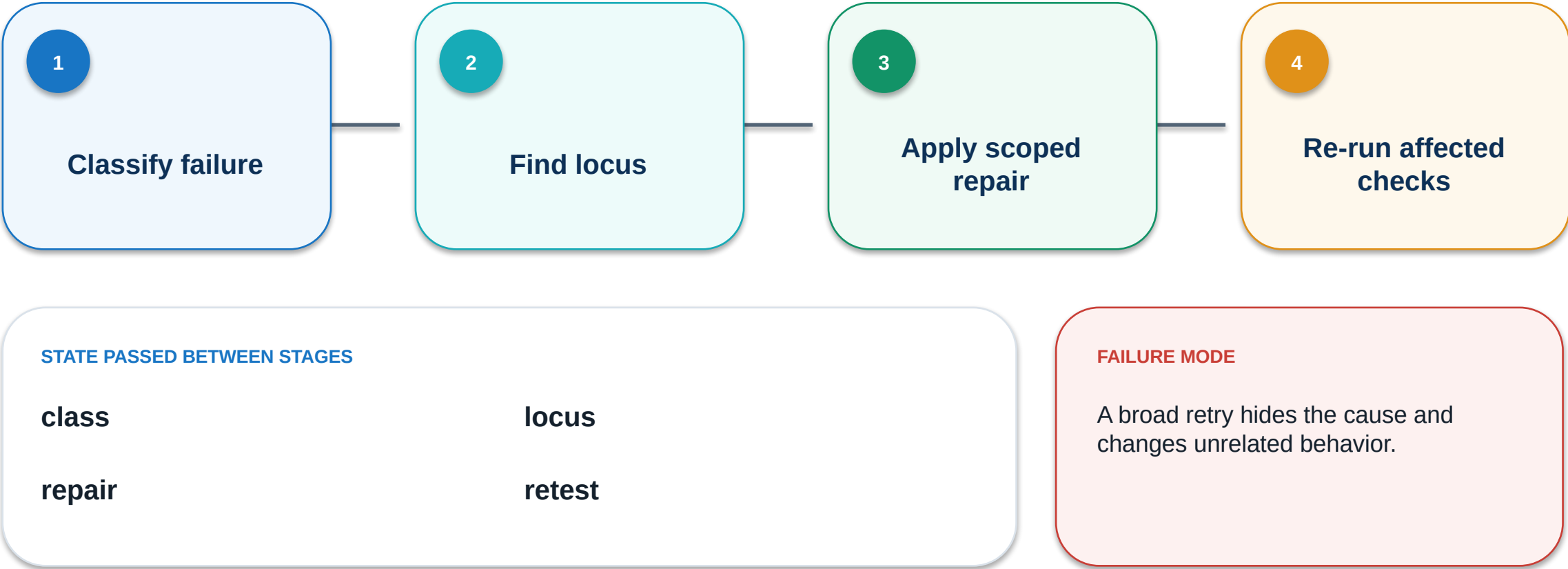
PASS / REPAIR

### OWNER

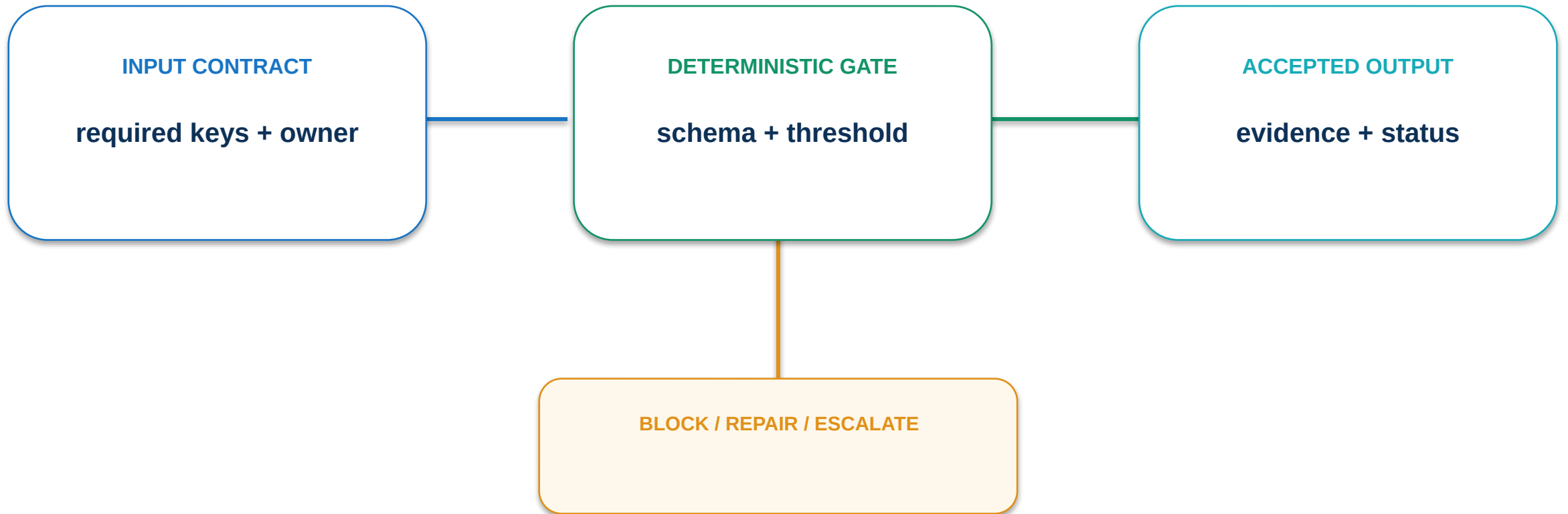
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 09 · Failure taxonomy and repair: execution path



## 09 · Failure taxonomy and repair: contract and control



Repair the smallest causal layer and re-test scope plus regression set.

## 09 · Failure taxonomy and repair: evidence and decision

### MEASURE

**Repairs that close the failed criterion without regression**

### ACCEPTANCE EVIDENCE

Failure record, patch and re-test result

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

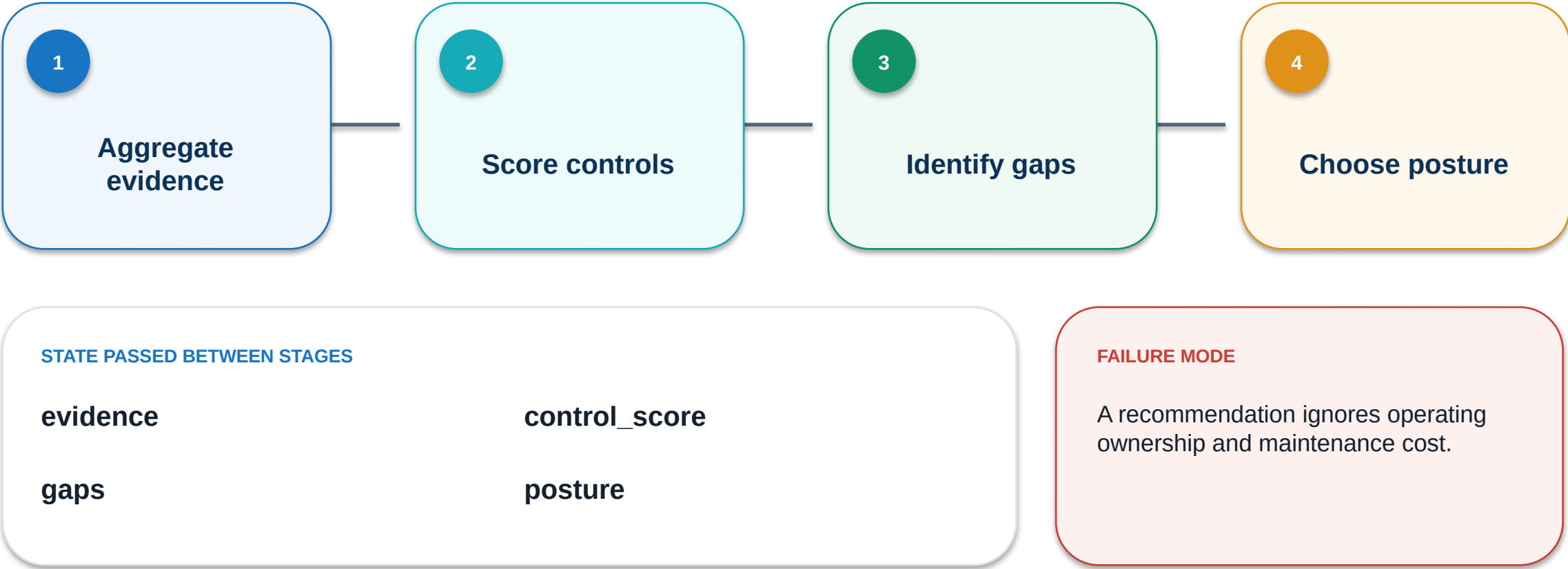
PASS / REPAIR

### OWNER

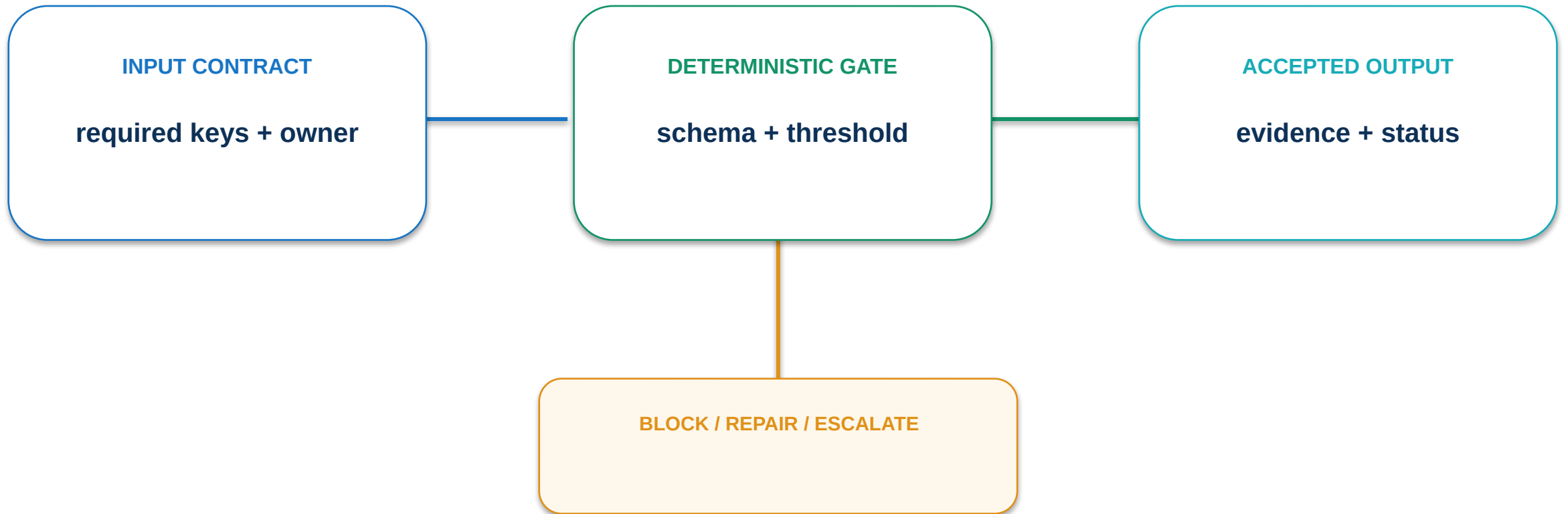
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 10 · Adoption readiness: execution path



## 10 · Adoption readiness: contract and control



Evaluate people, process, technology and governance together.

## 10 · Adoption readiness: evidence and decision

### MEASURE

**Readiness score supported by retrievable evidence**

### ACCEPTANCE EVIDENCE

Readiness scorecard and gap owners

### OBSERVED

machine value

### THRESHOLD

approved rule

### DECISION

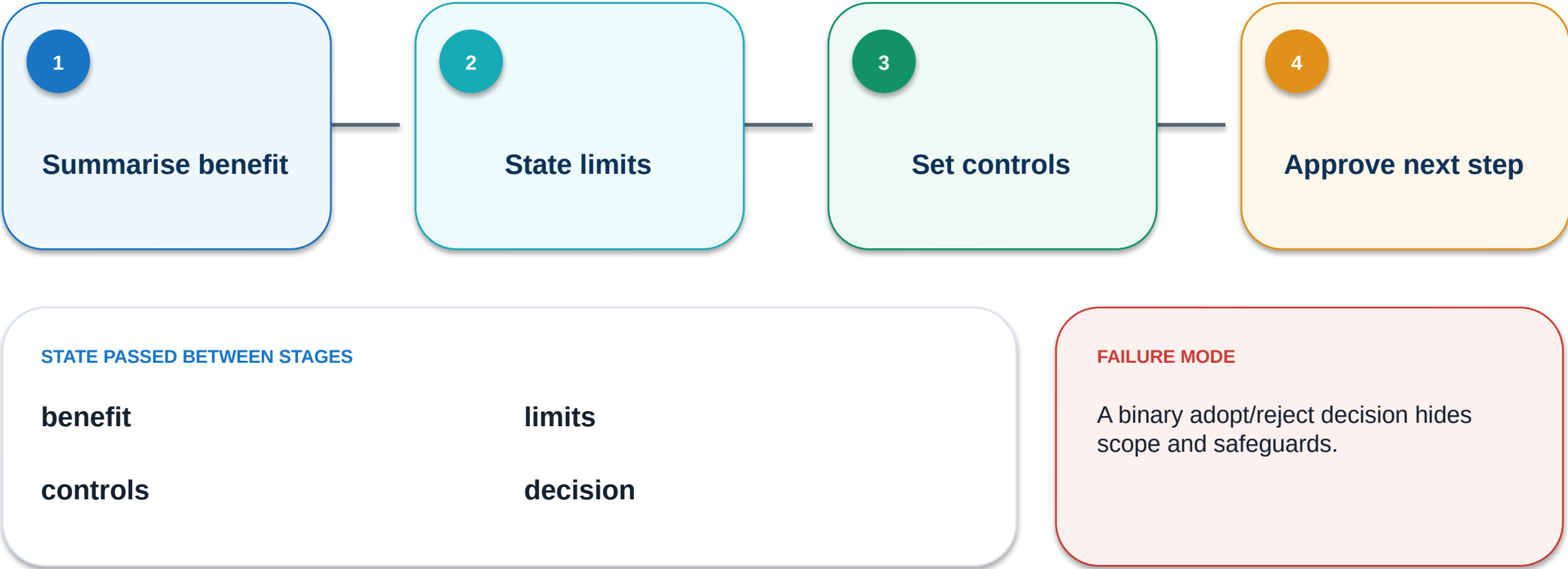
PASS / REPAIR

### OWNER

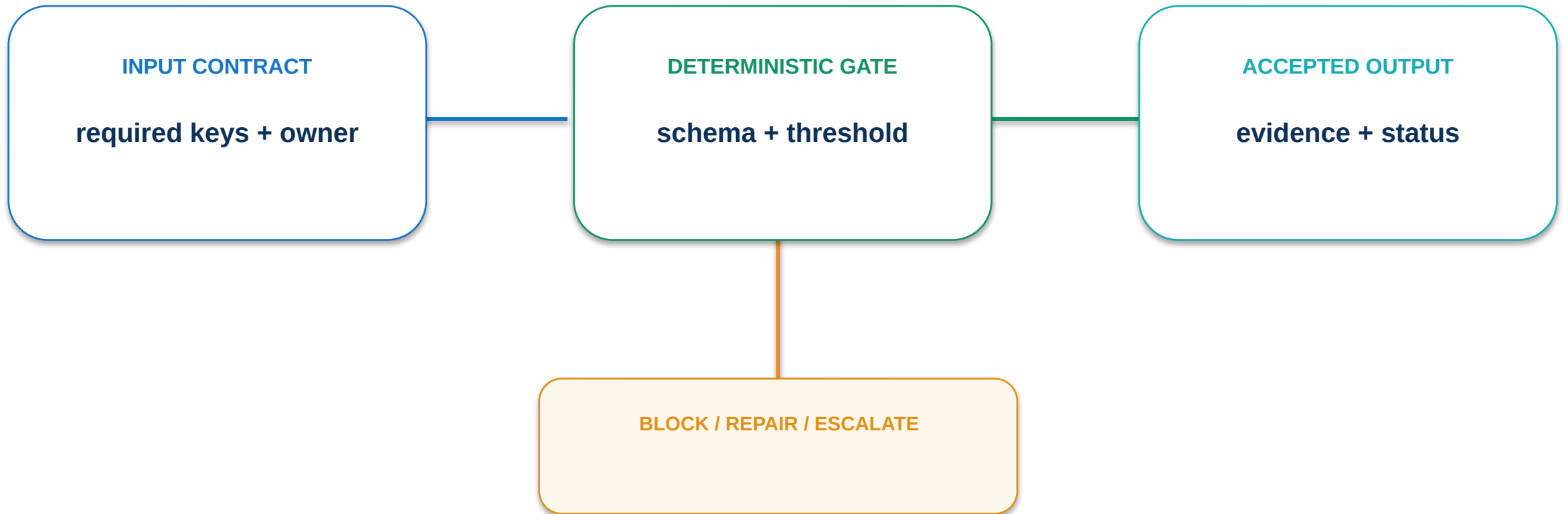
named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# 11 · Governed recommendation: execution path



## 11 · Governed recommendation: contract and control



**Recommend a bounded posture: pilot, expand, hold or stop.**

## 11 · Governed recommendation: evidence and decision

MEASURE

**Recommendation accepted with conditions  
and review date**

ACCEPTANCE EVIDENCE

Signed recommendation with conditions

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

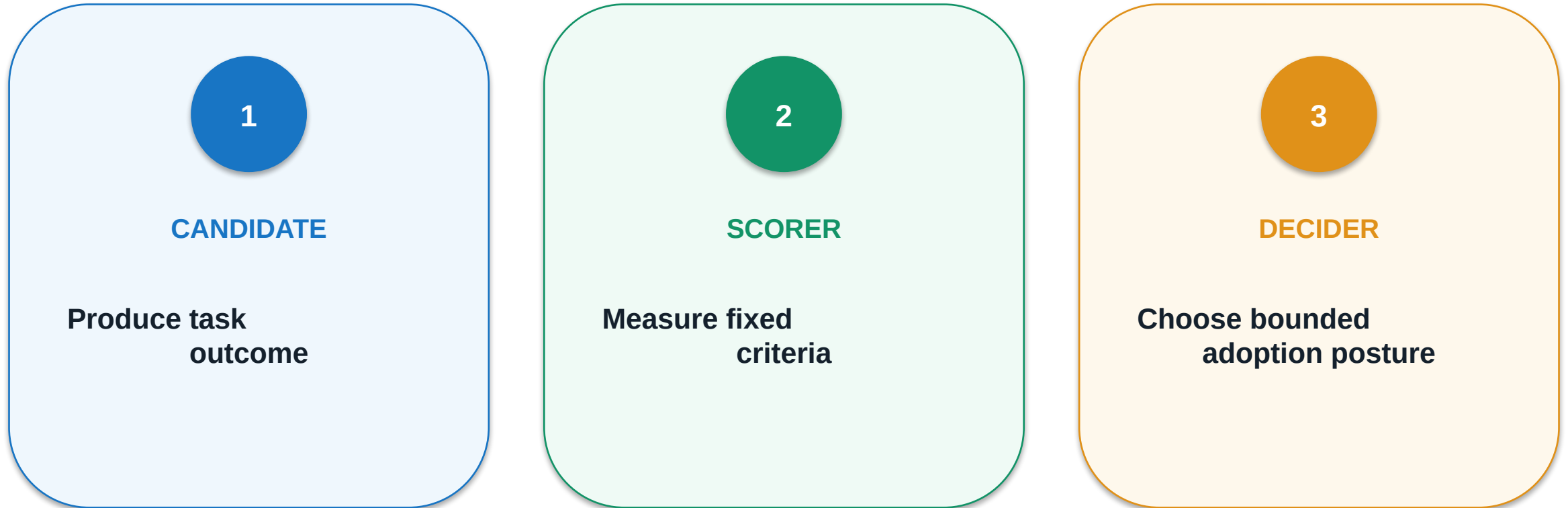
OWNER

named human

**Completion principle: evidence travels with the task, changed state and decision—not in an unsupported chat claim.**

# Evaluation boundary: model output, deterministic score and decision

---



# Adoption state: PILOT, EXPAND, HOLD and STOP

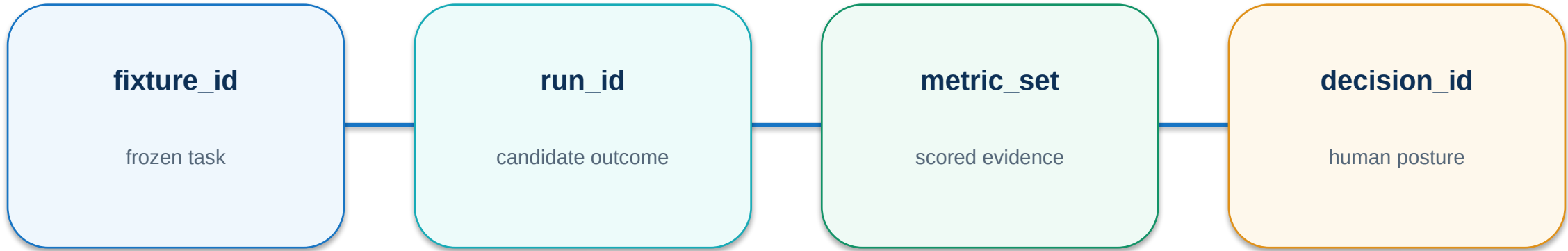
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The adoption posture records evidence, conditions, owner and review date.

## Evaluation chain: fixture → run → metric → recommendation

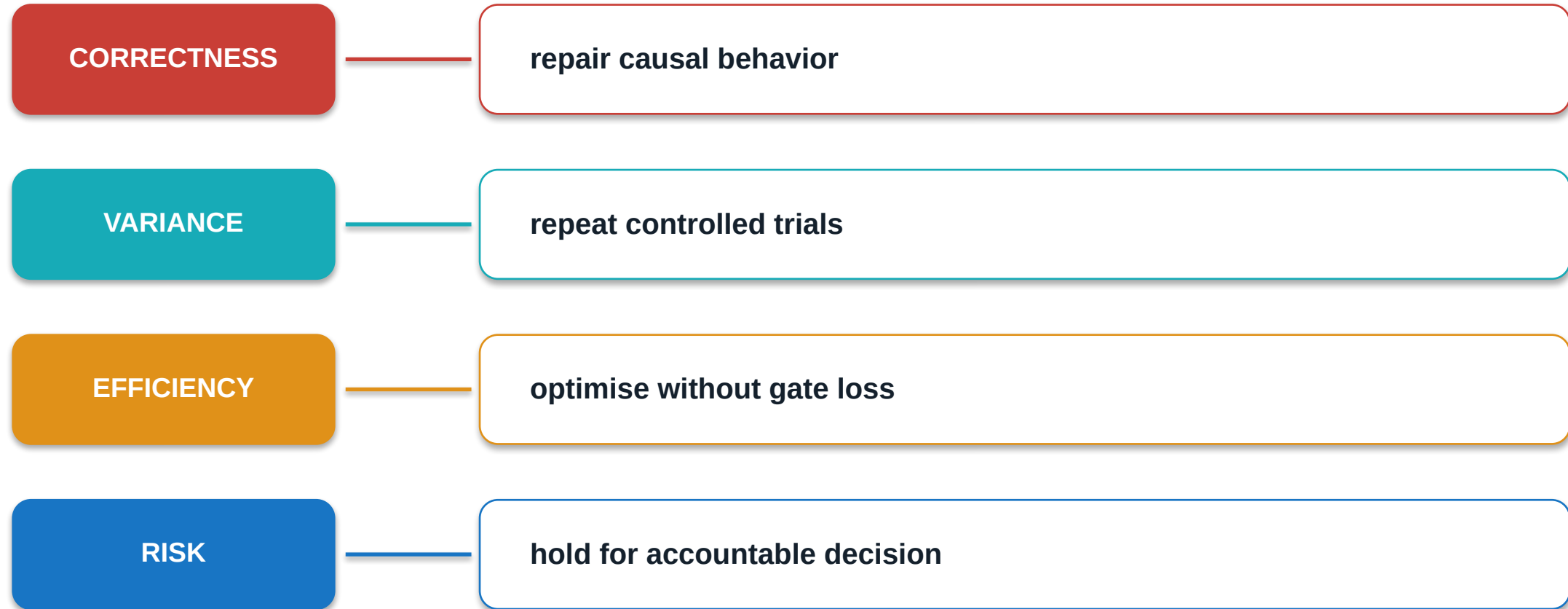
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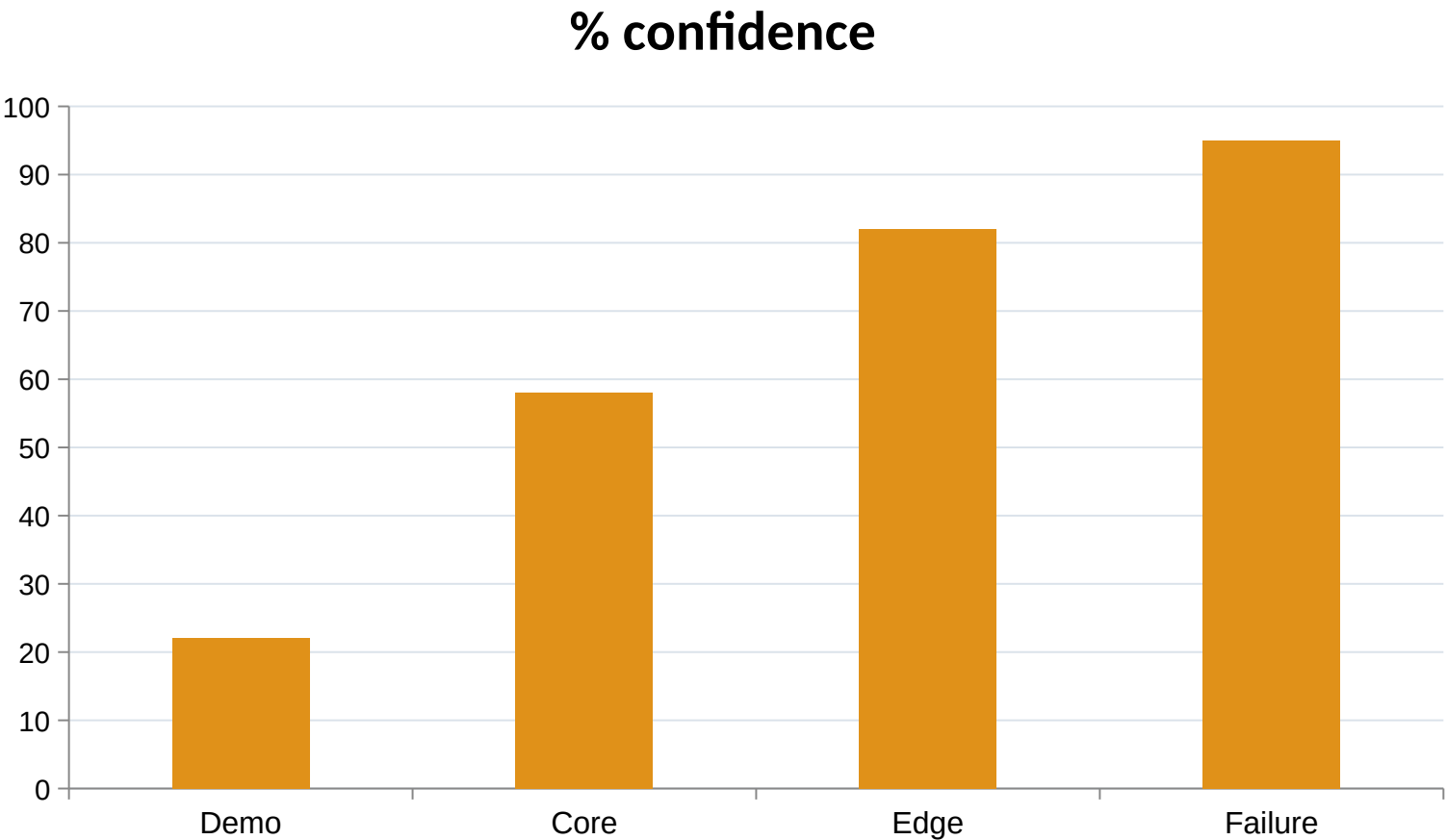
**A recommendation is defensible when every score resolves to a frozen task and run.**

# Failure taxonomy: correctness, variance, efficiency and risk

---



# Golden-task coverage reduces decision uncertainty

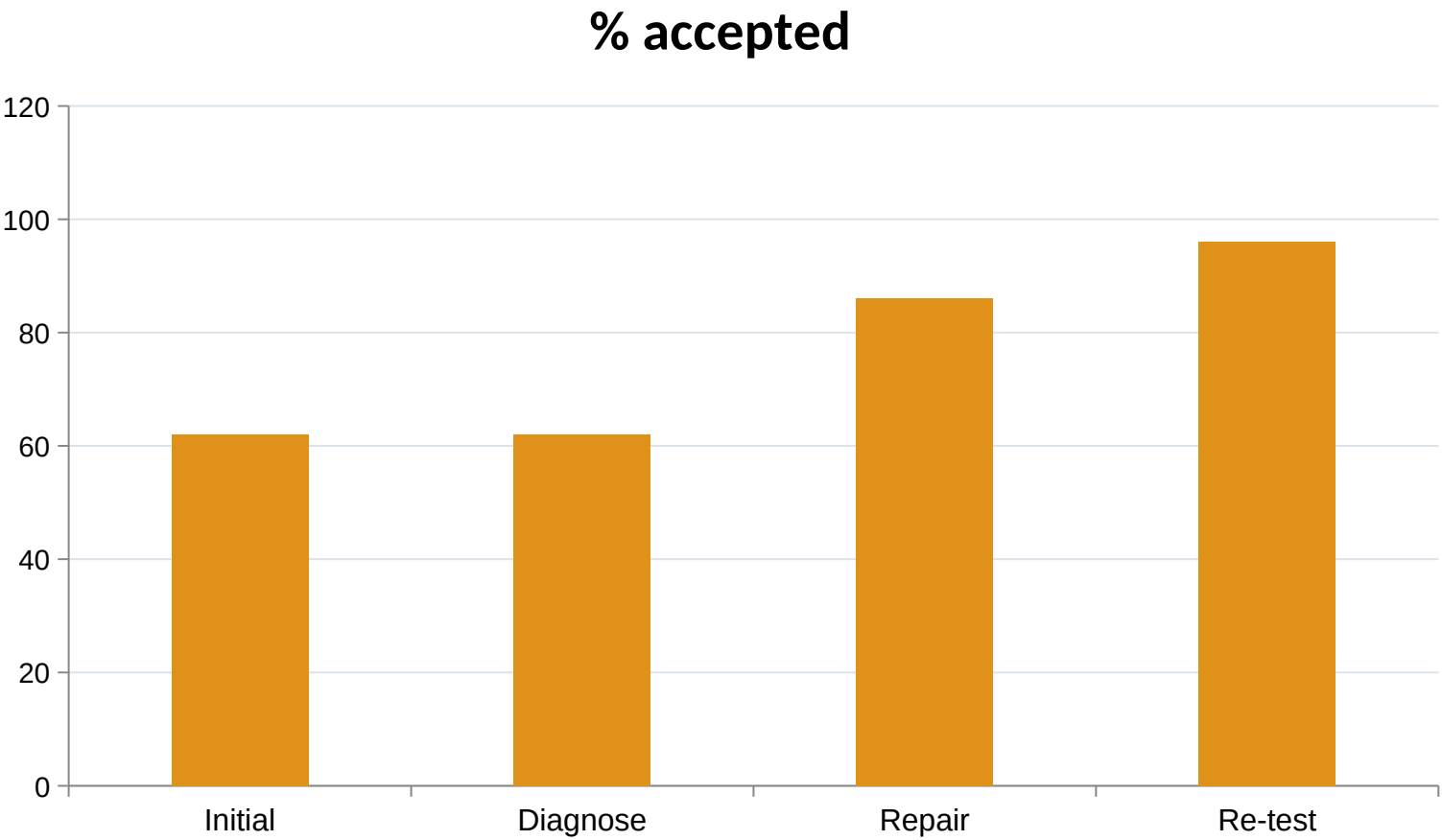


INSIGHT

**Representative core, edge and failure cases make an adoption decision more defensible.**

Illustrative synthetic values for mechanism explanation.

# Scoped repair raises accepted-task rate

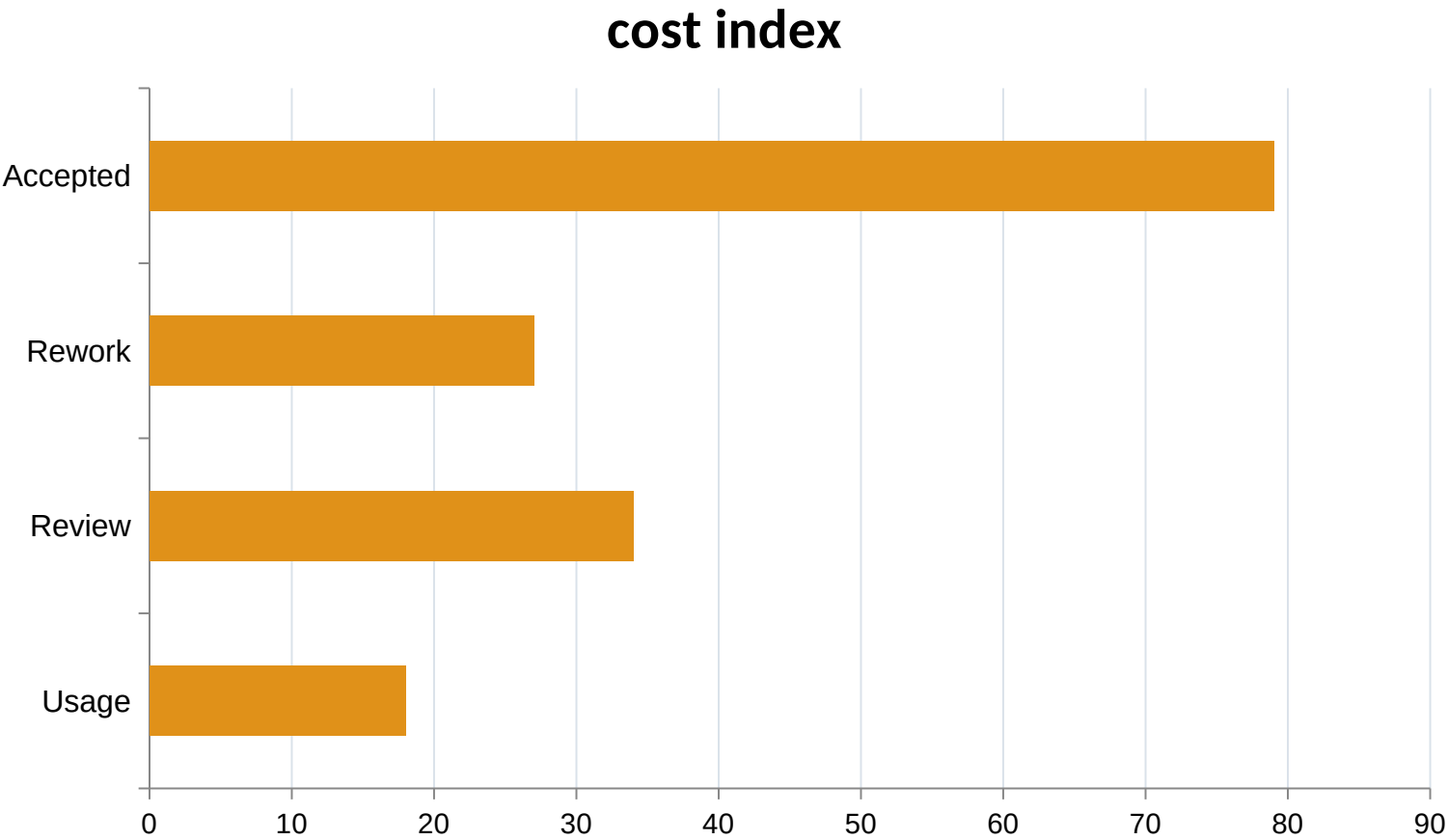


INSIGHT

**Targeted repair  
should improve the  
failed criterion  
without changing  
frozen expectations.**

Illustrative synthetic values for  
mechanism explanation.

# Fully loaded cost reveals hidden review work



INSIGHT

**Human review and rework belong in the cost of an accepted result.**

Illustrative synthetic values for mechanism explanation.

## Lab 09 · Golden-Task Evaluation Harness

---

### SCENARIO

**Run a repeatable golden-task suite and calculate correctness, reliability, latency and cost-per-accepted-task.**

### EXPECTED OUTPUT

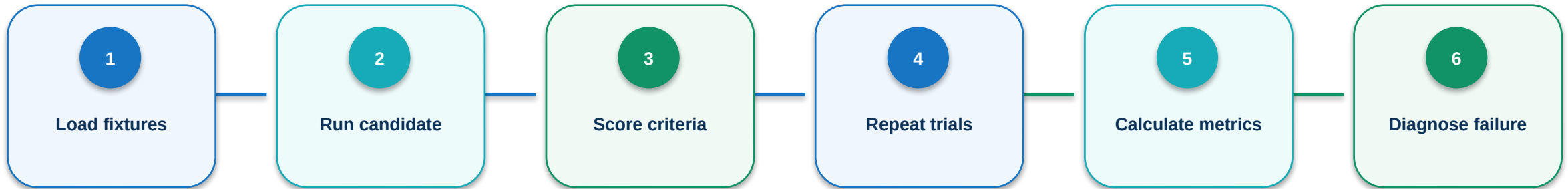
**Evaluation report with reproducible metrics**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-09-evaluation-harness/**

## Lab 09 · Codex workflow contract

---



**Each node consumes named keys, emits a status, and leaves retrievable execution evidence.**

## Lab 09 · evidence and failure gates

### PASS EVIDENCE

The report is reproducible, includes at least four decision-relevant metrics and links each failure to a fixture.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Lab 10 · Capstone: Governed Codex Adoption

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### SCENARIO

**Deliver a bounded TaskBoard improvement, evaluate it against the baseline and make a pilot, expand, hold or stop recommendation.**

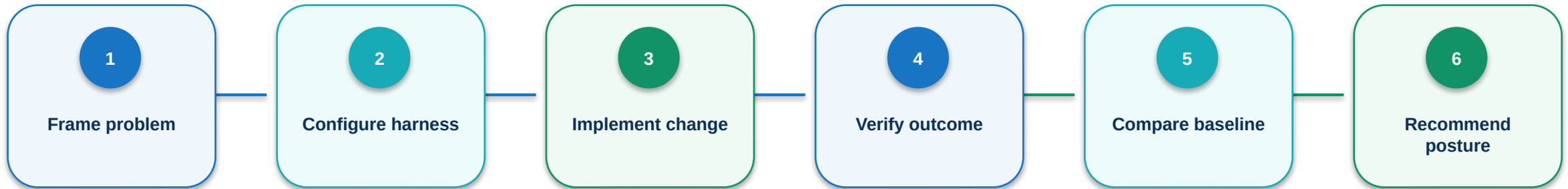
### EXPECTED OUTPUT

**Working change, evidence pack and governed recommendation**

**Detailed click paths, prompts, troubleshooting and acceptance checks are in the Learner Guide and lab PDF.**

**Self-contained folder: labs/lab-10-capstone-governed-adoption/**

# Lab 10 · Codex workflow contract



Consequential actions require a current diff, passing checks and a named decision.



## Lab 10 · evidence and failure gates

### PASS EVIDENCE

The solution works, the evidence pack covers correctness, efficiency and risk, and the recommendation states limits, controls, owner and review date.

### FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

**README.pdf**

detailed procedure

**starter/**

safe project state

**solution/**

reference outcome

**evidence/**

acceptance record

## Topic 4 technical checkpoint

---

1

**Which state field prevents a silent downstream assumption?**

2

**What evidence proves the mechanism completed successfully?**

3

**Which failure must stop rather than retry?**

4

**Who owns the consequential decision?**

## Topic 4 transfer challenge

---

1

**Change one input constraint and predict the affected nodes.**

2

**Name the identifier that preserves provenance across the change.**

3

**Define one machine threshold and one human decision.**

4

**Describe the minimum evidence required to resume the run.**

# Course achievement map

1

**LO1 Analyse agentic AI applications, their strengths, limitations and suitability for a work problem.**

2

**LO2 Correlate agent-control design with algorithm efficiency and evaluate the resulting process improvement.**

3

**LO3 Assess methods for evaluating the effectiveness, safety and reliability of Codex-enabled workflows.**

4

**LO4 Evaluate comparative effectiveness and recommend a governed Codex implementation using observable evidence.**

# Codex controls that must remain intact

---

## GATE

**Automation is complete only when governance travels with the change.**

- 1 Every instruction resolves to an applicable scope
- 2 Every reviewed change resolves to a diff and base state
- 3 Every consequential action has a named human owner
- 4 Every completion claim resolves to verification evidence

# Course feedback

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**FB**

**Share specific feedback that helps improve the course.**

**1**

**Technical depth**

**2**

**Lab usefulness**

**3**

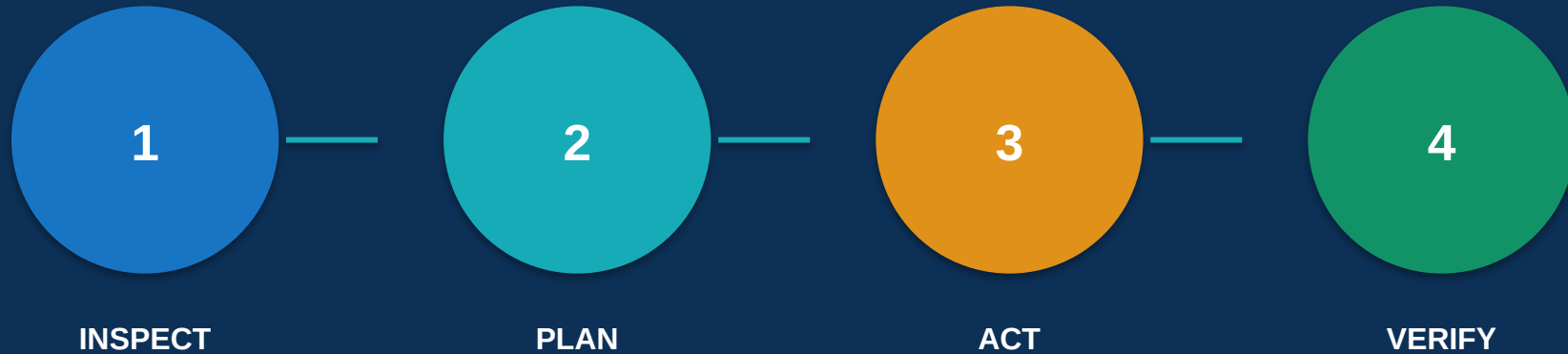
**Pacing and clarity**

**4**

**One improvement suggestion**

# Build Codex workflows that can prove what happened.

Evidence before completion • Human approval before consequence



Dr Alfred Ang | Tertiary Infotech Academy Pte. Ltd.

C695 • v1.0 • 20 September 2026